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Run Forecasts and Harvest Projections for 2025 Alaska Salmon Fisheries and Review of the 2024 Season

Edited by

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Alaska Department of Fish and Game

Divisions of Sport Fish and Commercial Fisheries



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Weights and measures (metric)		General		Mathematics, statistics	
centimeter	cm	Alaska Administrative Code	AAC	<i>all standard mathematical signs, symbols and abbreviations</i>	
deciliter	dL	all commonly accepted abbreviations	e.g., Mr., Mrs., AM, PM, etc.	alternate hypothesis	H _A
gram	g	all commonly accepted professional titles	e.g., Dr., Ph.D., R.N., etc.	base of natural logarithm	<i>e</i>
hectare	ha			catch per unit effort	CPUE
kilogram	kg			coefficient of variation	CV
kilometer	km	at compass directions:	@	common test statistics	(F, t, χ^2 , etc.)
liter	L			confidence interval	CI
meter	m			correlation coefficient	
milliliter	mL			(multiple)	R
millimeter	mm			correlation coefficient	
				(simple)	r
Weights and measures (English)		north	N	covariance	cov
cubic feet per second	ft³/s	south	S	degree (angular)	°
foot	ft	west	W	degrees of freedom	df
gallon	gal	copyright	©	expected value	<i>E</i>
inch	in	corporate suffixes:		greater than	>
mile	mi	Company	Co.	greater than or equal to	≥
nautical mile	nmi	Corporation	Corp.	harvest per unit effort	HPUE
ounce	oz	Incorporated	Inc.	less than	<
pound	lb	Limited	Ltd.	less than or equal to	≤
quart	qt	District of Columbia	D.C.	logarithm (natural)	ln
yard	yd	et alii (and others)	et al.	logarithm (base 10)	log
		et cetera (and so forth)	etc.	logarithm (specify base)	log₂, etc.
Time and temperature		exempli gratia		minute (angular)	'
day	d	(for example)	e.g.	not significant	NS
degrees Celsius	°C	Federal Information Code	FIC	null hypothesis	H₀
degrees Fahrenheit	°F	id est (that is)	i.e.	percent	%
degrees kelvin	K	latitude or longitude	lat or long	probability	P
hour	h	monetary symbols		probability of a type I error	
minute	min	(U.S.)	\$, ¢	(rejection of the null hypothesis when true)	α
second	s	months (tables and figures): first three letters	Jan,...,Dec	probability of a type II error	
Physics and chemistry		registered trademark	®	(acceptance of the null hypothesis when false)	β
all atomic symbols		trademark	™	second (angular)	"
alternating current	AC	United States		standard deviation	SD
ampere	A	(adjective)	U.S.	standard error	SE
calorie	cal	United States of America (noun)	USA	variance	
direct current	DC	U.S.C.	United States Code	population	Var
hertz	Hz			sample	var
horsepower	hp				
hydrogen ion activity (negative log of)	pH				
parts per million	ppm	U.S. state	use two-letter abbreviations		
parts per thousand	ppt, ‰		(e.g., AK, WA)		
volts	V				
watts	W				

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**RUN FORECASTS AND HARVEST PROJECTIONS FOR 2025 ALASKA
SALMON FISHERIES AND REVIEW OF THE 2024 SEASON**

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DEFINITION OF TERMS

Biological escapement goal	The number of salmon in a particular stock that the Alaska Department of Fish and Game has determined should be allowed to escape the fishery to spawn to achieve the maximum yield (human use). This determination is based on biological information about the fish stock in question. See 5 AAC 39.222(f)(3). (Also see <i>optimum escapement goal</i> .)
Commercial common property harvest	Harvests taken by traditional, competitive commercial fisheries (gillnet, purse seine, and troll), as opposed to commercial harvests resulting from hatchery cost recovery, fishing derbies, and sale of confiscated fish.
Commercial harvest	Harvests of fish that are used for commercial purposes. This includes fish caught by the commercial common property fishery and by hatchery operators for cost recovery; it excludes sport, subsistence, and personal use harvests.
Common property harvest	Harvests taken by the commercial common property fisheries, as well as the sport, subsistence, and personal use fisheries. This category excludes hatchery cost-recovery harvests.
Cost-recovery harvest	Harvests of salmon by hatchery operators in specially designated areas to fund the operation of hatcheries and other enhancement activities.
Enhanced salmon stock, runs	Hatcheries and other means of artificial propagation to create salmon runs or make existing salmon runs larger. Enhancement includes remote fish stocking, fertilization of lakes, and other techniques. See 5 AAC 39.222(f)(9).
Escapement, spawning population, or broodstock	The portion of a salmon run that is not harvested and survives to reach the spawning grounds or hatchery. See 5 AAC 39.222(f)(10)).
Harvest projections or harvest outlooks	Harvest outlooks are the best available estimates of upcoming harvest levels. Prepared by local biologists, outlooks are based on formal run forecasts when available. At other times outlooks are based on historical average catches, subjectively adjusted based on recent trends and local knowledge.
Optimal escapement goal	The number of salmon in a particular stock that should be allowed to spawn both to achieve sustainable runs based on biological needs of the stock and to meet social and allocative needs. See 5 AAC 39.222(f)(25).
Return	Return refers to an aggregation of salmon over several or more years that represent the surviving adult offspring from a single brood year. See 5 AAC 39.222(f)(30).
Run forecast	Forecasts of a run (harvest + escapement) are estimates of the fish that will return in a given year based on such information as parent-year escapements, subsequent fry abundance, and spring seawater temperatures. Run forecasts are generally thought to be more reliable than harvest outlooks, but run forecasts are provided only for selected areas.
Salmon run	Run refers to the total number of mature fish returning in a given year from ocean-rearing areas to spawn. See 5 AAC 39.222(f)(31).
Sustainable escapement goal	Sustainable escapement goal is defined as a level of escapement, indicated by an index or a range of escapement estimates, that is known to have provided for sustained yield over a 5- to 10-year period. A sustainable escapement goal is used in situations where a biological escapement goal cannot be estimated due to the absence of a stock-specific catch estimate. See 5 AAC 39.222(f)(36).

ABSTRACT

This report contains salmon run forecasts and harvest projections for 2025 as well as a detailed review of Alaska's 2024 commercial salmon season. The Alaska all-species salmon harvest for 2024 totaled approximately 103.5 million fish, about 32.2 million less fish than the preseason forecast of 135.7 million fish. This combined harvest was composed of approximately 244,000 Chinook (*Oncorhynchus tshawytscha*), 42.2 million sockeye (*O. nerka*), 1.8 million coho (*O. kisutch*), 40.0 million pink (*O. gorbuscha*), and 19.3 million chum salmon (*O. keta*). The Alaska Department of Fish and Game is expecting an increase in commercial salmon harvests in 2025. The 2025 total commercial salmon harvest (all species) projection of 214.6 million fish is expected to include approximately 144,000 Chinook salmon, 52.9 million sockeye salmon, 2.4 million coho salmon, 138.4 million pink salmon, and 20.8 million chum salmon. Compared to 2024 commercial harvests, the projected 2025 commercial harvests are expected to be as follows: 98.2 million more pink salmon, 10.8 million more sockeye salmon, 544,000 more coho salmon, and 876,000 more chum salmon.

Keywords: pink salmon, *Oncorhynchus gorbuscha*, sockeye salmon, *O. nerka*, chum salmon, *O. keta*, Chinook salmon, *O. tshawytscha*, coho salmon, *O. kisutch*, catch projection, run forecast, harvest projection, smolt outmigrations, sibling age classes, hatchery releases, fishing effort, salmon management

INTRODUCTION

This report contains salmon run forecasts and harvest projections for 2025 and a review of Alaska's 2024 commercial salmon season. Salmon escapement and harvest estimates reported in this document were summarized from the Alaska Department of Fish and Game (ADF&G) escapement and fish ticket databases. Data provided in this report supersede any data previously published. Detailed area-specific harvest and escapement summaries can be found in Area Management Reports.

ADF&G is expecting an increase in overall commercial salmon harvest in 2025, mostly due to an increase in pink salmon *Oncorhynchus gorbuscha* harvests compared to 2024. The 2025 total commercial salmon harvest (all species) projection of 214.6 million fish is expected to include approximately 144,000 Chinook salmon *O. tshawytscha*, 52.9 million sockeye salmon *O. nerka*, 2.4 million coho salmon *O. kisutch*, 138.4 million pink salmon, and 20.8 million chum salmon *O. keta*. Compared to 2024 commercial harvests, the projected 2025 commercial harvests are expected to be as follows: 98.2 million more pink salmon, 10.8 million more sockeye salmon, 544,000 more coho salmon, and 876,000 more chum salmon.

There is a great deal of uncertainty in forecasting pink salmon returns due to their fixed 2-year life history and therefore limited information to serve as the basis for predictions (i.e., no siblings returning during prior years). As a result, pink salmon harvest forecasts are generally based on harvests from previous brood years. A notable exception is Southeast Alaska where a joint ADF&G and National Oceanic and Atmospheric Administration (NOAA) survey and juvenile pink salmon outmigration index is the basis for predicting harvests the following year. During recent decades, Alaska-wide pink salmon returns have tended to be larger during odd years than during adjacent even years, although there is much regional variation to this trend.

Table 1 shows specific harvest projection numbers by species and fishing area, and the "Preliminary Forecasts" section of this document provides forecast details for specific runs by area. When the appropriate data were available, harvest forecasts were arrived at through quantitative projections based on information of previous spawning levels, smolt outmigrations, returns of sibling age classes, and survival rates for hatchery releases. Other projections were based on averages of recent harvests. Fishing effort influences harvests, and effort is partly determined

by market conditions in addition to the size of salmon runs. Therefore, these projections may not be indicative of actual harvests.

Except for the Southeast Alaska Chinook salmon fisheries, the South Peninsula June fisheries, and salmon fisheries in the Exclusive Economic Zone (EEZ) of Cook Inlet, Alaska salmon management will be based on inseason estimates of salmon run strength. Alaska fishery managers have the primary goal of maintaining spawning population sizes, not of reaching preseason harvest projections.

Salmon productivity, survival, and resulting returns are probably influenced by many freshwater and marine drivers including spawning escapements, hatchery releases, temperature, and complex trophic interactions such as prey availability, predation, and competition. Although a review of this topic is well beyond the scope of this report, NOAA's Ecosystem Status Reports provide a summary of recent physical and ecological conditions encountered by a variety of species in the North Pacific, including salmon during their marine phase (Ferriss 2024; Ortiz and Zador 2024; Siddon 2024; available online¹).

Table 1.—Projections of 2025 Alaska commercial salmon harvests, by fishing area and species, in thousands of fish.

Region & Area	Species					Total
	Chinook	Sockeye	Coho	Pink	Chum	
Southeast Alaska						
Natural production	—	782 ^a	866 ^a	28,401	1,395	31,444
Hatchery production ^b	—	104	595	599	14,101	15,399
Southeast Region total	103 ^c	886	1,461	29,000	15,496	46,946
Prince William Sound						
Natural production	6	2,251 ^d	232 ^e	16,788	443	19,720
Hatchery production ^f	—	1,020	62	45,143	1,958	48,182
Lower Cook Inlet						
Natural production	0 ^a	157 ^a	0	1,050	16 ^a	1,223
Hatchery production	—	218 ^g	0	731	0	949
Upper Cook Inlet	— ^h	— ⁱ	99 ^a	87 ^j	74 ^a	— ⁱ
Bristol Bay	7 ^a	36,400	46 ^a	10 ^j	331 ^a	36,794
Central Region total	14	44,975 ⁱ	439	63,808	2,821	112,058 ⁱ
Kodiak						
Natural production	8 ^a	1,311 ^k	165 ^a	21,005	372 ^a	22,861
Hatchery production ^l	—	264	37	10,786	268	11,355
Chignik	4 ^a	757 ^m	49 ^a	2,995 ^j	57 ^a	3,862
South Peninsula	14 ^a	2,621 ^a	168 ^a	10,639	1,142 ^a	14,584
North Peninsula	1 ^a	2,118 ^a	19 ^a	134 ^j	51 ^a	2,324
Westward Region total	27	7,071	438	45,559	1,890	54,986
Arctic-Yukon-Kuskokwim total	0	0	18	15	610	642
Statewide total	144	52,932 ⁱ	2,356	138,382	20,818	214,632 ⁱ

-continued-

¹ Ecosystem Status Reports For The Gulf Of Alaska, Bering Sea And Aleutian Islands: NOAA Fisheries: <https://www.fisheries.noaa.gov/alaska/ecosystems/ecosystem-status-reports-gulf-alaska-bering-sea-and-aleutian-islands> (accessed February 21, 2025)

Table 1.–Page 2 of 2.

Note: En dashes indicate no projection, and zeros indicate projection of <500 fish. Columns and rows may not total exactly due to rounding.

- ^a Average harvest of the previous 5 years (2020–2024).
- ^b Hatchery salmon projections made by Southern Southeast Regional Aquaculture Association, Northern Southeast Regional Aquaculture Association, Douglas Island Pink and Chum, Armstrong-Keta Inc., and Metlakatla Indian Community less broodstock (5-year average), and excess. Wild chum salmon catch estimated as 9% of total catch.
- ^c The allowable catch of Chinook salmon in Southeast Alaska is determined by the Pacific Salmon Commission.
- ^d Includes formal natural harvest estimates for Prince William Sound and Copper/Bering River Districts.
- ^e Average harvest of the previous ten years (2015–2024).
- ^f Hatchery salmon projections made by Prince William Sound Aquaculture Corporation and Valdez Fisheries Development Association. Gulkana Hatchery projection made by ADF&G, less broodstock (5-year average).
- ^g Hatchery salmon projections made by Cook Inlet Aquaculture Corporation minus broodstock (5-year average).
- ^h Upper Cook Inlet Chinook salmon harvest forecast is not available for 2025.
- ⁱ An Upper Cook Inlet sockeye salmon commercial harvest forecast is not available for 2025. Central Region and Statewide totals include 4.93 million fish available for harvest to all user groups.
- ^j Average harvest of the previous 5 odd years (2015–2023).
- ^k Total Kodiak harvest of natural run sockeye salmon includes projected harvests from formally forecasted systems, projected Chignik harvest at Cape Igvak, and projected harvest from additional minor systems.
- ^l Hatchery projections made by Kodiak Regional Aquaculture Association. Sockeye salmon hatchery projections include enhanced Spiridon Lake sockeye salmon run harvest forecast and other Kodiak Regional Aquaculture Association projections.
- ^m The Chignik sockeye salmon forecast is projected harvest of sockeye within Chignik Management Area.

The Alaska all-species salmon harvest for 2024 totaled approximately 103.5 million fish, about 32.2 million less fish than the preseason forecast of 135.7 million fish. This combined harvest was composed of approximately 244,000 Chinook, 42.2 million sockeye, 1.8 million coho, 40.0 million pink, and 19.3 million chum salmon. Table 2 shows 2024 harvest numbers by salmon species and fishing area in units of thousands of fish harvested, and Table 3 provides this information in units of thousands of whole pounds harvested. Tables 4–7 provide detailed information on the 2024 harvest by area and species.

Table 2.—2024 Alaska commercial salmon harvests, by fishing area and species, in thousands of fish.

Fishing area	Species					Total
	Chinook	Sockeye	Coho	Pink	Chum	
Southeast Region Total ^{a,b,c}	207	797	1,355	20,128	15,750	38,236
Prince William Sound ^{a,d}	10	3,204	167	9,952	1,702	15,035
Lower Cook Inlet ^{e,f}	0	298	0	17	3	319
Upper Cook Inlet ^{f,g}	0	1,871	25	42	74	2,011
Bristol Bay ^g	5	31,659	31	78	509	32,281
Central Region Total	15	37,033	223	10,088	2,288	49,647
Kodiak Area ^{e,f,g}	1	1,619	64	7,322	498	9,505
Chignik ^{g,h}	12	281	68	892	61	1,315
South Peninsula ^{f,g,h}	8	1,291	81	1,591	583	3,553
North Peninsula	1	1,173	8	17	67	1,266
Westward Region Total	22	4,365	221	9,822	1,209	15,639
Arctic-Yukon-Kuskokwim Region Total ^{f,g}	0	0	14	3	10	27
Total Alaska	244	42,195	1,812	40,041	19,257	103,549

Note: En dashes indicate no harvest, and zeros indicate harvest activity but <500 fish. Columns may not total exactly due to rounding. Confidential data omitted.

^a Chinook salmon adults and jacks are totaled.

^b Catch accounting period for the 2024 Chinook salmon troll season spans from October 1, 2023, to September 30, 2024.

^c Total includes fish that were confiscated, harvested in sport fisheries derbies and later sold, and harvested in test fisheries.

^d Total includes hatchery sales for operating expenses and broodstock harvests.

^e Total includes hatchery sales for operating expenses and hatchery donated fish but not broodstock.

^f Total includes commercially harvested fish retained for personal use.

^g Total includes commercial harvest that was discarded, confiscated, seized, or donated.

^h Total includes test fish sales.

Table 3.—2024 Alaska commercial salmon harvests, by fishing area and species, in thousands of whole lb.

Fishing area	Species					Total
	Chinook	Sockeye	Coho	Pink	Chum	
Southeast Region Total ^{a,b,c}	2,427	4,188	8,924	58,074	105,987	179,600
Prince William Sound ^{a,d}	129	17,293	1,294	32,853	10,835	62,404
Lower Cook Inlet ^{e,f}	1	1,354	1	69	25	1,449
Upper Cook Inlet ^{f,g}	2	8,938	105	136	305	9,486
Bristol Bay ^g	45	143,426	159	284	2,727	146,642
Central Region Total	177	171,010	1,559	33,342	13,892	219,980
Kodiak ^{e,f,g}	8	7,561	419	24,120	3,397	35,504
Chignik ^{g,h}	37	1,691	393	3,249	412	5,781
South Peninsula & Aleutians ^{f,g,h}	54	5,989	431	5,628	3,382	15,484
North Peninsula	8	5,775	43	53	462	6,342
Westward Region Total	108	21,016	1,286	33,050	7,652	63,112
Arctic-Yukon-Kuskokwim Region Total ^{f,g}	0	2	86	10	69	167
Total Alaska	2,712	196,216	11,855	124,477	127,600	462,859

Note: Dashes indicate no harvest, and zeros indicate harvest activity but <500 fish. Columns may not total exactly due to rounding. Confidential data omitted.

^a Chinook salmon adults and jacks are totaled.

^b Catch accounting period for the 2024 Chinook salmon troll season goes from October 1, 2023, to September 30, 2024.

^c Total includes fish that were confiscated, harvested in sport fish derbies and later sold, and harvested in test fisheries.

^d Total includes hatchery sales for operating expenses and broodstock harvests.

^e Total includes hatchery sales for operating expenses and hatchery donated fish but not broodstock.

^f Total includes commercially harvested fish retained for personal use.

^g Total includes harvest that was discarded, confiscated, seized, or donated.

^h Total includes test fish sales.

Inseason harvest information, postseason statistics, and other information about salmon in Alaska can be found online at <http://www.adfg.alaska.gov/index.cfm?adfg=fishingCommercial.main>.

ADF&G's 4 major commercial fishery management regions (Southeast, Central, Arctic-Yukon-Kuskokwim, and Westward) are shown in Figure 1. These regions supersede any references to ADF&G's former statistical regions.



Figure 1.—The 4 ADF&G fishery management regions (Southeast, Central, Arctic-Yukon-Kuskokwim, and Westward) of the Division of Commercial Fisheries.

REVIEW OF THE 2024 ALASKA COMMERCIAL SALMON FISHERIES

SOUTHEAST REGION

SOUTHEAST ALASKA AND YAKUTAT AREAS

The combined 2024 Southeast Alaska and Yakutat (SEAK) Area cumulative commercial salmon harvest, including hatchery cost recovery, was 38.2 million fish (Table 4). Of this total, commercial common property harvest contributed approximately 32.8 million fish (86% of the total harvest). The overall harvest in numbers of salmon was higher than the 2023 harvest by 6.6 million fish and a little more than half of the 2023 harvest. The total commercial harvest proportions by species were 1% Chinook, 2% sockeye, 4% coho, 53% pink, and 41% chum salmon. The all-species total harvest was 93% of the recent 10-year average (2014–2023) of 41 million fish and 93% of the long-term average (1962–2023) of 41.3 million fish.

Chinook Salmon

Harvest Summary: The 2024 Chinook salmon harvest of 206,000 fish was below both recent 10-year (254,000 fish) and long-term (288,000 fish) averages. Preliminary harvest estimates of coastwide Chinook salmon accountable under the Pacific Salmon Treaty included 144,000 fish by troll gear, 9,600 fish by purse seine gear, and under 1,500 fish by gillnet gear. In 2024, a total of 25,800 Alaska hatchery-origin Chinook salmon were commercially harvested in SEAK common property fisheries, and approximately 19,400 hatchery-origin Chinook salmon were harvested in private hatchery cost-recovery fisheries.

Sockeye Salmon

Harvest Summary: The 2024 commercial harvest of sockeye salmon of about 797,000 fish was below both the recent 10-year and long-term averages. The SEAK purse seine fishery harvest of 409,000 fish accounted for approximately 51% of the regional total sockeye salmon harvest. The drift gillnet fishery harvest of 250,000 fish accounted for 31% of the regional total sockeye salmon harvest. The set gillnet fishery harvest of 59,000 fish accounted for 7% of the regional total sockeye salmon harvest.

Coho Salmon

Harvest Summary: The 2024 coho salmon harvest of approximately 1.4 million fish was below the recent 10-year and long-term averages. The coho salmon harvest in the troll fishery was approximately 579,000 fish and accounted for 42% of the regional coho salmon harvest. The total drift gillnet harvest of 221,000 fish accounted for approximately 16% of the regional coho salmon harvest, and the set gillnet harvest of 110,000 fish accounted for 10% of the regional coho salmon harvest. The total purse seine harvests of 194,000 fish accounted for 14% of the regional coho salmon harvest.

Pink Salmon

Harvest Summary: The 2024 commercial pink salmon harvest was just over 20.1 million fish and was higher than the preseason forecast of 19.2 million fish. Approximately 52% of the total regional salmon harvest was composed of pink salmon. Most of the commercially harvested pink salmon in the SEAK Area were harvested in the purse seine fishery (18.4 million fish, 91%). The 2024 harvest and escapement continued an increasing trend in even-year pink salmon runs.

Chum Salmon

Harvest Summary: The 2024 SEAK chum salmon harvest was 15.7 million fish, 61% greater than the recent 10-year average of 9.8 million fish. Most commercial chum salmon fishing opportunities in the SEAK Area are attributable to hatchery production. During the 2024 commercial fishing season, 5.8 million chum salmon were harvested in the purse seine fishery (37% of the regional total), 3.9 million fish were harvested in the drift gillnet fishery (25%), and 4.6 million chum salmon were harvested through hatchery cost-recovery efforts.

Summary by Troy Thynes, Southeast Region Salmon/Herring Fisheries Management Coordinator, ADF&G, Petersburg.

Table 4.–2024 Southeast Region commercial salmon harvests, by fishing area and species in thousands of fish.

Fishery	Species					Total
	Chinook ^{a,b}	Sockeye	Coho	Pink	Chum	
Purse Seine						
Southern purse seine traditional	10	268	167	15,174	1,473	17,093
Northern purse seine traditional	1	125	22	2,918	1,097	4,162
Hatchery terminal	10	15	4	306	3,256	3,592
Total purse seine	20	409	194	18,399	5,825	24,847
Drift gillnet						
Tree Point	1	25	45	88	402	561
Prince of Wales	1	41	58	15	125	240
Stikine	1	16	10	3	88	117
Taku-Snettisham	1	89	33	7	828	957
Lynn Canal	0	57	62	11	814	944
Drift gillnet hatchery terminal	8	22	14	18	1,609	1,672
Total drift gillnet	12	250	221	142	3,866	4,491
Set gillnet (Yakutat)	0	59	110	8	0	177
Troll						
Hand troll						
Traditional	3	0	10	1	1	15
Hatchery terminal	0	0	0	1	15	16
Spring areas	0	0	0	0	2	3
Total hand troll	4	0	10	3	18	34
Power troll						
Traditional	131	1	569	57	803	1,561
Hatchery terminal	2	0	0	1	15	18
Spring areas	14	0	0	0	2	16
Total Power troll	147	1	570	58	819	1,595
Total Troll	151	1	579	61	837	1,629
Annette Island Reservation	3	18	30	1,065	525	1,641
Hatchery cost recovery	19	57	219	412	4,632	5,339
Miscellaneous ^c	1	5	1	40	65	112
Southeast Region Total	207	797	1,355	20,128	15,750	38,236

Note: En dashes indicate no harvest, and zeros indicate harvest activity but <500 fish. Columns may not total exactly due to rounding.

^a Chinook salmon adults and jacks are totaled.

^b Catch accounting period for the 2024 Chinook salmon troll season spans from October 1, 2023, to September 30, 2024.

^c Includes fish that were confiscated, harvested in sport fishery derbies and later sold, and harvested in test fisheries.

CENTRAL REGION

PRINCE WILLIAM SOUND AREA

The 2024 cumulative Prince William Sound (PWS) Area commercial common property fishery (CCPF) and hatchery cost-recovery salmon harvest was 15.0 million fish (Table 5). The overall harvest consisted of approximately 9,700 Chinook, 3.2 million sockeye, 167,000 coho, 9.9 million pink, and 1.7 million chum salmon. The CCPF accounted for 61% (9.2 million fish) of the total harvest; the remaining 39% (5.8 million fish) were attributed to hatchery cost-recovery fisheries and broodstock collection.

Chinook Salmon

Run and Escapement Summary: The 2024 Chinook salmon total run forecast was 47,000 fish. Preliminary Chinook salmon abundance estimates indicated spawning escapement was below the lower bound of the sustainable escapement goal (SEG) range of 21,000 fish.

Harvest Summary: The CCPF harvest of 9,200 Chinook salmon in the Copper River District was approximately 25% below the 10-year (2014–2023) average harvest of 12,200 fish.

Sockeye Salmon

Run and Escapement Summary: The 2024 Copper River sockeye salmon total run forecast was 2.00 million fish. Gulkana Hatchery represented 36,000 sockeye salmon in this total run forecast. The preseason commercial harvest forecast for the Copper River District sockeye salmon was 1.3 million fish.

The 2024 Miles Lake sonar passage was 948,206 salmon, which was above the inriver run goal of 627,000 fish. Considering preliminary projections of inriver harvest, the Upper Copper River sockeye salmon SEG range of 360,000–750,000 fish was likely achieved. Sockeye salmon escapement to the Copper River Delta was approximately 87,000 fish, which was within the SEG range of 55,000–130,000 fish.

The 2024 forecast of the sockeye salmon run to Coghill Lake was 273,000 fish, with 243,000 fish available for common property fishery (CPF) harvest. Approximately 87,000 sockeye salmon passed through the Coghill River weir, which was within the SEG range of 20,000–75,000 fish.

Prince William Sound Aquaculture Corporation (PWSAC) forecasted a run of approximately 864,000 Main Bay Hatchery enhanced sockeye salmon in 2024.

Harvest Summary: The 2024 Copper River District CCPF sockeye salmon harvest of 1.37 million fish was approximately 54% greater than the recent 10-year average harvest of 889,000 fish. Wild sockeye salmon accounted for 69% (952,000 fish) of the CCPF harvest in the Copper River District. Gulkana Hatchery commercial harvest was the second highest in the last 10 years, contributing 250,000 sockeye salmon, or 18% of the total commercial harvest. Main Bay Hatchery (MBH) contributed 172,000 sockeye salmon, or 13% of the Copper River sockeye salmon harvest.

The Coghill District sockeye salmon CCPF harvest of 99,000 fish was 75% wild fish, which was 45% below the recent 10-year average of 178,000 wild fish.

The CCPF harvest of sockeye salmon in the Eshamy District gillnet fishery (drift gillnet and setnet) was 1.17 million fish. The proportion of wild sockeye salmon harvested in the Eshamy District

CCPF was 7%. PWSAC harvested approximately 420,000 sockeye salmon for cost recovery and broodstock, almost 3.5 times the 10-year average of 127,000.

The Unakwik District CCPF gillnet harvest (drift and seine) of sockeye salmon was approximately 900 fish, well below the 10-year average harvest of 8,300 sockeye salmon.

Coho Salmon

Run and Escapement Summary: The Copper River District's 2024 preseason CCPF harvest forecast was 220,000 coho salmon. The Copper River Delta sum of peak escapement counts of approximately 40,000 fish was within the SEG range of 32,000–50,000 fish. The Bering River District coho salmon spawning escapement estimate of 29,000 fish was above the SEG range of 13,000–25,000 fish.

PWSAC forecasted a 2024 run of 62,000 coho salmon to Wally Noerenberg Hatchery.

The Valdez Fisheries Development Association (VFDA) enhanced coho salmon forecast was 54,400 fish, which yields a projected CCPF harvest of 13,000 fish.

Harvest Summary: The season total CCPF drift gillnet coho salmon harvest in the Copper River District of 117,000 fish was 42% below the 10-year average of approximately 201,000 fish. Similarly, the Bering River District CCPF drift gillnet harvest of approximately 42,000 coho salmon was 28% below the 10-year average harvest of 58,000 fish.

The Coghill District coho salmon CCPF harvest (purse seine and drift gillnet) was 500 fish. This harvest was 99% below forecast and marked the eighth time in the last 10 years that fewer than 15,000 coho salmon were harvested. Coho salmon harvest in the Eastern District was approximately 2,300 fish, much lower than the VFDA preseason forecasted harvest of 54,400 fish.

Enhanced coho salmon from VFDA are managed primarily through a recreational fishery, but the commercial fleet inadvertently harvests them throughout PWS. The CCPF harvest of VFDA coho salmon is unknown due to the absence of otolith sampling. However, it is assumed that VFDA coho salmon makes up a significant proportion of the coho salmon harvest in the Eastern District purse seine fishery.

Pink Salmon

Run and Escapement Summary: The 2024 pink salmon season was one of the worst on record. The pink salmon forecast was for a total run of 32.2 million fish, apportioned among 3 returns: 47% VFDA, 32% PWSAC, and 21% wild. After accounting for wild escapement, cost recovery, and broodstock, the projected harvest was 22.8 million pink salmon.

The total observed run in 2024 was 11.5 million pink salmon, 64% below forecast. The run composition was 49% VFDA, 27% PSWAC, and 24% wild. Composition estimates are based on otolith contributions. Pink salmon escapement goals were met in all districts.

Combined with an aerial escapement index of 1.4 million fish, the estimated wild pink salmon return in 2024 was approximately 2.7 million, below the 5-year, even-year (2014–2022) average return of 5.1 million wild fish. The 3.1 million PSWAC pink salmon run was 70% below the forecast and 62% below the even-year average of 8.2 million fish. In total, 96% (approximately 3.0 million fish) of the PWSAC run was collected for cost recovery and broodstock; PWSAC did not achieve its cost recovery revenue goal for 2024. The total run of 5.6 million VFDA pink salmon was 64% below forecast and 60% below the 5-year, even-year average of 14.3 million fish. In

total, 36% (2.0 million fish) of the VFDA run was collected for cost recovery and broodstock; VFDA did not achieve its egg-take goal for 2024.

Harvest Summary: The 2024 PWS pink salmon harvest (including all CCPF harvest, hatchery cost recovery, broodstock, and raceway sales) was just under 10.0 million fish, 62% below the 5-year, even-year average of 26.1 million fish. Approximately 4.9 million fish were harvested in the purse seine CCPF compared to the 5-year, even-year average CCPF purse seine harvest of 21.5 million fish. The number of permits fished in the PWS purse seine fishery decreased from 209 permits in 2023 to 177 permits in 2024. There were 28 dual permit operations in PWS for the 2024 season.

Pink salmon CCPF drift gillnet harvest in the Coghill District was 52,000 fish. The proportion of wild pink salmon in the Coghill District CCPF harvest was 85%.

In the Eshamy District, approximately 115,000 pink salmon were harvested in the CCPF gillnet (drift gillnet and setnet) fishery. The proportion of wild pink salmon in the Eshamy District CCPF harvest was 96%.

Chum Salmon

Run and Escapement Summary: The 2024 chum salmon total run forecast was 4.6 million fish, with a commercial harvest forecast of 2.8 million fish. Most of the total run forecast, 4.0 million fish (86%), was attributed to PWSAC hatchery production. The forecast for the 2024 hatchery chum salmon run to Wally Noerenberg Hatchery (WNH) was 2.82 million fish. PWSAC projected approximately 1.60 million fish for cost recovery and broodstock, leaving 1.22 million for commercial harvest. Approximately 240,000 fish were forecasted to return to the Armin F. Koernig hatchery (AFK), and 920,000 fish were forecasted to return to the Port Chalmers Subdistrict. Based on the department's wild stock chum salmon forecast of 639,000 fish, there was a total commercial harvest forecast of 469,000 wild chum salmon. Managing for each district's escapement goal, the department's objective was to secure a combined escapement total of 228,000 wild chum salmon.

Chum salmon escapement was met in 3 of the 5 districts surveyed for wild chum salmon escapement. Poor weather conditions impacted aerial surveys and prevented later season surveys in the districts that did not make their SEGs.

Harvest Summary: The 2024 CCPF harvest of chum salmon was approximately 1.7 million fish, 39% below the preseason harvest forecast and the 10-year (2014–2023) average harvest of 3.3 million fish. PWSAC harvested approximately 543,000 chum salmon for cost recovery and broodstock in the Coghill District. The drift gillnet harvest of chum in the Coghill District totaled 233,000, 81% below both the forecast and the 10-year average, with wild chum accounting for 4% of this commercial catch.

Based on contribution estimates, the CCPF harvest of enhanced chum salmon at the Port Chalmers remote release site was 656,000 fish, which is 29% below the preseason forecast of 920,000 fish. In the Southwestern District, the CCPF harvest of enhanced chum salmon at AFK was 109,000 fish, 55% below the preseason forecast of 240,000 fish.

Summary by Heather Scannell, Jeremy Botz, and Alex Niepoth, Area Management Biologists, ADF&G, Cordova.

LOWER COOK INLET AREA

The 2024 Lower Cook Inlet (LCI) Area commercial salmon harvest was 319,000 fish. The harvest consisted of approximately 112 Chinook, 298,000 sockeye, 98 coho, 16,700 pink, and 3,300 chum salmon (Table 5). Approximately 166,300 fish (52%) were harvested in the commercial fishery, and 155,100 fish (48%) were harvested through hatchery cost recovery.

Chinook Salmon

Harvest Summary: The 2024 LCI harvest of 112 Chinook salmon was below the 10-year average LCI harvest of 508 fish. The set gillnet Chinook salmon harvest for the Southern District was 102 fish, less than the 10-year average harvest of 379 fish. An additional 10 Chinook salmon were harvested by purse seine gear in the Southern and Outer Districts.

Sockeye Salmon

Run and Escapement Summary: Cook Inlet Aquaculture Association (CIAA) forecasted a run of 30,300 sockeye salmon to the Kirschner Lake remote release site. CIAA also forecasted a total run of 111,500 sockeye salmon to Resurrection Bay facilities, with all but 10,300 of these fish anticipated to be used for broodstock or cost-recovery purposes. An additional 71,000 fish were forecast to return to Kachemak Bay release sites with 64,000 of these fish anticipated to be used for cost recovery or brood harvest.

Sockeye salmon escapement in the Kamishak District was within the SEG ranges for Chenik Lake, and the Amakdedori River, and below the SEG range at Mikfik Lake.

Sockeye salmon escapements were within the SEG range for Delight Lake and above the SEG range for Desire Lake in the Outer District.

The estimated escapement of sockeye salmon through the Bear Creek weir of 13,894 fish was above the desired inriver passage goal range of 5,520–13,520 fish. This goal combines the SEG range (600–8,600 fish) and the estimated 4,920 fish required for broodstock for the CIAA Resurrection Bay sockeye salmon program at the Trail Lakes Hatchery. CIAA harvested fewer broodstock than anticipated (2,185 fish), allowing approximately 11,709 sockeye salmon to remain in the lake and spawn naturally. This estimate is above the SEG range of 600–8,600 fish for this system. The only other index stock in the Eastern District is Aialik Lake, where the final escapement of approximately 11,580 fish was above the SEG range of 3,200–5,400 fish.

Harvest Summary: The 2024 CCPF purse seine harvest for the Southern District was 79,460 sockeye salmon, which was higher than the previous 10-year average harvest of 59,743 fish. The set gillnet harvest for the Southern District was 23,065 fish, which was below the previous 10-year average harvest of 24,113 fish.

The Kamishak Bay District CCPF sockeye salmon harvest was confidential due to fewer than 3 permit holders reporting deliveries. The previous 10-year average harvest is 41,662 fish.

The Outer District harvest was 35,017 sockeye salmon, which was above the previous 10-year average harvest of 10,652 fish.

Due to small runs in recent decades, no wild sockeye salmon were forecast to be available for commercial harvest from the Eastern District. Portions of Resurrection Bay were open for commercial harvest from June 24 through July 12. The Eastern District commercial harvest was confidential due to fewer than 3 permit holders reporting deliveries.

Coho Salmon

Harvest Summary: The 2024 purse seine harvest for the Southern District was 31 coho salmon, which is below the 10-year average harvest of 1,248 fish. The set gillnet harvest for the Southern District was 42 fish, which is less than the previous 10-year average harvest of 2,417 fish.

The Kamishak Bay District CCPF coho salmon harvest was confidential due to fewer than 3 permit holders reporting deliveries. The previous 10-year average harvest was 1,321 fish.

The Outer District harvest was less than 25 coho salmon, which was lower than the 10-year average harvest of 380 fish.

Pink Salmon

Run and Escapement Summary: None of the aggregated pink salmon SEGs were achieved for the Southern, Outer, or Kamishak Bay Districts.

Harvest Summary: The 2024 purse seine harvest for the Southern District was 303 pink salmon, less than the previous 10-year average harvest of 181,882 fish. The set gillnet harvest for the Southern District was 1,128 fish, less than the previous 10-year average harvest of 21,288 fish.

The Kamishak Bay District CCPF pink salmon harvest was confidential due to fewer than 3 permit holders reporting deliveries. The previous 10-year average for pink salmon harvest from this district is 51,976 fish.

The Outer District pink salmon harvest was 7,091 fish, which was lower than the previous 10-year average harvest of 1.1 million fish.

Chum Salmon

Run and Escapement Summary: The aggregated chum salmon SEG was achieved for the Southern District, but was not reached for the Outer, or Kamishak Bay Districts.

Harvest Summary: The 2024 purse seine harvest for the Southern District was 352 chum salmon, which was lower than the previous 10-year average harvest of 1,247 fish. The set gillnet harvest for the Southern District was 1,219 chum salmon, which was less than the previous 10-year average harvest of 4,462 fish.

The Kamishak Bay District CCPF chum salmon harvest was confidential due to fewer than 3 permit holders reporting deliveries. The previous 10-year average annual chum salmon harvest from this district is 10,162 fish.

The Outer District harvest was 1,693 chum salmon, which was lower than the previous 10-year average harvest of 53,903 fish.

Summary by Glenn Hollowell, Area Management Biologist, ADF&G, Homer.

UPPER COOK INLET AREA

The 2024 Upper Cook Inlet (UCI) Area commercial salmon fishery harvest of 2.0 million salmon was 7% less than the recent 10-year average harvest of 2.1 million fish. The harvest was composed of approximately 170 Chinook, 1.87 million sockeye, 25,000 coho, 42,000 pink, and 74,000 chum salmon (Table 5). The 2024 season was the first season when the National Marine Fisheries Service directly managed the exclusive economic zone (EEZ) of Upper Cook Inlet. Harvest is reported separately for salmon caught in the Central District's EEZ and remaining State of Alaska waters (SOA) for the drift gillnet fishery.

Chinook Salmon

Run and Escapement Summary: In UCI, most Chinook salmon are harvested in 2 commercial fisheries. These include the set gillnet fisheries in the Northern District and the Upper Subdistrict of the Central District. For the 2024 season, Chinook salmon runs were expected to be below average across Southcentral Alaska. As expected, the run was below average and even lower than the preseason forecasts. This led to preseason conservation measures in UCI fisheries to reduce the harvest of Chinook salmon.

The preseason run forecast for Deshka River Chinook salmon of approximately 6,600 fish was below the biological escapement goal (BEG) range of 9,000–18,000 fish. The department issued preseason restrictions, closing sport fishing for Chinook salmon in the Susitna River drainage (2-KS-2-05-24). In accordance with the *Northern District King Salmon Management Plan*, the directed commercial Chinook salmon fishery in the Northern District was also closed preseason (EO 2-F-H-1-24) after the Deshka River Chinook salmon sport fishery was closed. The final escapement estimate of Chinook salmon in the Deshka River was approximately 3,445 fish, which did not fall within the BEG range of 9,000–18,000 fish. The Little Susitna River Chinook salmon SEG range of 2,100–4,300 fish was not achieved in 2024, with the incomplete weir count of under 1,005 Chinook salmon. Results of the aerial escapement goals for the various other Susitna drainage Chinook salmon systems are still preliminary and are pending data analysis to determine whether goals have been achieved.

Kasilof River and Kenai River late-run Chinook salmon are the primary Chinook salmon stocks harvested in the eastside set gillnet (ESSN) fishery; however, the ESSN fishery was closed to commercial salmon fishing in 2024 (EO 2-F-H-2-24). At the February 2024 Alaska Board of Fisheries (BOF) meeting, the Kenai River Late-Run King Salmon Stock of Concern Management Plan was created. Kenai River late-run Chinook salmon were managed to meet the recovery goal (RG) range of 14,250–30,000 large Chinook salmon (≥ 75 cm mid eye to tail fork length) established in the Stock of Concern Management Plan.

The total estimated passage at the river mile 14 sonar through August 19 was just under 6,984 large Chinook salmon, and the preliminary escapement was 6,959 large fish. The department applies harvest, catch-and-release mortality estimates, and spawning downstream of the sonar estimates to generate a preliminary spawning escapement estimate. The midpoint of the run occurred on July 29, which is 2 days later than the mean historical midpoint. The RG range of 14,250–30,000 large fish was not achieved, and the SEG range of 13,500–27,000 large fish was achieved. Although the SEG has been achieved in 4 of the last 8 years, the lower bound of the OEG goal has not been achieved since it was established in 2020. The 2024 preseason forecast was for a total run of approximately 13,600 large Kenai River late-run Chinook salmon.

Of the 3 southern Kenai Peninsula Chinook salmon systems, the SEG was achieved at the Anchor River, not achieved at the Ninilchik River, and not assessed at Deep Creek. The Anchor River preliminary escapement estimate was approximately 3,329 fish (SEG range 3,200–6,400 fish). The Ninilchik River wild run count was 676 fish (SEG range 750–1,300 fish). The Deep Creek Chinook salmon run was not assessed due to a lack of funding. The recreational fisheries at Deep Creek and Anchor River were closed during the preseason. Sport fishing for Chinook salmon was opened in the Anchor River during the last 4 days of the fishing season. The Ninilchik River was closed for sport fishing for wild Chinook salmon but was open to fishing for hatchery Chinook salmon during the season. During the season, bait was prohibited in the Ninilchik River Chinook salmon sport fishery.

Harvest Summary: The 2024 UCI commercial harvest of Chinook salmon was 169 fish, which was 97% less than the previous 10-year average of 4,969 fish. The drift gillnet fishery harvested just over 76 Chinook salmon of all sizes and all stocks. The estimated harvest of Chinook salmon in the set gillnet fishery of the Western, Kustatan, and Kalgin Island Subdistricts and the commercial dip net fishery in the Upper Subdistrict was 89 fish.

The directed Chinook salmon set gillnet fishery in the Northern District was closed preseason (EO 2-F-H-1-24). However, 4 Chinook salmon were harvested in the Northern District during other fisheries in July.

Sockeye Salmon

Run and Escapement Summary: The 2024 UCI preseason total run forecast of 5.72 million sockeye salmon included a harvest estimate (sport, personal use, and commercial) of 3.72 million fish. The 2024 preliminary total run estimate (harvest and escapement) of 6.41 million sockeye salmon was 0.69 million fish greater, or 12% more than the preseason forecast. Sockeye salmon run abundance to the Kenai River was more than forecasted by approximately 550,000 fish; the run abundance to the Kasilof River exceeded the preseason forecast by 350,000 fish. The number of sockeye salmon returning to Fish Creek was 1,400 fish less than forecasted (an incomplete escapement count of just under 38,000 fish was within the 15,000–45,000 fish SEG range). The Susitna River sockeye salmon run estimate was near the forecast. For all other systems combined (minor systems), inseason abundance was 189,000 fish less than forecasted. The final passage estimated at the river mile 19 sonar of approximately 1.9 million sockeye salmon exceeded the Kenai River sockeye salmon middle tier inriver goal range (1,100,000–1,400,000 fish). The peak day of sockeye salmon passage in the Kenai River occurred on July 17 with a count of over 191,000 fish. During the previous 20 years, the average date when 50% of the sonar passage occurred in the Kenai River was July 28. In 2024, the midpoint of total sockeye salmon passage occurred on July 21, which is 7 days earlier than the previous 20-year average. Approximately 12% of the sockeye salmon run arrived in the Kenai River during the month of August.

The Kasilof River sockeye salmon sonar count of approximately 1.04 million fish exceeded the Kasilof River BEG range of 140,000–320,000 fish and the OEG range of 140,000–370,000 fish. The passage midpoint for Kasilof River sockeye salmon occurred on July 17, which matched the average midpoint of July 17. Peak daily Kasilof River sockeye salmon passage of just over 99,000 fish occurred on July 17.

Harvest Summary: The UCI commercial combined-gear harvest of approximately 1.87 million sockeye salmon was 9% more than the 10-year average annual harvest of 1.7 million fish.

From June 20 through August 8, the drift gillnet fleet fished a total of 37 days broken up as follows: 4 days in the Expanded Kasilof Section only; 3 days in the Expanded Kasilof and Expanded Kenai sections only; 14 days in the Expanded Kenai, Expanded Kasilof, and Anchor Point sections only; 7 days in Drift Gillnet Area 1 with some or all the expanded sections open; and 8 days districtwide, excluding the Chinitna Bay Subdistrict. Beginning August 8, all Monday/Thursday regulatory drift gillnet fishing periods were restricted to Drift Gillnet Areas 3 and 4 due to less than 1% of the seasons total sockeye salmon harvest being caught for 2 consecutive fishing periods. The total UCI drift gillnet harvest in State of Alaska waters was approximately 1.4 million sockeye salmon. The total UCI drift gillnet harvest in EEZ waters was approximately 325,294 sockeye salmon. In 2024, 353 drift gillnet permits made deliveries in SOA fisheries for a season average harvest of approximately 3,966 sockeye salmon per permit. The peak day of harvest for the drift gillnet fleet occurred on Thursday, July 18, with 264,034 sockeye salmon harvested.

Approximately 41,000 sockeye salmon were harvested by set gillnetters in the Western Subdistrict. This was 23% more than the 10-year average annual harvest of approximately 36,000 fish during the previous 10 years.

Approximately 8,200 sockeye salmon were harvested in the Kustatan Subdistrict, of which almost 1,800 fish were harvested during the Big River fishery. The 2024 sockeye salmon harvest for the Kustatan Subdistrict was much larger than the recent 10-year average harvest of over 6,600 fish.

In 2024, approximately 44,500 sockeye salmon were harvested from the Kalgin Island Subdistrict, with just over 5,000 fish taken during the Big River sockeye salmon fishery. The average annual sockeye salmon harvest in the Kalgin Island Subdistrict during the previous 10 years was approximately 47,600 fish.

In 2024, approximately 12,000 sockeye salmon were harvested in the Northern District. This harvest was 24% less than the 10-year average annual harvest of almost 55,600 sockeye salmon. As in past years, restrictions to the Northern District limited the number of nets allowed from July 20 until the end of the season on August 8. The season was closed early by emergency order for coho conservation.

Coho Salmon

Run and Escapement Summary: In UCI, there are 4 coho salmon systems with escapement goals. Weirs are used to assess escapement on Fish Creek, Little Susitna River, and Deshka River; McRoberts Creek is assessed using foot surveys.

The Little Susitna weir was inundated by floodwaters for the majority of the coho salmon season. The projected escapement was below the SEG range when the weir was removed on September 3. Even though the season total escapement is incomplete, the SEG range of 9,200–17,700 fish likely would not have been attained. The weir count of approximately 964 fish is considered an incomplete and minimum count.

The Deshka weir was removed due to high-water conditions on August 16. Through this date, just over 642 coho salmon were counted, which is considered an incomplete and minimum count. The weir at Fish Creek was pulled on September 4, with 235 coho salmon passed, which did not achieve the SEG of 1,200–4,400 fish. The SEG range for Jim Creek of 250–700 coho salmon is assessed

postseason by a foot survey of McRoberts Creek, a small spawning tributary within the Jim Creek system. The survey resulted in almost 376 coho salmon, which is within the SEG range.

Harvest Summary: The 2024 commercial harvest estimate of approximately 24,700 coho salmon was 85% less than the previous 10-year average of 167,000 fish. The drift gillnet harvest in SOA waters was 6,709 fish and in EEZ waters was 4,443 fish. The Northern District set gillnet fishery harvested 8,725 coho salmon, which was 80% less than the recent 10-year average of 44,500 fish.

Pink Salmon

Harvest Summary: Pink salmon runs in UCI are even-year dominant, with odd-year average harvests typically less than one-sixth of even-year harvests. The 2024 UCI commercial pink salmon harvest was estimated to be 42,000 fish, 88% lower than the average annual harvest of 340,000 fish from the 10-year, even-year (2002–2022) average harvest.

Chum Salmon

Run and Escapement Summary: An aerial survey of Chinitna River and Clearwater Creek was conducted on August 20. This survey produced an estimate of approximately 800 chum salmon, which was below the SEG range (3,500–8,000 fish). With the escapement goal not being met, Chinitna Bay was not opened to drift gillnet fishing. Regularly scheduled Monday and Thursday drift gillnet fishing periods for Drift Gillnet Areas 3 and 4 began August 8.

Harvest Summary: The 2024 harvest of 74,000 chum salmon was 44% lower than the 10-year average annual harvest of 133,000 fish.

Summary by Colton Lipka and Lucas Stumpf, Area Management Biologists, ADF&G, Soldotna.

BRISTOL BAY AREA

The 2024 commercial salmon harvest in the Bristol Bay Area totaled almost 32.3 million salmon (Table 5). Sockeye salmon accounted for most of this harvest at 31.6 million fish. Harvest for the other species were approximately 5,000 Chinook, 509,000 chum, 78,000 pink, and 31,000 coho salmon.

Chinook Salmon

Run and Escapement Summary: Bristol Bay Chinook salmon harvest and abundance remained low in 2024. The Nushagak River Chinook salmon in-river run index at Portage Creek sonar was 48,000 fish, below the sustainable escapement goal (SEG) range of 55,000–120,000 fish and the in-river goal of 95,000 fish.

Harvest Summary: Bristol Bay Chinook salmon harvest and abundance continued to be low in 2024. Chinook salmon harvested in Bristol Bay were incidentally caught during directed sockeye salmon fishing periods. Overall, the 2024 Chinook salmon harvests were below average in all districts of Bristol Bay, with a total of approximately 4,600 Chinook salmon harvested bay-wide (Table 5). Nushagak Chinook salmon is the main source of Chinook harvest in Bristol Bay and is a stock of concern. The Nushagak District was managed according to the Nushagak Chinook salmon action plan, with sockeye salmon optimum escapement goal (OEG) triggers that delay fishing to allow for increased Chinook salmon escapement. The Nushagak District Chinook harvest was approximately 2,400 fish, well below average.

Sockeye Salmon

Run and Escapement Summary: The 2024 Bristol Bay sockeye salmon run of 51.6 million fish was 36% above the preseason inshore forecast of 37.9 million fish. Runs to every district were larger than their preseason forecasts. Sockeye salmon escapement goals were exceeded on the Ugashik, Igushik, and Togiak Rivers. The Kvichak, Naknek, Alagnak, and Egegik Rivers were within their escapement goal ranges. Nushagak and Wood River sockeye salmon escapements were above the OEG ranges adopted in 2023 by the Alaska Board of Fisheries for Chinook salmon conservation. Overall, run timing was 2 days late compared to the 20-year average.

In 2024, the sockeye salmon run made up a higher proportion of younger and smaller fish than forecasted. Eighty percent of the 2024 Bristol Bay sockeye salmon run consisted of fish with 2 years of ocean growth. The 1.2 age class composed roughly 72% of the total run and came in well over its preseason forecast of 50%. The 2.2 age class contributed 8%, which was well below the forecasted 13%. Age 1.3 fish contributed 16%, or approximately half of the forecasted 31%. The high proportion of younger fish resulted in the smallest average sockeye salmon weight on record at 4.53 pounds.

Harvest Summary: The 2024 harvest of 31.7 million sockeye salmon was 5% lower than the recent 20-year average of 33.2 million fish between all districts.

Coho Salmon

Harvest Summary: The commercial harvest of coho salmon was approximately 31,000 fish. The Nushagak District is typically the largest producer of coho salmon and accounted for 22,000 of the coho salmon harvested in 2024. Approximately 7,000 coho salmon were also harvested in the Egegik District. Harvests of coho salmon can be variable from year to year, depending on processor availability, market conditions, and overall fishing effort. In 2024, fishing ended early

due in part to bad weather and low market interest in purchasing coho salmon. Coho salmon escapement is not monitored in Bristol Bay.

Pink Salmon

Harvest Summary: Pink salmon in Bristol Bay are predominantly an even-year species. Pink salmon harvest was incidental to the sockeye salmon fishery, totaling 78,000 fish. Pink salmon escapement is not monitored in Bristol Bay.

Chum Salmon

Escapement Summary: The Nushagak River sonar project is the only chum salmon escapement assessment project in Bristol Bay. The escapement of 311,000 fish was above the lower-bound SEG of 200,000 fish.

Harvest Summary: In 2024, the baywide commercial harvest of 509,000 chum salmon was well below the recent 20-year average of 1.0 million fish.

Summary by Tim Sands, Travis Elison, and Stacy Vega, Area Management Biologists, ADF&G, Dillingham, and Anchorage.

Table 5.—2024 Central Region commercial salmon harvests, by area and species, in thousands of fish.

Fishing area	Species					Total
	Chinook	Sockeye	Coho	Pink	Chum	
Drift gillnet						
Bering River District	0	58	42	—	0	100
Coghill Dist.	0	97	0	52	233	382
Copper River Dist.	9	1,374	117	5	31	1,536
Eshamy Dist.	0	857	1	96	62	1,016
Montague Dist.	—	—	—	—	—	—
Unakwik Dist.	—	0	—	0	0	0
Purse Seine						
Coghill District	0	2	0	11	0	14
Eastern District	0	4	2	4,148	22	4,177
Montague District	0	4	0	140	683	828
Northern District	0	1	0	90	1	92
Northwestern District	—	5	0	143	3	151
Southwestern District	0	70	0	252	109	432
Southeastern District	0	0	0	116	1	118
Unakwik District	0	1	—	0	0	1
Set gillnet						
Eshamy District	0	310	0	19	13	342
Hatchery ^a	—	420	4	4,880	543	5,847
Prince William Sound total ^b	10	3,204	167	9,952	1,702	15,035
Southern District purse seine	0	79	0	0	0	80
Southern District set gillnet	0	23	0	1	1	26
Southern District total	0	103	0	1	2	106
Kamishak and Eastern District	—	—	—	—	—	—
Outer District	0	35	0	7	2	44
Hatchery ^c	—	147	0	8	0	155
Lower Cook Inlet total ^d	0	298	0	17	3	319
Central District Drift Gillnet - EEZ	0	325	4	6	29	365
Central District Drift Gillnet - SOA	0	1,360	7	31	40	1,439
Central District Set Gillnet	0	94	5	1	2	102
Central District Dip Net	0	28	0	0	0	28
Central District Other Gear	0	22	0	0	0	22
Central District Total	0	1,829	16	39	71	1,956
Northern District	0	42	9	2	3	56
Upper Cook Inlet total ^{d,e}	0	1,871	25	42	74	2,011
Naknek-Kvichak District	1	9,251	1	4	69	9,326
Nushagak District	2	12,300	22	40	317	12,682
Egegik District	0	5,287	7	1	33	5,328
Ugashik District	0	4,245	0	0	43	4,289
Togiak District	1	575	1	33	48	657
Bristol Bay total ^e	5	31,659	31	78	509	32,281
Central Region total	15	37,033	223	10,088	2,288	49,647

Notes: En dashes indicate no harvest or confidential data, and zeros indicate harvest activity but <500 fish. Columns may not total exactly due to rounding. EEZ is Economic Exclusive Zone. SOA is State of Alaska.

^a Total includes hatchery sales for operating expenses and broodstock harvests.

^b Chinook salmon adults and jacks are totaled.

^c Total includes cost-recovery and hatchery-donated fish but not broodstock.

^d Total includes commercially harvested fish retained for personal use.

^e Total includes harvest that was discarded, confiscated, seized, or donated.

ARCTIC-YUKON-KUSKOKWIM REGION

The Arctic-Yukon-Kuskokwim (AYK) Region 2024 harvests totaled approximately 27,000 fish (Table 6). The cumulative all-gear commercial harvest included approximately 300 sockeye, 14,000 coho, 10,000 chum, and 3,000 pink salmon, all harvested in the Norton Sound and Kotzebue areas.

KUSKOKWIM AREA

The 2024 Kuskokwim River Chinook salmon forecast was for a range of 108,000–160,000 fish. The drainagewide Chinook salmon SEG range was 65,000–120,000 fish. If the run came in as projected, the drainagewide and tributary escapement goals were expected to be achieved with a limited subsistence harvest. The 2024 season was managed in accordance with the *Kuskokwim River Salmon Management Plan* (5 AAC 07.365) with input from the Kuskokwim River Salmon Management Working Group (Working Group). It was the intent of the department to manage all Kuskokwim River salmon stocks in a conservative manner, consistent with the *Policy for the Management of Sustainable Salmon Fisheries* under 5 AAC 39.222, to meet escapement goals and the subsistence priority. Preseason management actions that were intended to achieve escapement goals included early season subsistence fishing closures, tributary closures, time and area restrictions, gillnet mesh size and length restrictions, and live-release requirements.

Between June 1 and August 16, a Federal Special Action (FSA) closed the Kuskokwim River subsistence gillnet fishery to federally qualified users within the boundary of the Yukon Delta National Wildlife Refuge (YDNWR). The U.S. Fish and Wildlife Service (USFWS) determined that adopting the FSA was warranted due to the expectation that Chinook and coho salmon runs had been forecast to be below average, and that the chum salmon run forecast indicated a poor run.

Upstream of the YDNWR boundary at Aniak to the Kuskokwim River headwaters, subsistence Sections 4 (from the refuge boundary at Aniak to the Holitna River mouth) and 5 (Holitna River mouth to headwaters), fishing was not subject to the FSA. Beginning June 1, subsistence fishing in Sections 4 and 5 was closed to gillnets (drift and set) with additional restrictions that required live release of Chinook and chum salmon captured in selective gears. On June 7, fishing with set gillnets was opened in Sections 4 and 5 with live release of Chinook and chum salmon captured in selective gears required. Beginning June 12, subsistence Sections 4 and 5 were opened to subsistence fishing until further notice with 6-inch or less mesh gillnets (set or drift), 25 fathoms in length, and live release of chum salmon caught in selective gears.

On August 16, the USFWS rescinded the FSA and the associated fishing closures that had been in place in the Kuskokwim River within the YDNWR boundaries. Based on preliminary inseason run assessment data, the department determined that ending restrictions to fishing with gillnets was warranted. On August 17, the department resumed management of Sections 1–3 and subsequently opened subsistence fishing with gillnets until further notice from the mouth of the Kuskokwim River upstream to its headwaters. The Bethel Test Fishery (BTF) cumulative catch per unit effort (CPUE) on August 10 for coho salmon was almost 2,000 fish, and the total coho salmon passage past the sonar was 198,000 fish. These BTF and sonar data indicated that the coho salmon escapement goals at the Kwethluk River and Kogrukluk River weirs would likely be met. On average, approximately 58% of the coho salmon run has passed through Bethel by August 10. By August 10, an average of 100% of the Chinook salmon run, 100% of the sockeye salmon run, and 99% of the chum salmon run will have passed through Bethel.

Chinook Salmon

Run and Escapement Summary: The preliminary Kuskokwim River total run estimate was 176,619 Chinook salmon (95% CI = 131,529–237,166 fish), and an estimated 143,919 Chinook salmon (95% CI = 98,829–204,466 fish) escaped Kuskokwim River fisheries, which exceeded the drainagewide SEG range of 65,000–120,000 fish.

One escapement goal for Chinook salmon was assessed within the Kuskokwim River drainage in 2024. Escapement at the George River weir was approximately 2,400 Chinook salmon, which fell within the SEG range of 1,800–3,300 fish. Chinook salmon escapements at the remaining weir projects were not assessed because of extended periods of missed passage due to high water. Eight aerial surveys were carried out for Chinook salmon in 2024. The Chinook salmon aerial survey estimate at the Salmon (Pitka Fork) River was 702 fish, which was within the SEG range of 470–1,600 fish.

One escapement goal for Chinook salmon was assessed within the Kuskokwim Bay drainages in 2024. The Chinook salmon aerial survey SEG range of 3,900–12,000 fish at the Kanektok River (District 4) was achieved with an estimated 4,700 fish. The Chinook salmon aerial survey SEG range of 640–3,300 fish at the North Fork Goodnews River was not assessed this season due to poor weather during the peak spawning time.

Harvest Summary: Postseason subsistence harvest surveys have been completed for 2024. The preliminary subsistence harvest estimate for Kuskokwim River Chinook salmon is 32,493 fish, which is above the recent 10-year average subsistence harvest of 26,943 fish. Final subsistence harvest estimates are anticipated to be available in spring 2025.

Sockeye Salmon

Run and Escapement Summary: Sockeye salmon escapement was estimated at 1 weir in 2024. The Telaquana River weir counted over 111,000 fish, which is below the most recent 20-year average (2014–2023) of 148,000 fish. Sockeye salmon escapements at the other weirs were not assessed due to extended period of high water. No escapement goals for Kuskokwim River sockeye salmon were assessed in 2024.

One of 2 escapement goals for sockeye salmon were assessed within Kuskokwim Bay drainages in 2024. The sockeye salmon aerial survey SEG range of 15,300–41,000 fish at the Kanektok River (District 4) was exceeded with 179,000 fish counted. The sockeye salmon aerial survey SEG range of 9,600–18,000 fish at the North Fork Goodnews River (District 5) was not assessed due to poor weather conditions during the peak spawning time.

Harvest Summary: Postseason subsistence harvest surveys have been completed for 2024. The preliminary subsistence harvest estimate for Kuskokwim River sockeye salmon is 43,563 fish, which is below the recent 10-year average subsistence harvest of 46,119 fish. Final subsistence harvest estimates are anticipated to be available in spring 2025.

Coho Salmon

Run and Escapement Summary: Coho salmon escapement was estimated at 1 weir in 2024. The escapement estimate at the Kogrukluk River weir was approximately 14,500 coho salmon, which fell within the SEG of 13,000–28,000 fish. Escapements at the other weir projects in the area were not assessed because of extended periods of missed passage due to high water.

Harvest Summary: Postseason subsistence harvest surveys have been completed for 2024. The preliminary subsistence harvest estimate for Kuskokwim River coho salmon is 19,659 fish, which is below the recent 10-year average subsistence harvest of 30,513 fish. Final subsistence harvest estimates are anticipated to be available in spring 2025.

Chum Salmon

Run and Escapement Summary: Chum salmon escapement was estimated at 1 weir in 2024. Escapement at the George River weir was 20,243 chum salmon. Chum salmon escapements were not estimated at the remaining weir projects due to extended periods of missed passage. No escapement goals for Kuskokwim River chum salmon were assessed in 2024.

Harvest Summary: Postseason subsistence harvest surveys have been completed for 2024. The preliminary subsistence harvest estimate for Kuskokwim River chum salmon is 22,772 fish, which is below the recent 10-year average subsistence harvest of 35,798 fish. Final subsistence harvest estimates are anticipated to be available in spring 2025.

Summary by Sam Decker, Kuskokwim Area Management Biologist, and Sean Larson, Kuskokwim Area Research Biologist, ADF&G, Anchorage.

YUKON AREA

No commercial fishing occurred in the Yukon Management Area during the summer or fall seasons in 2024 (Table 6).

The Yukon River summer season spans from early May through July 15 in District 1. On July 16, management in District 1 transitions to the fall season and becomes focused on fall chum and coho salmon entering the mouth of the Yukon River. However, summer season management continues beyond this date in upper river districts as Chinook and summer chum salmon migration progresses upstream. The subsequent transition of upriver districts to the fall season is based on the migration timing of fall chum salmon.

Summer Season

Chum Salmon

Run and Escapement Summary: Approximately 758,300 summer chum salmon were estimated to have passed Pilot Station sonar (90% CI = 719,012–797,508 fish), which was within the Yukon River Drainage SEG range of 500,000–1,200,000 fish. The estimated run size was within the preseason forecast of 550,000–1,800,000 fish.

Three pulses of summer chum salmon passed the sonar project; the largest group consisted of approximately 256,500 fish between July 5 and July 10. The first quarter point, midpoint, and third quarter point of the summer chum salmon run at the Pilot Station sonar were June 26, July 2, and July 7, respectively. Based on the midpoint at the sonar project, this indicated that the summer chum salmon run was probably 4 days later than average.

The 2024 total summer chum salmon passage estimated at Pilot Station sonar through July 18 was a conservative estimate consistent with historical data; however, the true total run size in 2024 was probably somewhat larger. Genetic mixed stock analyses in 2024 indicated summer chum salmon continued to pass Pilot Station sonar beyond the administrative summer season cutoff date of July 18. In 2024, 89% (CI = 79–96%) of chum salmon arriving at the Pilot Station sonar from

July 19 to July 28 were genetically summer chum salmon. Overall, an estimated 38,000 summer chum salmon came in during the fall season.

There are 3 established escapement goals for summer chum salmon in the Yukon area: the drainagewide BEG range of 500,000–1,200,000 fish, the East Fork Andreafsky River lower-bound SEG of 40,000 fish, and the Anvik River BEG range of 350,000–700,000 fish. The estimated drainagewide escapement was 775,800 summer chum salmon, meeting the drainagewide BEG. Passage estimates for the Andreafsky River weir (operated by the USFWS), were affected by high water and should be considered minimum estimates (190 summer chum salmon). Anvik Sonar missed several days of counts due to persistent high-water conditions. The expanded estimate, including missed days, was 134,700 summer chum salmon, which is below the project median of 450,200 fish.

Harvest Summary: No commercial fishing occurred in the Yukon Management Area during the summer or fall seasons. Subsistence fishing for summer chum salmon opened with selective gear types during a portion of the summer chum salmon run in the Coastal District and District 1 through Subdistrict 5-C. Fishing remained closed in Subdistrict 5-D because summer chum salmon do not migrate that far upriver. Subsistence fishers were permitted to retain summer chum, pink, and sockeye salmon, and nonsalmon species, but were required to release Chinook salmon alive.

Chinook Salmon

Run and Escapement Summary: The cumulative passage estimate at the Pilot Station sonar was approximately 64,200 Chinook salmon (90% CI = 50,105–78,291 fish). This passage was the fourth lowest recorded for the project and was within the forecast range (45,000–68,000 fish). Most Chinook salmon entered the river in 2 pulses, and the run appears to have been 7 days later than average based on the midpoint passage through the sonar project.

Several assessment projects were affected by funding and high water during the summer season and there were several late starts or missed days of counts. Gisasa and Henshaw did not operate due to lack of funding. Aerial surveys of the East and West Forks of the Andreafsky River, Anvik River, and Nulato River were conducted under various conditions. Counts were below average for Chinook and summer chum salmon.

Passage estimates for the Andreafsky River weir (operated by the USFWS) were affected by high water and should be considered minimum estimates (24 Chinook salmon).

The Chena and Salcha River escapement projects are operated by the ADF&G Sport Fish Division. Due to high water, the sonar equipment was pulled for part of the season on the Chena River, and the tower was unable to count for several days. The preliminary estimates are 336 Chinook salmon, which was below the BEG range of 2,800–5,700 fish.

The Salcha River escapement project operated from June 27 to July 28. The preliminary estimates are 719 Chinook salmon which was below the BEG range of 3,300–6,500 fish. Due to high water, the tower counts were unavailable, and only sonar counts were used for part of the season.

Based on genetic mixed stock analysis and passage, 28,600 (90% CI = 21,308–36,182) Canada-origin salmon were estimated to have passed the Pilot Station sonar in 2024. The preliminary cumulative passage estimate at the Eagle sonar was almost 24,200 Chinook salmon (90% CI = 23,888–24,478 fish). This estimate did not meet the U.S./Canada border passage objective of 71,000 fish. Fishing for Chinook salmon remained closed all season throughout the Yukon Area;

however, a small number of Chinook salmon were harvested through test fisheries or retained as incidental harvest using nonsalmon gear.

Harvest Summary: For the 17th consecutive year, no commercial periods targeting Chinook salmon were allowed in the Yukon Management Area during the summer season.

Fall Season

Fall Chum Salmon

Run and Escapement Summary: In 2024, the preliminary estimate of the drainagewide total run size was 165,000 fall chum salmon. The drainagewide escapement (after preliminary harvest estimates from U.S. and Canada were removed) was estimated to be 161,000 fall chum salmon, which was below the SEG range of 300,000–600,000 fish. The 2024 run was the 2nd lowest fall chum salmon run on record with 5 consecutive low years from 2020–2024.

Three fall chum salmon escapement goals were assessed in the Yukon Area during the 2024 fall season. Approximately 58,000 fall chum salmon were estimated to have passed through the Teedriinjik sonar project (estimate includes expansion beyond project termination), which is well below the SEG range of 85,000–234,000 fish. The Delta River, a tributary of the Tanana River drainage, was monitored using replicate ground (foot) surveys. The surveys resulted in an escapement estimate of 17,000 fall chum salmon, which is within the SEG range of 7,000–20,000 fish. The Sheenjek River sonar project, which was reestablished in 2022, resulted in an estimated passage of approximately 14,000 fall chum salmon (estimate includes expansion beyond project termination). The Sheenjek River was previously monitored from 1974–2012 with an average passage of 96,000 fall chum salmon.

Several escapement monitoring projects were also operated in Canada. On the Porcupine River, approximately 8,800 fall chum salmon passed the sonar. Although there is no established escapement goal for this location, this abundance estimate indicated that treaty obligations for the upper Porcupine River would not be met. In the Fishing Branch River, a weir/sonar project produced an escapement estimate of 5,900 fall chum salmon. This level of escapement in the Fishing Branch River was well below the lower end of the IMEG range of 22,000–49,000 fish.

The largest component of Canada-origin fall chum salmon pass into Canada via the mainstem Yukon River. The fall chum salmon passage estimate at the mainstem Yukon River sonar project near Eagle was under 14,000 fish (SE = 152 fish) for the dates September 1 through October 6. Due to continued passage after the termination of the project, the fall chum salmon estimate was subsequently adjusted to approximately 16,200 fish. The preliminary escapement for the mainstem Yukon River in Canada is derived by subtracting the upstream U.S. and Canadian harvests from the expanded sonar estimate. The preliminary mainstem Yukon River escapement estimate (16,000 fall chum salmon), was the lowest on record since 1980 and was below the IMEG range (70,000–104,000 fish). As expected, based on the extremely low total drainagewide run size, neither of the Yukon River treaty obligations with Canada were achieved in 2024.

Stock composition estimates, provided by USFWS Conservation Genetics Laboratory, were determined using tissue samples collected from chum salmon captured in the mainstem Yukon River sonar test net fishery. Chum salmon genetic samples processed and reanalyzed to include 5 strata, between July 19 and September 7 (fall season), indicated stocks represented approximately 18% summer fish. Within the fall components, the 3 main stock groupings were 35% Border U.S. (Teedriinjik/Sheenjek/Draanjik), 15% Canadian, and 51% Tanana. In 2024, the upper Yukon U.S.

was near the average of 36%, the Tanana River stocks were well above the average of 34%, and the Canadian component was lower than the average of 30%, where the average includes 2004–2023. Analysis often reviews the mainstem Yukon to Canada separately because of sample size issues for the genetic Porcupine component. In 2024, the mainstem Yukon Canada component of fall chum salmon set a new record low at 7% compared to the average (2004–2023) of 27%.

Harvest Summary: No commercial fishing occurred in the Yukon Management Area during the summer or fall seasons. Subsistence fishing for fall chum salmon remained closed in the Yukon Area. However, fall chum salmon were caught primarily on test fisheries with some incidental harvest in nonsalmon gear. The preliminary 2024 harvest estimate from these sources was approximately 3,600 fall chum salmon.

Coho Salmon

Run and Escapement Summary: There are few coho salmon spawning escapement assessment projects in the Yukon River drainage because of funding limitations and late timing relative to onset of winter. The sonar in the mainstem Yukon River near Pilot Station was operated through September 7. The project estimated passage through Pilot Station was under 78,000 coho salmon (90% CI = 70,000–85,000 fish), which is well below the historical average of 145,000 fish. A boat survey conducted on the Delta Clearwater River in late October resulted in an escapement estimate of 1,455 coho salmon, which is the third lowest count since monitoring began in 1972. Escapement count estimates for coho salmon were conducted by aerial survey in the Nenana River drainage, and all spawning areas monitored were below the 1974–2023 average.

Harvest Summary: No commercial fishing occurred in the Yukon Management Area during the summer or fall seasons. Subsistence fishing for coho salmon in the Yukon Area closed August 26 due to poor returns. Coho salmon were caught primarily in test fisheries, in selective gear such as dip nets before the closure, or as incidental harvest in nonsalmon gear. The preliminary 2024 harvest estimate from these sources was approximately 1,730 coho salmon.

Summary by Deena Jallen, Area Management Biologist, and Bonnie Borba, Research Biologist, ADF&G, Fairbanks.

NORTON SOUND AREA

Weak runs of salmon, especially in southern Norton Sound, resulted in low volumes of commercial salmon harvest this season. Escapement goals were achieved for 1 of 5 chum salmon stocks, 3 of 3 coho salmon stocks, all 3 pink salmon stocks, and both sockeye salmon stocks. Escapement goals were not achieved for Chinook. Total commercial salmon harvests were approximately 300 sockeye, 13,800 coho, 3,000 pink, and 4,000 chum salmon (Table 6).

Chinook Salmon

Harvest Summary: Commercial fishing targeting Chinook salmon was prohibited in the Norton Sound Area during the 2024 season.

Sockeye Salmon

Harvest Summary: Sockeye salmon harvest accounts for a small portion of the overall harvest in Norton Sound and is incidental harvest when targeting other species. Despite higher harvest totals in recent years, the 2024 commercial harvest of approximately 300 fish was well below the recent 5-year (2019–2023) average harvest of just over 2,800 sockeye salmon.

Coho Salmon

Harvest Summary: The 2024 coho salmon harvest of 13,800 fish was 16% of the recent 5-year average (87,000 fish).

Pink Salmon

Harvest Summary: The 2024 pink salmon run was weak, unlike the record runs experienced in Norton Sound in recent years. Small runs resulted in a harvest of about 3,000 pink salmon, which was 4% of the 5-year, odd-year average of almost 92,000 pink salmon. Pink salmon harvest in Norton Sound is highly dependent on market availability, and directed pink salmon fisheries have been sporadic.

Chum Salmon

Harvest Summary: The 2024 chum salmon harvest of approximately 4,000 fish was 4% of the recent 5-year average (under 92,000 fish).

Summary by Kevin Clark, Area Management Biologist, ADF&G, Nome.

KOTZEBUE SOUND AREA

There were 24 permit holders that commercially harvested fish in the Kotzebue Sound Area in 2024. The highest fishing effort this season occurred on July 29 and again on August 7 and 8 when 12 permit holders made landings. Total commercial salmon harvests were approximately 1 Chinook and 5,400 chum salmon (Table 6).

Chum Salmon

Run and Escapement Summary: No aerial surveys were conducted in 2024, and the Kobuk River test fish project was discontinued in the 2024 season due to budget cuts.

Harvest Summary: The Kotzebue Sound chum salmon harvest of 5,400 fish were the poorest on record when a buyer was available in the Kotzebue District.

Summary by Kevin Clark, Area Management Biologist, ADF&G, Nome.

Table 6.—2024 Arctic-Yukon-Kuskokwim Region commercial salmon harvests, by fishing area and species, in thousands of fish.

Fishing area	Species					Total
	Chinook	Sockeye	Coho	Pink	Chum	
Kuskokwim River	—	—	—	—	—	—
Kuskokwim Bay	—	—	—	—	—	—
Kuskokwim Area total	—	—	—	—	—	—
Lower Yukon River	—	—	—	—	—	—
Upper Yukon River	—	—	—	—	—	—
Yukon River total	—	—	—	—	—	—
Norton Sound ^a	0	0	14	3	4	22
Kotzebue Sound ^a	0	0	0	0	5	5
AYK Region total	0	0	14	3	10	27

Note: En dashes indicate no harvest or confidential data, and zeros indicate harvest activity but <500 fish. Columns may not total exactly due to rounding. Confidential data omitted.

^a Total includes commercial harvest that was discarded, confiscated, seized, donated, or retained for personal use.

WESTWARD REGION

KODIAK AREA

The 2024 commercial harvest in the Kodiak Management Area (KMA) was approximately 1,486 Chinook salmon, 1.6 million sockeye salmon, 64,000 coho salmon, 7.3 million pink salmon, and 498,000 chum salmon (Table 7). The total KMA salmon harvest of 9.5 million fish was below the 2024 forecast and the previous 10-year average of approximately 23.5 million fish.

Commercial fishing effort was below average compared to recent years. Of the 583 eligible commercial salmon permits, 225 permits were used, including 117 purse seine and 107 set gillnet permits.

Chinook Salmon

Run and Escapement Summary: The total Chinook salmon escapement (434 fish) was a record low, and below the previous 10-year average of 5,127 fish. Escapement goals for Chinook salmon have been developed for the Karluk and Ayakulik Rivers, and escapements are estimated using fish counting weirs.

The Chinook salmon escapement through the Karluk River weir was a record low (76 fish) and below the biological escapement goal (BEG) range of 3,000 to 6,000 fish. Chinook salmon escapement through the Ayakulik River weir was a record low (354 fish) and below the BEG range of 4,800 to 8,400 fish.

Harvest Summary: There are no directed Chinook salmon commercial fisheries in the KMA, but incidental commercial harvest occurs during targeted sockeye and pink salmon fisheries. The Karluk and Ayakulik Rivers both support small populations of Chinook salmon. By regulation, non-retention of Chinook salmon over 28 inches was implemented for the purse seine fleet areawide for the entire season. The 2024 commercial harvest of Chinook salmon in the KMA totaled 1,486 fish, which was below the previous 10-year average (8,247 fish) and the 2024 forecast (9,500 fish). Due to commercial salmon restrictions because of weak early run sockeye runs and restrictive pink salmon fishing periods, the 2024 harvest of Chinook salmon of 1,486 was the lowest in the past 42 years (excluding the 1989 Exxon Valdez year).

Sockeye Salmon

Run and Escapement Summary: Sockeye salmon runs in many systems in the KMA were average to above average. Most sockeye salmon systems in the Kodiak Area met their established escapement goals, except for the early runs to Karluk and Upper Station. The KMA estimated sockeye salmon escapement of 1.4 million fish was below the previous 10-year average of 1.5 million fish.

Harvest Summary: The 2024 commercial harvest of sockeye salmon totaled 1.6 million fish. The harvest was below the recent 10-year average (2.5 million fish) and the preseason forecast.

Early-season management for much of the westside of Kodiak Island is driven by Karluk River early-run sockeye salmon. The 2024 Karluk River early run was weak. Only 2 mandatory 33-hour fishing periods were allowed in June along much of the westside of Kodiak Island until the management focus turned to pink salmon (July 6). A total of 117,045 sockeye salmon were harvested in early-season westside areas based on Karluk River early-run sockeye salmon and the beginning of the pink salmon fishery (June 1–July 15). Eighty-nine percent of those sockeye

salmon were harvested after July 5. Westside sockeye salmon numbers include an estimated contribution of 22,274 sockeye salmon from the enhanced Spiridon Lake sockeye salmon run harvested outside of the Spiridon Bay Special Harvest Area.

Late-season management for much of the westside of Kodiak is driven by the Kodiak Island pink and chum salmon fishery and the Karluk River late-run sockeye salmon run. Most of the wild pink salmon runs to the westside of Kodiak Island were below average, and very short weekly fishing periods were allowed in July and early August. The Karluk River late-run sockeye salmon run was average, and liberal fishing time was allowed in August and September. A total of 698,481 sockeye salmon were commercially harvested in late-season westside areas opened based on Karluk River late-run sockeye and westside Kodiak pink and chum salmon abundance. Westside sockeye salmon numbers include an estimated contribution of 90,875 sockeye salmon from the enhanced Spiridon Lake sockeye salmon run harvested outside of the Spiridon Bay Special Harvest Area.

During early-run sockeye salmon management, the Ayakulik River early-run sockeye salmon run was weak, and very few fishing days were allowed in the Outer Ayakulik and Halibut Bay Sections of the Southwest Kodiak District. A total of 117,399 sockeye salmon were harvested in areas managed based on the abundance of Ayakulik River early-run sockeye salmon.

The Ayakulik River late-run sockeye salmon run was also weak. A total of 62,707 sockeye salmon were harvested in areas managed based on Ayakulik River late-run sockeye. A total of 180,106 sockeye salmon were harvested from Westside sections managed based on abundance of Ayakulik River sockeye salmon.

Upper Station early-run sockeye salmon escapement was weak, and the traditional fishing areas of the Alitak District (i.e., purse seine and set gillnet sections) were open for very few weekly fishing periods in June. The Frazer Lake sockeye salmon escapement was below average, but escapement was adequate to allow an Alitak District early-run sockeye salmon commercial salmon fishery through July 15. The Alitak District early-run sockeye salmon harvest was approximately 3,337 fish.

Commercial set gillnet effort in the Alitak District was very low, and despite liberal fishing time, Upper Station late-run sockeye salmon escapement was above average. The Alitak District late-run sockeye salmon harvest was 236,859 fish. The total harvest of the Alitak District sockeye salmon was 240,196 fish.

Coho Salmon

Run and Escapement Summary: Although substantial coho salmon runs occur in the KMA, the department no longer has funds to monitor most stocks. The only established coho salmon escapement goals occur in the Northeast Kodiak and Eastside Kodiak Districts for the following systems: American (400 fish), Olds (1,000 fish), Buskin (4,700 to 9,600 fish), and the Pasagshak Rivers (1,200 fish). Coho salmon runs in many systems in the KMA were weak in 2024. Both Pasagshak and American Rivers achieved their escapement goals. The Olds River did not achieve its escapement goal, and the inseason coho count was incomplete for the Buskin River.

Harvest Summary: The commercial coho salmon harvest of 63,928 fish was well below the forecast and well below the previous 10-year average (337,506 fish). The majority of the coho salmon were harvested on the Westside Kodiak fishery in August and September.

Pink Salmon

Run and Escapement Summary: The KMA pink salmon escapement of 3.5 million fish was below the previous 10-year average of 5.5 million fish. However, this should be seen as a minimal estimate given the lack of aerial surveys conducted due to an insufficient budget and increased costs. Pink salmon escapement goals have been established as aggregate goals for the entire Kodiak Archipelago and the Mainland District. The 2024 escapement for the Kodiak Archipelago (3,383,009 fish) was within the escapement goal range of 3.0 to 7.0 million fish. The Mainland District pink salmon escapement of only 158,000 fish was below average and below the established escapement goal range (250,000 to 1,000,000 fish).

Harvest Summary: The 2024 KMA pink salmon harvest (including cost recovery) of 7,322,327 fish was below the forecast and the previous 10-year average harvest of 20,406,587 fish. Pink salmon harvested in the areas managed on KMA wild stocks totaled 4,781,450 fish, which was slightly below the forecast.

The Kitoi Bay Hatchery pink salmon run was also below average, with 2.5 million pink salmon harvested in sections near the hatchery and below the 2024 forecast. Kitoi-bound pink salmon are probably harvested along the west and east sides of Kodiak and Afognak Islands. Likewise, additional wild stock salmon are probably harvested in areas associated with Kitoi Bay Hatchery. The department does not have a stock separation program for pink salmon and is unable to differentiate the KMA wild and hatchery stocks. Cost-recovery fish harvested by Kodiak Regional Aquaculture Association accounted for 39% of the harvest, or 978,032 fish. An additional 293,412 pink salmon carcasses were also sold but not included in the totals.

Chum Salmon

Run and Escapement Summary: The overall KMA chum salmon escapement of 158,371 fish was very weak and below the previous 10-year average (373,482 fish). A peak indexed escapement goal based on 17 streams on Kodiak Island has been established; the 2024 peak indexed escapement of 72,200 fish was below the goal of 101,000 fish.

Harvest Summary: Most of the KMA wild chum salmon are harvested during the directed pink and chum salmon fisheries in July. Most of the 2024 KMA chum salmon runs were weak. The 2024 KMA chum salmon harvest of 498,035 fish was below the 2024 forecast. Additionally, the Kitoi Bay Hatchery chum salmon harvest of 376,272 chum salmon was well above the forecast.

Summary by James Jackson Area Management Biologist, ADF&G, Kodiak.

CHIGNIK AREA

Commercial salmon fishing effort in the Chignik Management Area (CMA) was average, with 53 permits participating. From 2018 to 2021, the early run of sockeye salmon into the Chignik River watershed failed to develop, and both the early and late runs failed to develop in 2018 and 2020. Despite the stronger run in 2023, the recent weak CMA sockeye salmon runs are the major contributing factor to low participation. Prior to 2018, between 50 and 70 permits participated in CMA commercial fisheries on any given year.

The 2024 commercial harvest in the CMA was approximately 12,000 Chinook, 281,000 sockeye, 68,000 coho, 892,000 pink, and 61,000 chum salmon (Table 7).

On July 4, commercial salmon fishing was allowed in the Eastern, Central, Chignik Bay, Western, and Perryville Districts for 48 hours to assess the development of incoming sockeye salmon. Through July, commercial salmon fishing in the Chignik Bay District was restricted to 2 days a week to protect the Chignik River Chinook salmon run. The Eastern, Central, Western, and Perryville Districts were open for the majority of July. During the month of August, the Chignik Bay District was kept open, barring mandatory weekend regulatory closures. The Eastern and Central Districts were primarily kept closed, and the Western and Perryville Districts were intermittently open due to low pink and chum escapement. Harvest effort ceased on August 26 when processors concluded fish purchasing operations.

Chinook Salmon

Run and Escapement Summary: The Chignik River is the only major Chinook salmon producing stream within the CMA and one of the largest Chinook salmon streams on the South Alaska Peninsula. The BEG range for Chinook salmon into the Chignik River watershed is 1,300–2,700 fish. The Chinook salmon run has failed to consistently develop since 2017 and was declared a stock of concern in 2023. The Chignik River Chinook salmon escapement (above the weir) of 1,116 fish in 2024 did not meet the escapement goal. State subsistence and sport fishery harvest of Chinook salmon will not be known until permits and questionnaires are returned and tabulated.

Harvest Summary: A total of approximately 12,000 Chinook salmon were harvested during the 2024 season, which was above recent 5- and 10-year averages. Most of the harvest occurred in the Western District with Chinook salmon harvest being incidental to fisheries targeting other species. Chinook salmon escapements into the Chignik River system have been near recent averages but below the BEG despite additional restrictions on commercial salmon harvest opportunities. Throughout the entire commercial salmon fishing season, Chinook salmon 28 inches or greater could not be retained in the commercial salmon fishery in the entire Chignik Management Area. Commercial salmon openings in the Chignik Bay District were also limited to 48 hours a week to protect Chinook salmon.

Sockeye Salmon

Run and Escapement Summary: The Chignik River watershed supports 2 genetically distinct sockeye salmon runs that traditionally provide most directed harvest opportunities within the CMA: an early run and a late run. The early run starts in May, continues through June, and decreases in July, whereas the late run of sockeye salmon starts in June, continues through July, and decreases throughout August and into September. The early-run escapement was approximately 373,000 fish and was within the early-run OEG range of 300,000–400,000 fish. The

late-run estimated escapement of 355,000 fish was also within the late-run OEG range of 240,000–360,000 fish.

Harvest Summary: Harvest opportunity targeting sockeye salmon in the CMA is based upon the escapement of both early and late-run sockeye salmon in the Chignik River watershed. Due to the significant overlap of early and late runs of sockeye salmon, management must allow opportunity to harvest surplus fish without jeopardizing either run.

The 2024 Chignik River early run of sockeye salmon did not develop until late June, and no directed sockeye salmon commercial fishing periods were scheduled until July 4. The 2024 CMA sockeye salmon harvest of approximately 281,000 fish was below recent 5-, 10-, and 20-year average sockeye salmon harvests. Most sockeye salmon harvest came from the Chignik Bay District. Sockeye salmon harvest in the CMA occurred primarily in July.

Coho Salmon

Run and Escapement Summary: Coho salmon start to enter CMA drainages in mid-August and generally continue through November. In 2024, the weir ceased operations after August 24, at which point approximately 1,100 coho salmon had passed the weir. Late season coho salmon stream surveys were not conducted in the CMA in 2024 due to staff departure from Chignik.

Due to late season run timing and limited directed effort, escapement goals for coho salmon have not been established in the CMA.

Harvest Summary: A total of approximately 68,000 coho salmon were harvested during the 2024 commercial salmon fishing season. Coho salmon harvest was incidental to efforts targeting sockeye and pink salmon. The most recent 10-year average harvest of coho salmon in the CMA is approximately 106,000 fish.

Pink Salmon

Run and Escapement Summary: The CMA has an even-year areawide aggregate pink salmon SEG range of 170,000–280,000 fish. The aggregate pink salmon escapement is the sum of pink salmon escapement into 8 different area index streams. In 2024, the aggregate pink salmon escapement was 262,000 fish, which was within the SEG range. Escapements into most other CMA streams were monitored via aerial surveys to assess areawide run timing and distribution.

Pink salmon were observed migrating past the Chignik River weir starting in mid-July. Pink salmon continued to migrate past the weir until August 24 when the weir was removed. The pink salmon escapement should be considered a minimum because runs continue through August and into September. A total of over 2,800 pink salmon were observed migrating past the Chignik River weir in 2024.

Chum Salmon

Run and Escapement Summary: Chum salmon escaping into CMA streams were estimated via aerial surveys, except for the Chignik River, which was enumerated using the Chignik River weir. The chum salmon SEG range of 45,000–110,000 fish is based on escapement of 6 total index streams within 4 of the 5 districts. The 2024 estimated total peak chum salmon escapement for the 6 index streams was 83,000 fish, which was within the established SEG.

At the Chignik River weir, a total of 58 chum salmon were observed passing in 2024.

Harvest Summary: Approximately 61,000 chum salmon were harvested in the CMA, below the recent 10-year average of 140,000 fish. The majority of chum salmon harvest occurred in the Western District.

Summary by Carl Burnside, Area Management Biologist, ADF&G, Kodiak.

ALASKA PENINSULA, ALEUTIAN ISLANDS, AND ATKA-AMLIA ISLANDS AREAS

The 2024 commercial salmon harvest in the Alaska Peninsula, Aleutian Islands, and Atka-Amlia Islands Management areas (Area M) totaled approximately 9,000 Chinook, 2.5 million sockeye, 89,000 coho, 1.6 million pink, and 649,000 chum salmon (Table 7).

Chinook Salmon

Run and Escapement Summary: Nelson River is the only river in Area M with a Chinook salmon escapement goal. At the Nelson River weir, a total of approximately 3,542 Chinook salmon escaped, meeting the BEG range of 2,400–5,000 fish. The total Northern District Chinook salmon escapement of over 4,800 fish was below the most recent 10-year average of approximately 12,200 fish.

Harvest Summary: On the North Alaska Peninsula, approximately 690 Chinook salmon were harvested, which was below the 10-year average harvest of over 1,950 fish. The total commercial harvest for the South Peninsula fishery (including the Southeastern District Mainland [SEDM] from July 26 to October 31) was approximately 8,000 Chinook salmon.

Sockeye Salmon

Run and Escapement Summary: The Orzinski Lake sockeye salmon escapement of approximately 14,570 fish was within the SEG range of 14,000–28,000 fish.

The South Peninsula sockeye salmon escapement of almost 46,000 fish was below the management objective range of 48,200–86,400 fish. Escapement into Mortensen Lagoon was not observed in 2024 due to poor survey conditions. Escapement into Thin Point Lagoon (just under 27,000 fish) was within the SEG range of 14,000–28,000 fish.

The 2024 sockeye salmon escapement in the Northwestern District was within the escapement objective range of 52,600–106,000 fish, with a total escapement of almost 95,000 fish.

In the Northern District, the 2024 North Creek escapement estimate of over 500 sockeye salmon was below the SEG range of 7,500–10,000 fish. Surveys for much of the season could not be conducted due to poor weather and stream conditions.

The Nelson Lagoon Section was open for all regularly scheduled weekly fishing periods and was open continuously starting on June 26. The Nelson (Sapsuk) River total run of approximately 980,000 sockeye salmon (including harvest and escapement) was above the estimated forecast of 177,000 fish. From the total run, just over 204,000 fish were harvested in Nelson Lagoon, and almost 776,000 fish escaped in the Nelson River. The 2024 escapement exceeded the BEG range of 97,000–219,000 fish.

The Bear River early-run (through July 31) sockeye salmon escapement of 208,000 fish was within the SEG range of 176,000–293,000 fish. The Bear River late-run (after July 31) sockeye salmon escapement of approximately 245,768 fish was above the SEG range of 117,000–195,000 fish. The Bear River season sockeye salmon escapement was just over 454,000 fish, within the combined early- and late-run sockeye salmon SEG range of 293,000–488,000 fish.

The Sandy River sockeye salmon escapement of almost 38,000 fish met the escapement goal range of 34,000–74,000 fish.

The Ilnik River system sockeye salmon escapement through the weir was approximately 99,700 fish, exceeding the Ilnik River SEG range of 40,000–60,000 fish.

The McLees Lake sockeye salmon escapement project was not operated in 2024 due to funding.

Aerial escapement surveys of the Meshik River began on June 20. Subsequent surveys occurred throughout the season, and the final sockeye salmon escapement into the Meshik River system was just over 52,500 fish, within the SEG range of 48,000–86,000 fish. This estimate includes escapement into the Meshik River and tributaries, as well as Red Bluff and Yellow Bluff Creeks.

The total Cinder River (including Mud Creek) sockeye salmon escapement estimate of almost 37,700 fish was within the SEG range of 36,000–94,000 fish.

Harvest Summary: The total commercial harvest for the South Peninsula fishery (including the SEDM from July 26–October 31), was approximately 1.3 million fish.

On the North Peninsula, approximately 1.1 million sockeye salmon were harvested, which was below the 10-year average of 2.2 million fish.

Coho Salmon

Run and Escapement Summary: No coho salmon surveys were flown on the South Peninsula streams in 2024. A lack of escapement information for coho salmon is due to the departure of management staff from the South Peninsula region prior to peak coho salmon runs and poor weather conditions.

During the 2024 season, the final aerial surveys on the North Peninsula occurred on August 23. No coho surveys were conducted in September and October due to poor weather conditions and budgetary and logistical constraints. Escapement estimates that are tabulated from aerial surveys are considered minimum estimates. No coho salmon were documented by aerial surveys in 16 North Alaska Peninsula streams during 2024. Nelson and Ilnik Rivers are the only systems on the North Alaska Peninsula that have a coho salmon lower-bound SEG. At Nelson River, no aerial survey was conducted during a time at which coho are typically observed, and it is unknown if the lower bound of the escapement goal was achieved (19,000–29,000 fish). The Ilnik River was surveyed in August, and 0 coho were observed during these surveys; it is unknown if the lower-bound escapement goal of 9,000 fish was achieved.

Harvest Summary: The total commercial harvest for the South Peninsula fishery (including the SEDM from July 26–October 31) was almost 81,000 coho salmon. A total of approximately 8,000 coho salmon were harvested in the North Peninsula.

Pink Salmon

Run and Escapement Summary: The South Peninsula pink salmon total indexed escapement of 2.5 million fish was within the SEG range of 1,750,000–4,000,000 fish.

There are no escapement goals for pink salmon on the North Peninsula. In some years, depending on market conditions, a directed pink salmon fishery may occur; however, this did not happen in 2024. A total of almost 9,000 pink salmon were observed in the Northern District in 2024, and over 72,000 pink salmon were observed in the Northwestern District.

Harvest Summary: The total commercial harvest for the South Peninsula fishery (including the July 26 to October 31 SEDM fishery) was 1.6 million fish. A total of 17,000 pink salmon were commercially harvested in the North Peninsula Area fisheries.

Chum Salmon

Run and Escapement Summary: The South Peninsula chum salmon indexed total escapement of over 382,000 fish was within the cumulative district SEG range of 218,300–410,600 fish.

In the Northwestern District, the chum salmon escapement of 126,000 fish was within the SEG range of 100,000–215,000 fish.

The Northern District escapement goal is determined by an aggregate of 18 index streams. In 2024, approximately 41,000 chum salmon were observed in the 18 index streams, below the escapement goal range of 49,000–132,000 fish. A total of almost 52,000 chum salmon escaped into Northern District streams in 2024.

Harvest Summary: The total commercial harvest for the South Peninsula fishery (including the SEDM from July 26 to October 31) was 583,000 chum salmon. During the 2024 commercial chum salmon fishery, over 67,000 fish were harvested in the North Peninsula.

Summary by Matt Keyse, Geoff Spalinger, Annie Brewster, Charles Russell, and William Middleton Area Management Biologists, ADF&G, Kodiak.

Table 7.—2024 Westward Region commercial salmon harvests, by fishing area and species, in thousands of fish.

Fishing area	Species					Total
	Chinook	Sockeye	Coho	Pink	Chum	
Kodiak ^{a,b}	1	1,619	64	7,322	498	9,505
Chignik ^{b,c}	12	281	68	892	61	1,315
South Peninsula ^{b,c,d}	8	1,291	81	1,591	583	3,553
North Peninsula	1	1,173	8	17	67	1,266
Alaska Peninsula total	9	2,465	89	1,608	649	4,819
Westward Region total	22	4,365	221	9,822	1,209	15,639

Note: Columns may not total exactly due to rounding.

^a Total includes hatchery cost recovery.

^b Total includes commercial harvest that was discarded, confiscated, seized, or donated.

^c Total includes test fish sales.

^d Total includes commercially harvested fish retained for personal use.

SALMON SPECIES CATCH AND PROJECTIONS

Figures 2–6 show actual catch and projected catch for Chinook, sockeye, coho, pink, and chum salmon.

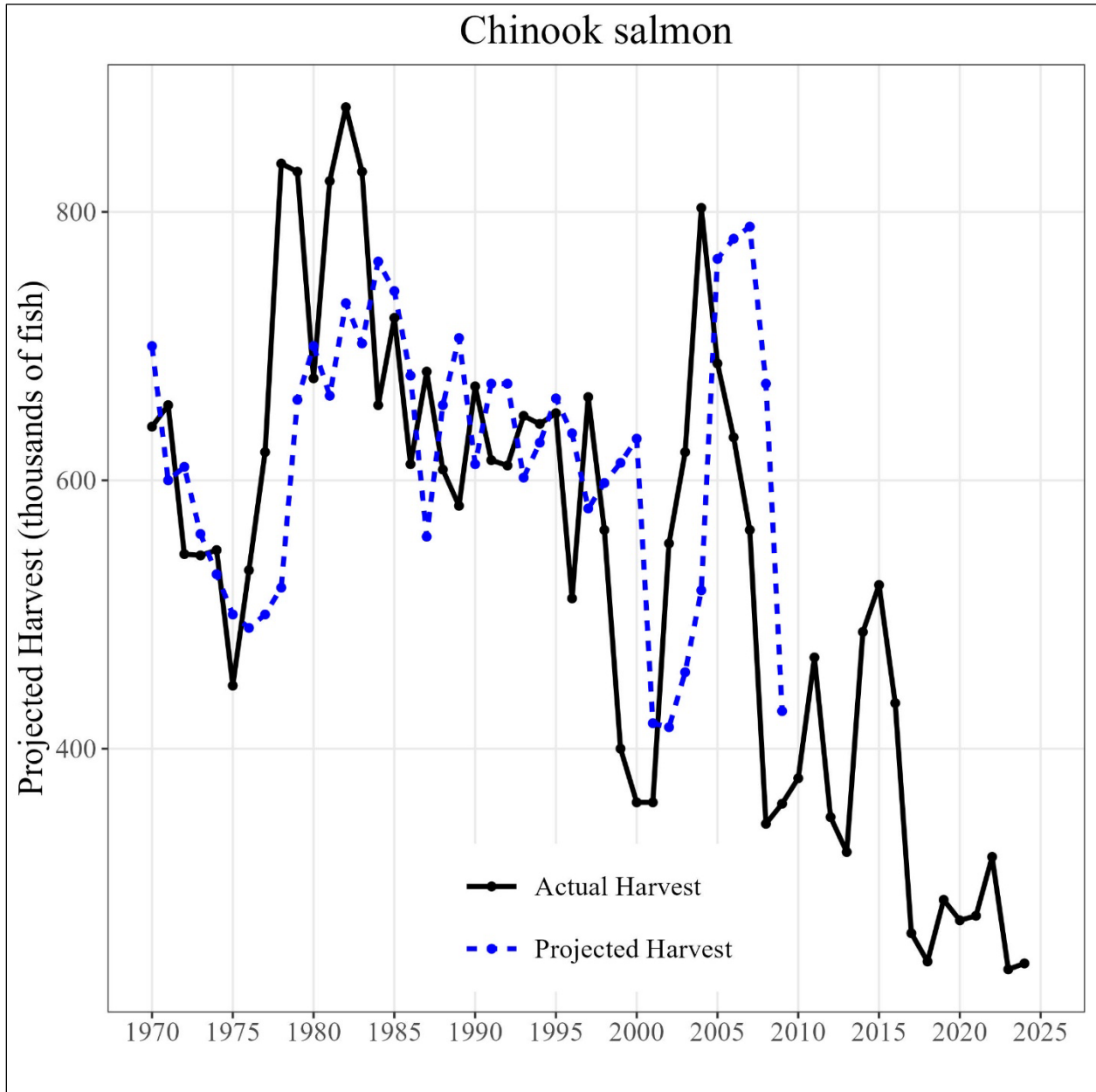


Figure 2.—Relationship between actual catch and projected catch in millions, for Alaska Chinook salmon fisheries from 1970 to 2024; 2010–2025 projections are not available.

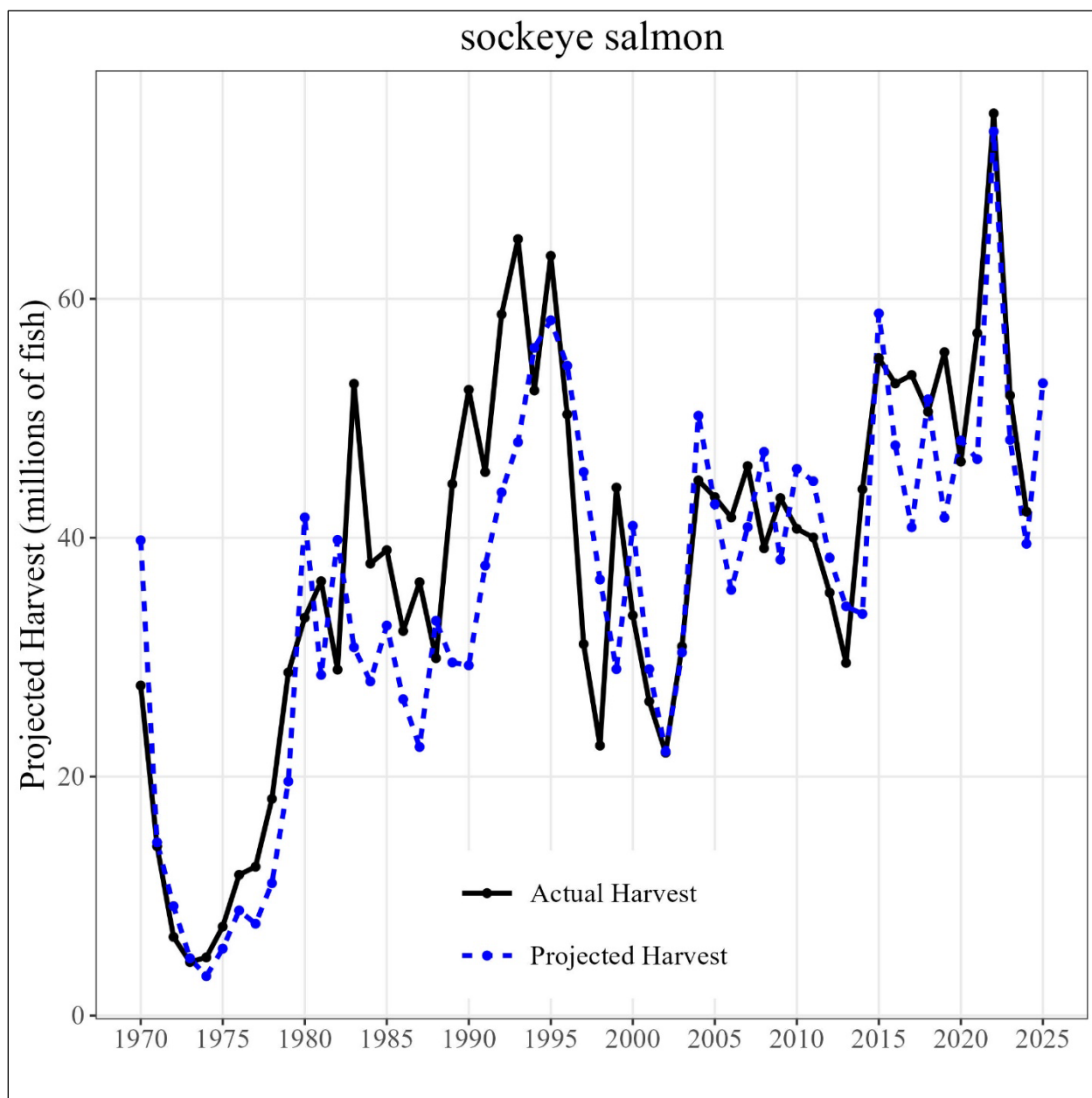


Figure 3.—Relationship between actual catch and projected catch in millions, for Alaska sockeye salmon fisheries from 1970 to 2024, with the 2025 projection.

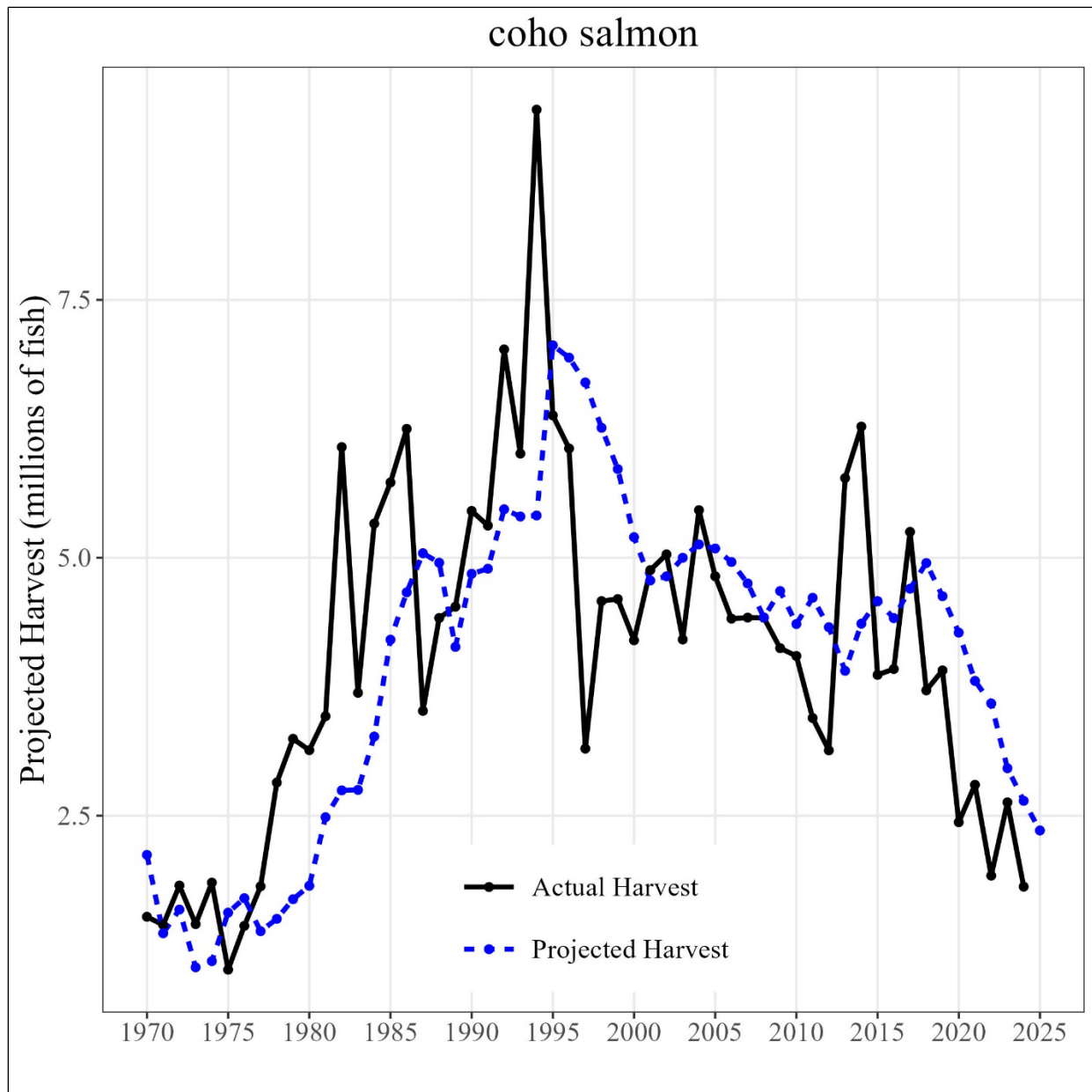


Figure 4.—Relationship between actual catch and projected catch in millions, for Alaska coho salmon fisheries from 1970 to 2024, with the 2025 projection.

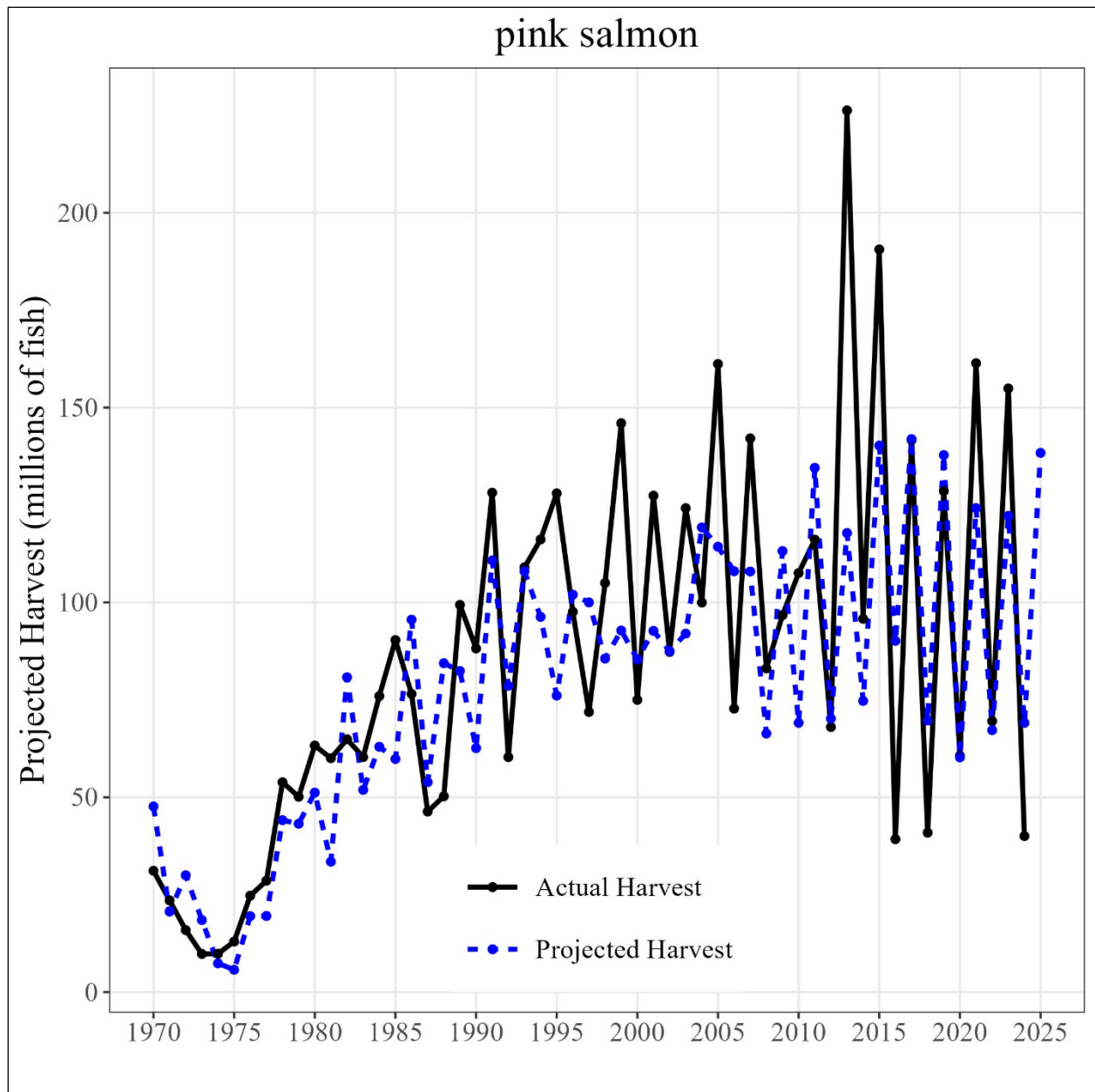


Figure 5.—Relationship between actual catch and projected catch in millions, for Alaska pink salmon fisheries from 1970 to 2024, with the 2025 projection.

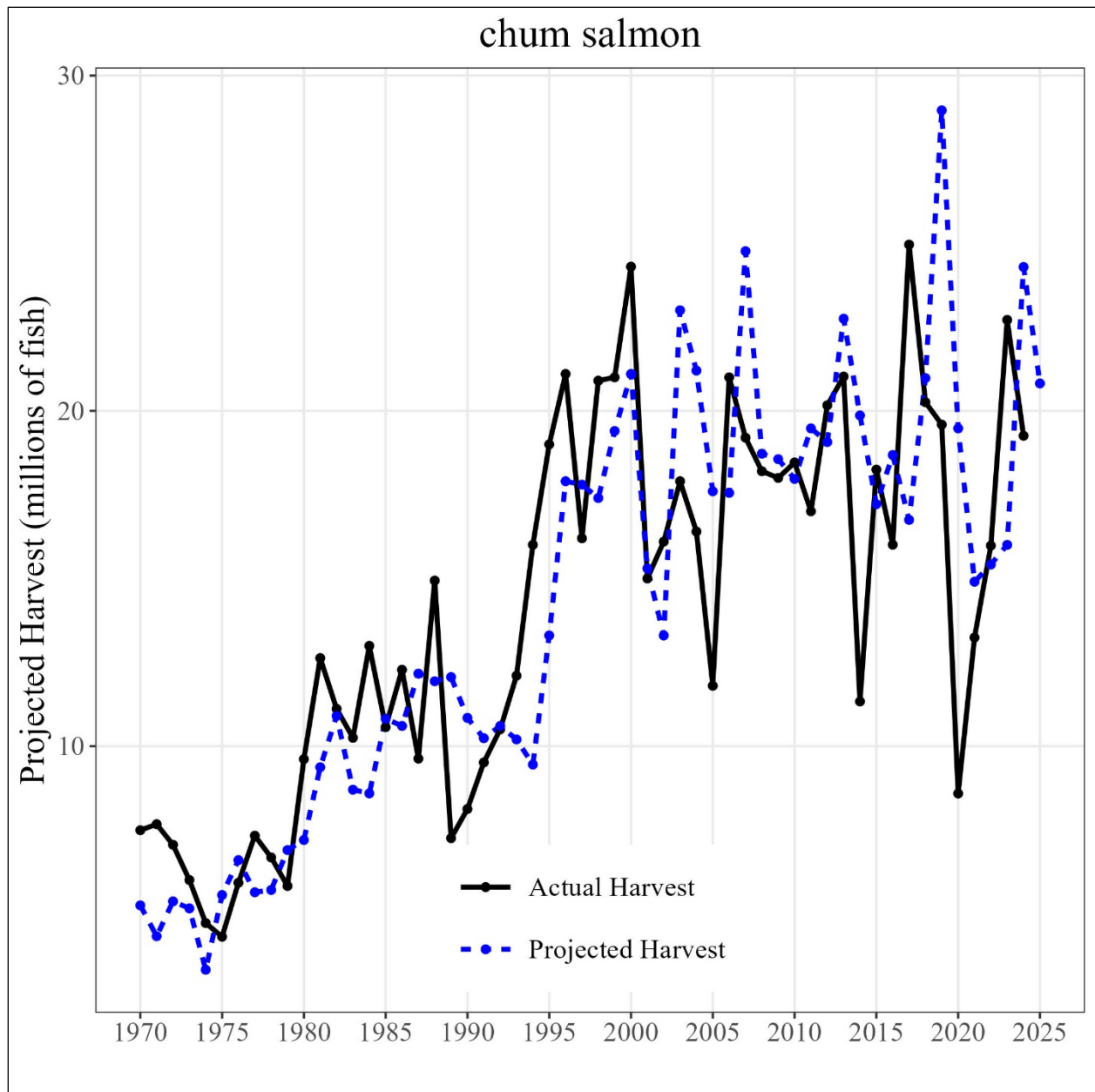


Figure 6.—Relationship between actual catch and projected catch in millions, for Alaska chum salmon fisheries from 1970 to 2024, with the 2025 projection.

PRELIMINARY FORECASTS OF 2025 SALMON RUNS TO SELECTED ALASKA FISHERIES

ADF&G prepares forecasts for salmon runs that affect major fisheries around the state. Salmon runs to be forecast are selected using several criteria, including economic importance, feasibility, compatibility with existing programs, and management needs (Table 8).

Table 8.—Forecast fisheries for the 2025 fishing year.

Fishing area	Targeted species
Southeast Region	Chinook salmon pink salmon
Central Region	
Copper River and Prince William Sound	wild Chinook, sockeye, pink, and chum salmon
Upper Cook Inlet	sockeye salmon
Lower Cook Inlet	pink salmon
Bristol Bay	sockeye salmon
Arctic-Yukon-Kuskokwim Region	
Yukon Management Area	fall chum salmon
Westward Region	
Kodiak Management Area	pink salmon
Ayakulik River	sockeye salmon
Karluk River	sockeye salmon
Alitak District (Frazer Lake and Upper Station)	sockeye salmon
Spiridon Lake	sockeye salmon
Chignik Management Area	sockeye salmon (early and late runs)
Alaska Peninsula/Aleutian Islands	sockeye salmon
Bear Lake	sockeye salmon (late run)
Nelson River	sockeye salmon
South Alaska Peninsula aggregate	pink salmon

A variety of information is used to forecast salmon runs and associated harvests. The escapement of parental stocks is an important first determinant of future run strength potential; however, other information that might be considered in forecasts includes outmigrating smolt numbers, returns from sibling age classes of the projected return, and environmental conditions. A range of run possibilities are predicted for each forecast fishery. In general, the actual run can be expected to fall within the range (between the lower and upper limits) less than half the time.

Harvest projections based on quantitative forecasts of salmon runs generally reflect potential harvests and are made for most of the major sockeye salmon fisheries and pink salmon fisheries in SEAK, PWS, Cook Inlet, Kodiak, and the Alaska Peninsula. Forecasts for large hatchery runs including sockeye, pink, and chum salmon runs to the SEAK, PWS, and Kodiak areas are provided by private nonprofit operators. For other fisheries, the harvest projections are made based on harvests levels from previous years and are reflective of fishing effort; thus, harvests are reflective of both market conditions and salmon run strength.

SOUTHEAST REGION

CHINOOK SALMON

Forecast Methods: Chinook salmon forecasts are typically produced for 5 stocks returning to SEAK rivers. The stocks of Chinook salmon originating in the Chilkat and Unuk Rivers have annual forecasts of total run, and terminal run forecasts are produced for stocks originating in the Situk, Taku, and Stikine Rivers. Methods described in Bernard and Jones (2014) are used to create forecasts of ocean-age-3 and ocean-age-4 fish. Ocean-age-5 fish are not forecast because these fish are rarely observed in escapements or harvests in SEAK. The age-specific forecasts use brood year data that are estimated for each stock annually. For example, the forecast of ocean-age-3 fish in year t , and from brood year $(t-5)$, is based on the return of ocean-age-2 fish in year $t-1$; the forecast of ocean-age-4 fish in year t , from brood year $(t-6)$, is based on the returns of ocean-age-2 fish in year $t-2$ and ocean-age-3 fish in year $t-1$. The brood year forecasting method incorporates more data for older age classes than a straight sibling regression and recognizes that previous returns from a given brood year have proven to be a good indicator of overall strength of an age class and early life-stage survival (Bernard and Jones 2014).

Run projections are created for the 2 principal brood years returning as ocean-age-3 and ocean-age-4 fish in the forecast year by applying previous abundance estimates at age to recently observed maturation rates. Once a brood year run projection is created, recent brood year return-at-age estimates are applied to the run projection to forecast the return of ocean-age-3 or ocean-age-4 fish. Using the previous 8 or 9 years of brood year information has proven to be the most accurate in forecasting, as described and investigated by the Chinook Technical Committee of the Pacific Salmon Commission (PSC 2016).

The uncertainty in return-at-age estimates is incorporated into the forecast by creating a normal distribution around the year $(t-1)$, ocean-age-2 abundance estimates, and standard error, which is then sampled 10,000 times to create a year (t) projection of ocean-age-3 fish. Similarly, a normal distribution using the year $(t-2)$, ocean-age-2 abundance estimate, and year $(t-1)$, ocean-age-3 abundance estimate, and standard errors, is created and sampled 10,000 times to create a year (t) projection of ocean-age-4 fish. A 5-year hindcast is then analyzed to determine forecast errors using the geometric mean, mean percent error (MPE), or mean absolute percent error (MAPE).

Most Chinook salmon systems in SEAK have been experiencing maturation rate changes for recent brood years, and this current trend includes a loss of older age fish (i.e., reduction of ocean-age-4 fish) in escapements and harvests. The loss of ocean-age-4 Chinook salmon has resulted in a slight increase of ocean-age-3 fish and an even greater increase in the ocean-age-2 fish. Because hindcast error for a specific year is calculated as (forecast-actual)/actual, hindcast errors of ocean-age-4 fish are generally greater than errors in forecasts of ocean-age-3 fish; however, this is quite variable among Chinook salmon systems. Overall, the 2025 forecasts are for total or terminal runs near or above the midpoint of the escapement goal range for the Situk, Chilkat, and Unuk stocks, above the upper bound of the escapement goal range for the Taku stock, and below the lower bound of the escapement goal range for the Stikine River stock (Table 9).

Table 9.—The 2025 run forecasts, average runs from 2015–2024, and 2024 runs and escapements for 3 Southeast Alaska Chinook stocks originating in the Chilkat, Taku, and Unuk Rivers.

Run type	Chinook stock	Situk Terminal	Chilkat total	Taku Terminal	Stikine Terminal	Unuk total
2025 Run forecast		750	3,000	40,125	9,921	2,600
Avg Run (2015–2024)		654	2,124	15,007	12,528	2,648
2024 Total/terminal run		517	2,182	23,338	9,921	2,293
2024 Escapement		517	2,070	24,518	9,835	1,980
5-year hindcast error	3-ocean	42.9%	19.2%	11.0%	-	16.9%
	4-ocean	79.8%	12.6%	37.3%	-	43.8%
Escapement Goal Range	Lower	450 ^a	1,750	19,000	14,000	1,800
	Upper	1050 ^a	3,500	36,000	28,000	3,800

^a ADF&G goal; Chinook Technical Committee of the Pacific Salmon Commission escapement goal range is 500 to 1,000 fish.

Forecast Discussion: Previous forecasting methods used traditional sibling regressions, which failed to capture current trends in return-at-age and resulted in inaccurate forecasts. Inaccurate forecasts impaired management’s ability to properly manage mixed stock and terminal area fisheries to pass adequate numbers of Chinook salmon to escapement. Overforecasting resulted in some very high harvest rates in poor run years and therefore several stocks failed to meet management objectives (Figure 7; Table 10). The methods outlined in Bernard and Jones (2014) helped improve forecast accuracy and have resulted in more conservative management in recent years.

Forecasts for SEAK Chinook salmon are produced by December 1 annually. The quality of data used in these forecasts is paramount in creating accurate and precise run projections. The department conducts full stock assessment projects on the Chilkat, Taku, Stikine, and Unuk Rivers, which includes marking emigrating Chinook salmon smolt with adipose fin clips and tagging with coded wire tags; estimating escapement either through mark–recapture experiments, index surveys, or both; and estimating stock-specific harvests in mixed stock fisheries using coded wire tag or genetic stock identification methodologies. The high-quality stock assessment data used to generate annual forecasts directly leads to more accurate forecasts.

Return year 2024 escapement and run estimates came in below forecast expectations for the Chilkat stock, and above forecast for the Taku and Unuk River stocks, and all 3 of these systems achieved the lower bound of the BEG. Most systems not only observed increased spawning abundance of large fish, but also a tremendous abundance of ocean-age-2 fish, which was close to the highest observed estimates in the time series.

Because run year 2024 showed a high abundance of ocean-age-2 fish from the 2020 brood year in most or all monitored systems for which age information is available, expectations for 2025 are higher than previous years. Ocean-age-2 abundance is usually a driver in assessing the strength of a particular brood year; however, management is taking a cautious approach due to over a decade of poor Chinook returns in Southeast Alaska. Management of sport, commercial, and subsistence fisheries in Southeast Alaska is anticipated to resemble previous years in 2025.

Table 10.—Estimated harvests, escapements, total runs, and annual harvest percentages, 2014–2024, along with the 2025 forecasts for the Situk, Chilkat, Taku, Stikine, and Unuk stocks of Chinook salmon in Southeast Alaska.

Stock	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	10-yr Average	Forecast 2025
HARVEST													
Situk	138	21	64	25	2	—	—	70	12	—	—	19	
Chilkat	1,077	713	437	239	208	87	79	127	44	82	128	214	
Taku	5,755	5,297	3,758	2,706	1,777	627	1,143	838	1,599	1,273	1,267	2,028	
Stikine	6,646	7,039	6,244	2,421	260	1,656	995	1,181	1,982	1,524	4,041	2,734	
Unuk	1,423	3,632	1,471	544	856	955	597	577	731	1,123	463	1,095	
ESCAPEMENT													
Situk	475	176	337	1,190	421	620	1,197	1,064	890	144	517	656	
Chilkat	1,529	2,452	1,380	1,173	873	2,028	3,180	2,038	1,582	2,234	2,070	1,901	
Taku	23,532	23,567	9,177	8,214	7,271	11,558	15,593	11,341	12,722	14,755	24,518	13,872	
Stikine	24,374	21,597	10,554	7,335	8,603	13,817	9,753	8,376	9,090	12,795	9,835	11,176	
Unuk	1,691	2,623	1,463	1,203	1,971	3,115	1,135	2,666	1,304	2,072	1,980	1,953	
RUN													
Situk	613	197	401	1,215	423	620	1,197	1,134	902	144	517	675	750
Chilkat	2,606	3,165	1,817	1,412	1,081	2,115	3,259	2,165	1,626	2,316	2,198	2,115	3,000
Taku	29,287	28,864	12,935	10,920	9,048	12,185	16,736	12,179	14,321	16,028	25,785	15,900	40,000
Stikine	31,020	28,636	16,798	9,756	8,863	15,473	10,748	9,557	11,072	14,319	13,876	13,910	10,000
Unuk	3,114	6,255	2,934	1,747	2,827	4,070	1,732	3,244	2,035	3,195	2,443	3,048	2,600
HARVEST RATE													
Situk	23%	11%	16%	2%	0%	0%	0%	6%	1%	0%	0%	3%	
Chilkat	41%	23%	24%	17%	19%	4%	2%	6%	3%	4%	6%	10%	
Taku	20%	18%	29%	25%	20%	5%	7%	7%	11%	8%	5%	13%	
Stikine	21%	25%	37%	25%	3%	11%	9%	12%	18%	11%	29%	20%	
Unuk	46%	58%	50%	31%	30%	23%	34%	18%	36%	35%	19%	36%	

Note: En dashes indicate no harvest.

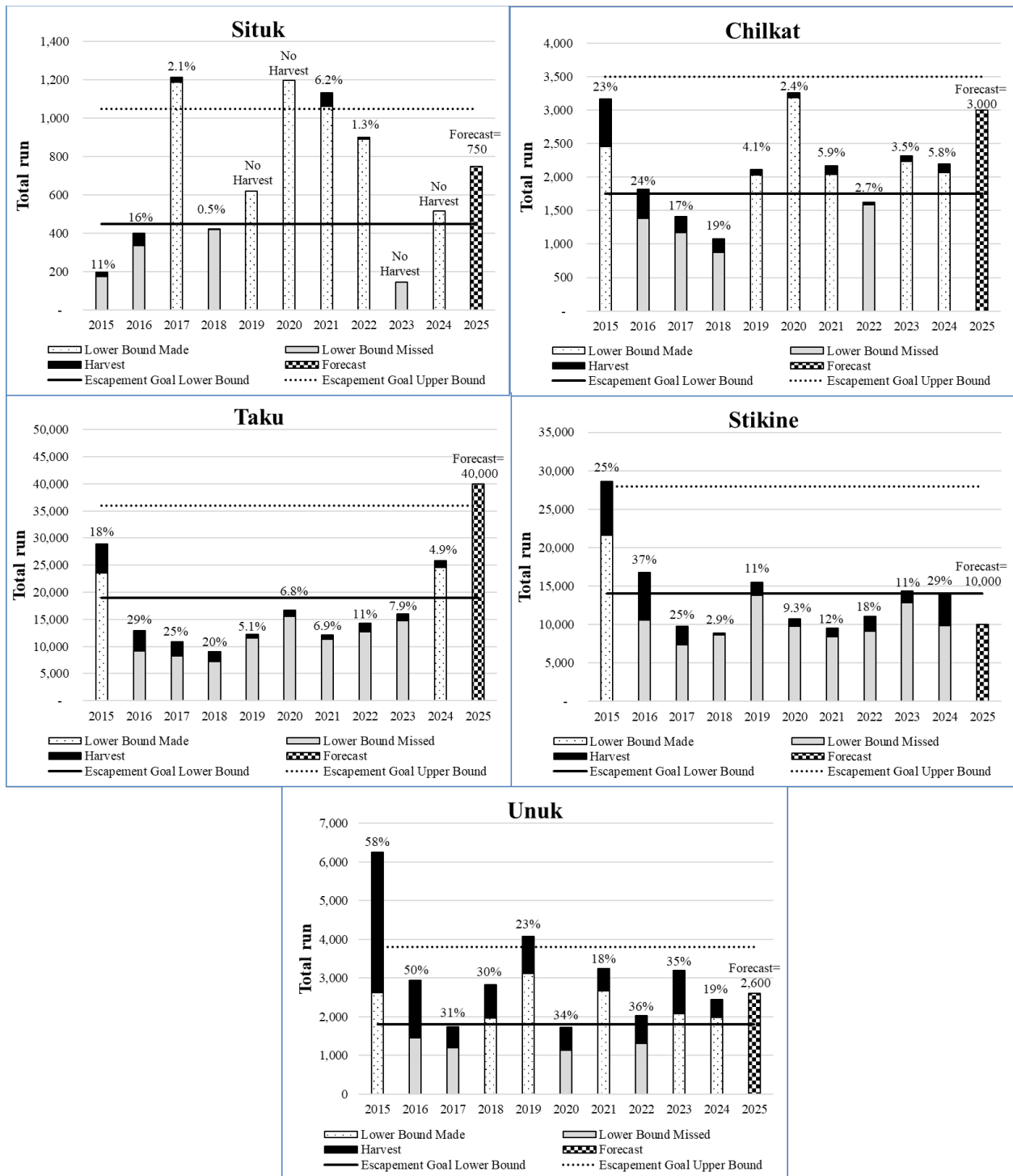


Figure 7.—Escapement estimates in relation to escapement goals, harvest estimates, and harvest percentages, expressed as a calendar year harvest rate of the total run, for the Situk, Chilkat, Taku, Stikine, and Unuk stocks of Chinook salmon in Southeast Alaska, 2015–2024, with 2025 forecasts depicted with checkered bars.

PINK SALMON

The Southeast Alaska (SEAK) pink salmon harvest in 2025 is predicted to be in the *average* range with a point estimate of 29 million fish (80% prediction interval: 16–53 million fish). The categorical ranges of pink salmon harvest in SEAK were formulated from the 20th, 40th, 60th, and 80th percentiles of historical harvest over the 64-year period 1960–2023 (Table 11).

Table 11.—Southeast Region pink salmon harvest categories.

Category	Range (millions)	Percentile
Poor	Less than 11	Less than 20th
Weak	11 to 19	20th to 40th
Average	19 to 33	40th to 60th
Strong	33 to 48	60th to 80th
Excellent	Greater than 48	Greater than 80th

Forecast Methods: The NOAA Alaska Fisheries Science Center, Auke Bay Laboratories (NOAA) initiated the Southeast Alaska Coastal Monitoring (SECM) project in 1997 to better understand the effects of climate and nearshore ocean conditions on year-class strength of salmon and other ecologically related species (Orsi et al. 2000). Since 2018, the SECM project has been conducted cooperatively by NOAA and the Alaska Department of Fish and Game (ADF&G) using the ADF&G research vessel *Medeia*, and the 2 agencies have combined efforts to produce a joint pink salmon harvest forecast using SECM data (Piston et al. 2019). We plan to continue coordination between agencies and explore ways to further expand the SECM survey to provide a wide variety of valuable information to the fishing industry.

The 2025 SEAK pink salmon harvest forecast (Figures 8 and 9) was primarily based on juvenile pink salmon abundance indices collected by the SECM project in northern SEAK inside waters. These data were obtained from systematic surface trawl surveys conducted annually in June and July in upper Chatham and Icy Straits and have been shown to be highly correlated with the harvest of adult pink salmon in the following year (Wertheimer et al. 2011). The 2024 juvenile pink salmon abundance index based on the average vessel-calibrated, log-transformed standardized catch per unit effort (CPUE) in either June or July, whichever month had the highest average in a given year (standardized juvenile catches are based on 20-minute trawl sets), was 1.66. Using only the years with data collected on the research vessel *Medeia* (2018–2024), this CPUE is the highest index since 2020 and falls just above average for juvenile even-year CPUE indices (*Medeia* average 1.61; *Medeia* range 1.2–2.2).

The forecasts for 2025 were created using a modified approach initially described by Wertheimer et al. (2006) and later adapted by Orsi et al. (2016) and Murphy et al. (2019). The current forecast assumes a log-normal error structure (Miller et al. 2022) and is based on a multiple regression model. This model integrates vessel-calibration coefficients estimated within the model, applied to raw CPUE as a proxy for abundance, along with an odd/even year factor to account for potential cyclical abundance patterns, and temperature data from the SECM survey (Piston et al. 2021) or satellite sea surface temperature (SST) data (Huang et al. 2017). This year, the CPUE calculation differs from prior forecasts: the raw CPUE term was set to the maximum value of the log-transformed, standardized catch (from 20-minute trawl sets) for either June or July, depending on which month yielded the highest value in each year. The raw CPUE was then adjusted by a model-derived vessel calibration coefficient. A total of 18 models were considered for the 2025 forecast,

with model performance evaluated based on the one-step-ahead MAPE for the last 5 years, along with AICc values (small-sample corrected Akaike Information Criterion; Burnham and Anderson 2004), parameter significance, and adjusted R-squared values. The optimal model included vessel-calibration coefficients, raw CPUE, an odd/even year factor, and the satellite SST variable from northern SEAK in May. Based on this model, the 2025 forecast falls within the average range, with a point estimate of 29 million fish (80% prediction interval: 16 to 53 million fish).

Forecast Discussion: The 2025 harvest forecast of 29 million pink salmon is slightly above the recent 10-year average harvest of 26 million but approximately 60% of the parent-year (2023) harvest of 48 million. Parent-year escapement indices exceeded the biological escapement goal ranges in the Southern Southeast and Northern Southeast Inside Subregions and were near the upper bound of the escapement goal range in the Northern Southeast Outside Subregion. Juvenile pink salmon in the 2024 SECM survey trawls were smaller than average in length but had higher than average energy density (1997–2024).

Poor forecast accuracy in recent odd years (2021 and 2023) led to a review of odd/even year treatment in the forecast model and potential issues with vessel-calibration coefficients. Historically, separate odd- and even-year models were avoided due to limited years in the dataset and similar trends observed in both broodlines until 2006. Including an odd/even year factor now aligns better with the 1-ocean-year life history of pink salmon, accounting for recent odd/even year harvest and escapement differences. The model also addressed errors potentially introduced by observed vessel-calibration coefficients by replacing them with model-estimated coefficients, based on 25 years of vessel-specific raw CPUE data, which better accounts for variations in vessel efficiency over the SECM survey's long history.

Although uncertainties are inherent in salmon forecasts, the NOAA/ADF&G joint pink salmon harvest forecast has maintained a strong track record (Figure 9), despite the unique forecasting challenges for pink salmon (Haeseker et al. 2005). For the 2025 season, the department will manage the commercial purse seine fisheries based on inseason data. Aerial escapement surveys and fishery performance data will continue to be key tools for making inseason management decisions.

Forecast by Teresa Fish, Andy Piston, and Sara Miller, ADF&G; and Wesley Strasburger and Emily Fergusson, NOAA, Auke Bay Lab, Alaska Fisheries Science Center.

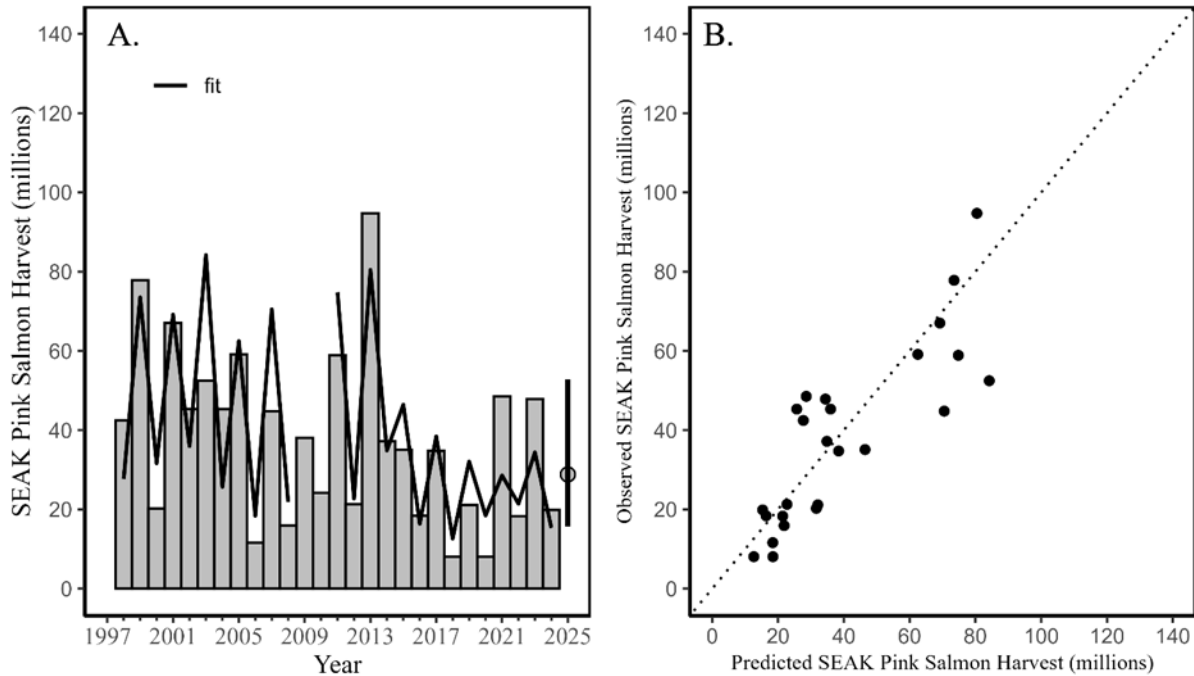


Figure 8.—Forecast model fit (hindcasts) for total Southeast Alaska (SEAK) pink salmon harvest, 1998–2024 by year (A) and by the fitted values (B) for the model based on the model-estimated vessel-calibration coefficients, raw CPUE, an odd- and even-year factor, and May satellite sea surface temperature readings in northern Southeast Alaska inside waters. In panel A, the 2025 forecast is shown as a grey circle with the 80% prediction interval as a black vertical line. The observed SEAK pink salmon harvest is represented by the grey bars and the model fit is shown by the black line in panel A. In panel B, the dotted line represents a one-to-one line; circles above the line represent hindcasts that produced a point estimate lower than the actual harvest and circles below the line represent hindcasts that produced a point estimate higher than the actual harvest.

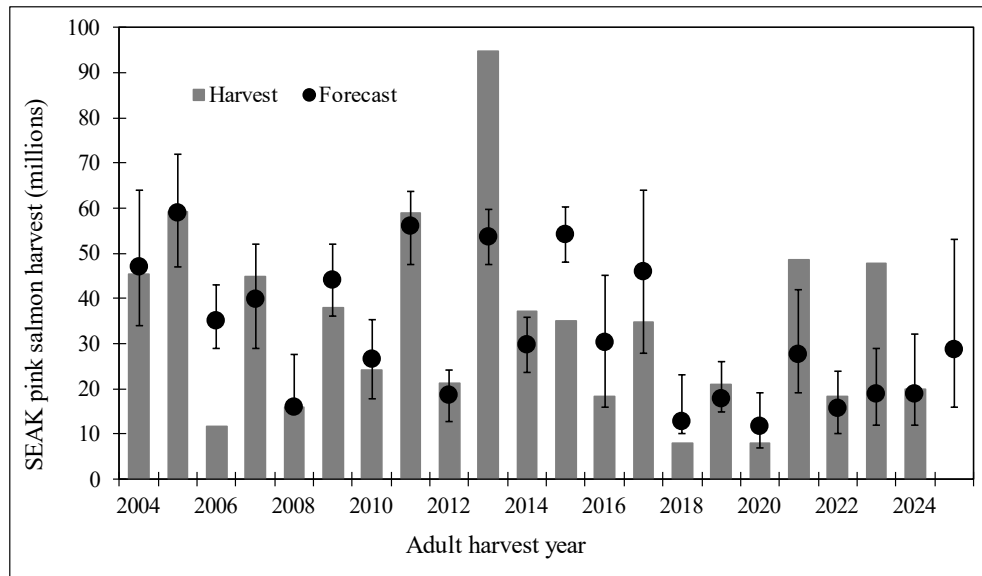


Figure 9.—Preseason forecasts compared to the annual SEAK pink salmon harvest, 2004–2024. The error bars represent either 80% confidence or 80% prediction intervals of the forecasts, depending on the modeling method used.

CENTRAL REGION

COPPER RIVER AND PRINCE WILLIAM SOUND

Forecasts of total run were calculated for Copper River Chinook salmon, Copper River wild sockeye salmon, Gulkana Hatchery sockeye salmon, Coghill Lake sockeye salmon, and for wild Prince William Sound (PWS) pink and chum salmon (Table 12). Prince William Sound Aquaculture Corporation and Valdez Fisheries Development Association provide additional forecasts for hatchery-specific stocks. The categorical ranges of total run strength were formulated for each stock from the 20th, 40th, 60th, and 80th percentiles of the recent 10 years for Chinook, chum, and sockeye salmon and 10 odd-years (2005–2023) for pink salmon (Table 13). Salmon forecasts are inherently uncertain and are primarily used to gauge the general magnitude of expected runs and set early-season harvest management strategy. In 2025, the department will manage PWS and Copper River area commercial salmon fisheries inseason based on the strength of salmon abundance indices, including sonar counts, weir passage, aerial escapement surveys, and fishery performance data.

Table 12.—2025 Prince William Sound Area salmon run forecast summary (thousands of fish) and percentile category.

Area/run type	Salmon species	Forecast point estimate	Forecast range	% Above/below 10-year average	Total run 10-year average	Category
Copper River						
<i>Wild</i>	Chinook	36	25–51	25% Below	48	Weak
<i>Wild</i>	Sockeye	2,558	2,205–2,910	55% Above	1,646	Excellent
<i>Gulkana Hatchery</i>	Sockeye	80	69–91	27% Below	110	Average
<i>Total Run</i>	Sockeye	2,638	2,274–3,002	50% Above	1,757	Excellent
Coghill Lake						
<i>Wild</i>	Sockeye	331	224–439	57% Above	211	Excellent
Prince William Sound						
<i>Wild</i>	Pink	18,626	8,500–28,753	8% Above	17,204	Strong
<i>Wild</i>	Chum	613	285–941	18% Above	520	Strong

Table 13.—Copper River and Prince William Sound categorical ranges of total run.

Category	Percentile
Poor	Less than 20th
Weak	20th to 40th
Average	40th to 60th
Strong	60th to 80th
Excellent	Greater than 80th

Copper River Chinook Salmon

The 2025 Copper River Chinook salmon total run forecast point estimate is *weak* at 36,000 fish (80% prediction interval: 25,000–51,000 fish. This is 25% below the recent 10-year average (2015–2024) total run of 48,000 fish (Figure 10). Because the sustainable escapement goal (SEG)

for Copper River Chinook salmon is 21,000 to 31,000 fish, this forecast supports conservative management.

Forecast Methods: For 2025, the Copper River Chinook salmon state-space model was chosen as the forecast method. This model simultaneously reconstructs runs and fits a spawner-recruit model to estimate total return, escapement, and recruitment of Copper River Chinook salmon from 1999 to 2024. Methods and details of this analysis are covered in separate reports (Savereide et al. 2018; Joy et al. 2021). The model uses harvest, age composition, and direct measures of inriver run abundance to estimate parameters that describe the spawner-recruit relationship for this stock. Uncertainty from the run reconstruction is passed through to the spawner-recruit analysis, and all relevant data are considered and weighted by their precision. The model accommodates missing data, measurement error in the data, and changes in age at maturity.

Several forecast methods were examined for the 2025 Copper River Chinook salmon total run forecast, including exponential smoothing; 2-, 3-, and 5-year running averages of total run; and projections from the Copper River Chinook salmon state-space model. The state-space model outperformed the exponential and average-based models by having a lower mean absolute percentage error (MAPE) and mean percentage error (MPE) when compared retrospectively (Table 14) and used more biological information to predict future runs. Total run size in prior years was calculated as the sum of commercial and subsistence harvests of Chinook salmon below Miles Lake and the mark-recapture point estimate of Chinook salmon inriver abundance. There are currently 25 years (1999–2024) of inriver mark-recapture abundance estimates and 46 years (1980–2024) of harvest, escapement and age composition data available for this analysis. The 80% prediction intervals were calculated from the posterior distributions of the model parameters, including the predicted run size for 2025.

Table 14.–2025 Copper River Chinook salmon forecast model performance summary. Model selected as the run forecast, lowest mean absolute percentage error (MAPE), is bold.

Stock/model	Forecast point estimate	80% prediction interval	MAPE
State-space	35,906	24,783-51,202	32%
Exponential	43,140	26,369-59,911	39%
2-year	46,586	19,567-73,604	40%
3-year	48,352	23,190-73,514	44%
5-year	41,642	19,740-63,545	39%

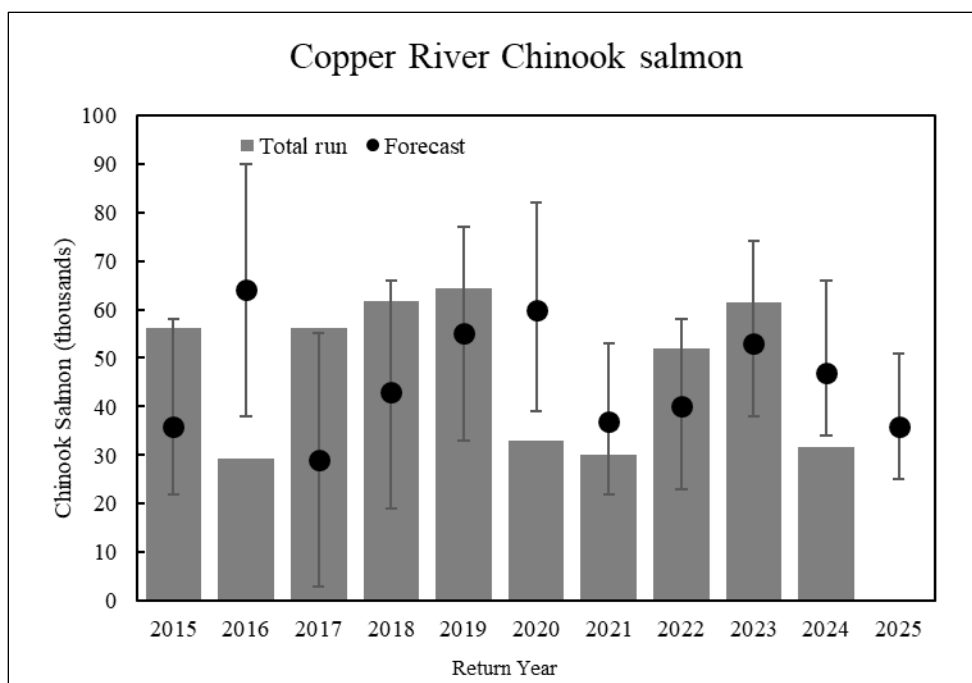


Figure 10.—Total run of Copper River Chinook salmon compared to preseason total run forecasts, 2015–2024, and the 2025 forecast. Error bars represent 80% prediction intervals of forecasts.

Copper River Sockeye Salmon

The 2025 wild Copper River sockeye salmon total run forecast point estimate is *excellent* at 2,558,000 fish (80% prediction interval: 2,205,000–2,910,000 fish). Gulkana Hatchery sockeye salmon total run forecast is *average* at 80,000 fish (80% prediction interval: 69,000–91,000 fish), for an *excellent* total Copper River sockeye salmon run (wild + hatchery production) forecast of 2,638,000 fish (80% prediction interval: 2,274,000–3,002,000 fish). This is 50% above the recent 10-year average (2015–2024) total run of 1,757,000 fish (Figure 11). The total Copper River sockeye salmon harvest estimate (all fisheries) is predicted to be 2,126,000 (80% prediction interval: 1,834,000–2,418,000 fish), with a commercial harvest of 1,920,000 fish (80% prediction interval: 1,628,000–2,212,000 fish).

Forecast Methods: The 2025 forecast of wild sockeye salmon to the Copper River is the sum of individual forecasts for six age classes. Linear regression models with log-transformed data were used to forecast returns for age-1.2, -1.3, -2.2, and -2.3 sockeye salmon. Forecasts of these 4 age classes were developed from the relationship between returns of each age class and returns of the age class 1 year younger from the same brood year (sibling model, Table 15). The forecast return of age-1.1, and -0.3, sockeye salmon were calculated as the 5-year (2020–2024) mean return of these age classes. The 2025 run to Gulkana Hatchery was estimated as the recent 10-year weighted average fry-to-adult survival estimate (0.83%).

The total harvest point estimate (all fisheries) was calculated by subtracting the Gulkana Hatchery broodstock, hatchery surplus, and wild stock escapement goal needs (upriver and Copper River Delta) from the total run forecast. The commercial harvest estimate was calculated by subtracting Copper River inriver goal categories (5 AAC 24.360(b)) and the lower bound of the Copper River

Delta spawning escapement goal, from the total run forecast. An estimated exploitation rate of 70% was used to project the total harvest of Gulkana Hatchery stocks in 2025. There are currently 60 years (1965–2024) of harvest, escapement, and age composition data available for this analysis. The total run of 80% prediction intervals was calculated from the mean squared error of the retrospective forecast predictions.

Table 15.—2025 Prince William Sound wild sockeye salmon forecast model summary. Models selected for inclusion in the run forecast (lowest MAPE) are bold.

Stock/age class	Brood year	Model	Forecast ^s	MAPE
Copper River wild sockeye				
0.3	2021	5-year mean	44,389	
1.1	2022	5-year mean	3,698	
1.2	2021	log 1.2 R/S x BYE	260,885	67%
		log 1.2 x log 1.1	257,003	58%
1.3	2020	1.3 x BYE	1,006,671	50%
		log 1.3R/S x BYE	990,140	42%
		log 1.3 x log 1.2	1,966,689	34%
		log 1.3 x log 0.3	692,649	43%
		1.3 x 1.2	2,863,064	52%
2.2	2020	log 2.2 x BYE	23,419	71%
		log 2.2 x log 1.2	63,809	66%
2.3	2019	log 2.3 x log 2.2	221,991	61%
		log 2.3 x log 1.3	65,850	70%
Total			2,557,579	
Coghill Lake sockeye				
1.1	2022	10-year mean	7,419	
1.2	2021	log 1.2 R/S x BYE	25,366	183%
		log 1.2 x log 1.1	33,163	129%
1.3	2020	log R/S 1.3 x BYE	99,979	189%
		log 1.3 x log 1.2	270,227	74%
2.2	2020	10-year mean	11,986	
2.3	2019	10-year mean	8,663	
Total			331,457	

Note: R/S = return per spawner; BYE = brood year escapement.

^a Age class forecasts do not always sum to total run forecast due to rounding error.

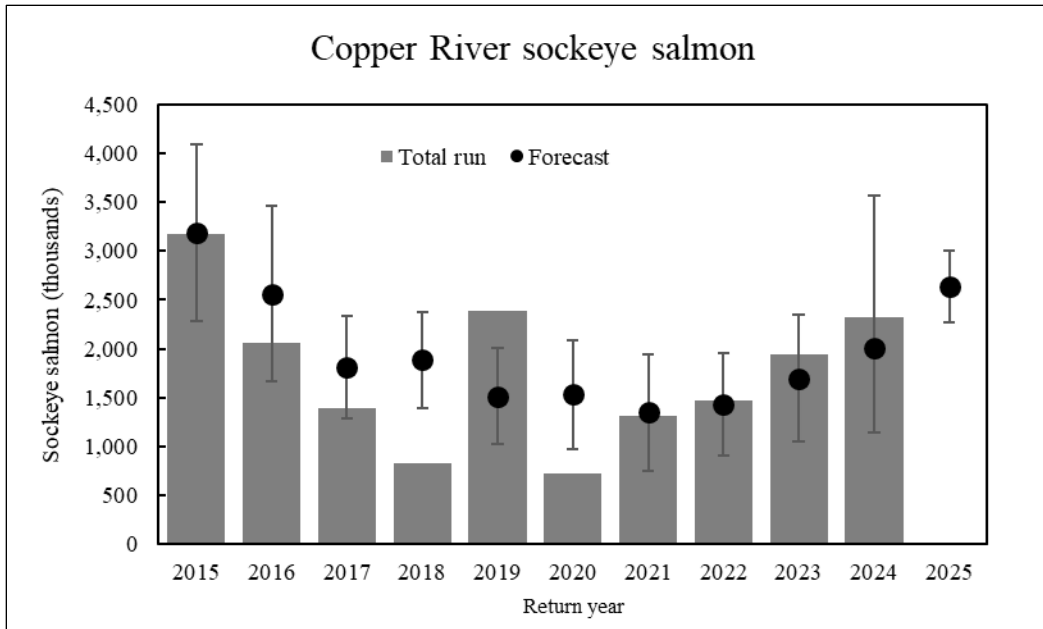


Figure 11.—Total run of Copper River sockeye salmon compared to preseason total run forecasts, 2015–2024, and the 2025 forecast. Error bars represent 80% prediction intervals of forecasts.

Coghill Lake Sockeye Salmon

The 2025 Coghill Lake sockeye salmon total run forecast point estimate is *excellent* at 331,000 fish (80% prediction interval: 224,000–439,000 fish). This is 57% above the 10-year average (2015–2024) total run of 211,000 fish (Figure 12). Subtracting the escapement target of 30,000 fish from the total run forecast results in a harvest point estimate (all fisheries) of 301,000 fish (range: 194,000–409,000 fish).

Forecast Methods: The 2025 sockeye salmon run forecast to Coghill Lake is the total of estimates for five age classes. Linear regression models with log-transformed data were used to predict returns of age-1.3 and -1.2 sockeye salmon (Table 15). These linear regression models were parameterized using the historical relationship between returns of age-1.3 sockeye salmon and returns of age-1.2 fish one year previous and returns of age-1.2 sockeye salmon and returns of the age-1.1 fish one year previous (sibling models). For example, the model chosen to forecast the return of age-1.3 sockeye salmon in 2025 used the return of age-1.2 fish in 2024 as the input parameter. Forecast returns of age-1.1, -2.2, and -2.3 sockeye salmon were calculated as the 2015–2024 mean return of that age class.

Harvest, escapement, and age composition data have been available for Coghill Lake sockeye salmon runs since 1962; however, the inclusion of escapements prior to the installation of a full weir in 1974 reduces forecast reliability. Therefore, only data collected since 1974 were used. Total run by year was estimated as the total commercial harvest contribution combined with the Coghill River weir escapement count. The 80% prediction intervals for the Coghill Lake sockeye salmon total run were calculated using the squared deviations between the 2015–2024 forecasts and actual runs as the forecast variance.

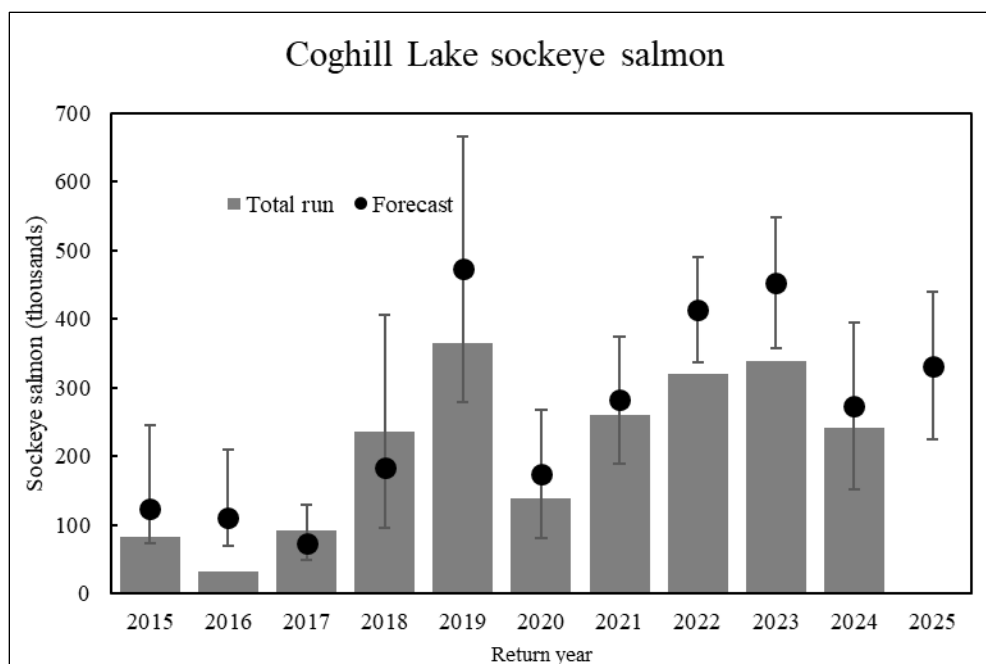


Figure 12.—Total run of Coghill Lake sockeye salmon compared to preseason total run forecasts, 2015–2024, and the 2025 forecast. Error bars represent 80% prediction intervals of forecasts.

PWS Odd-Year Wild Pink Salmon

The 2025 PWS wild pink salmon total run forecast point estimate is *strong* at 18,626,000 fish (80% prediction interval: 8,500,000–28,753,000 fish). This is 8% above the recent 10-odd-year average (2005–2023) PWS wild pink salmon total run of 17,204,000 fish (Figure 13). Subtracting the midpoint of the odd-year sustainable escapement goal, 1,838,000, from the total run forecast results in a harvest point estimate of 16,788,000 fish (range: 6,662,000 to 26,915,000).

Forecast Methods: Several models were examined for the 2025 PWS wild pink salmon total run forecast, including exponential smoothing and 2-, 3-, and 5-year running averages of past odd-year total runs. The 2-year running average forecast was selected for 2025 because it outperformed other forecast models by having the lowest MAPE and median symmetrical accuracy (Table 16). The 80% prediction intervals were calculated from the mean squared error of the retrospective forecast predictions.

Total wild run of pink salmon by year was estimated as the total wild (non-hatchery) contribution to commercial harvests combined with stream escapement index. The stream escapement index is calculated as the area under the curve of weekly aerial escapement surveys, adjusted for estimates of stream life. Hatchery and wild stock contributions were determined from thermal marked otolith recoveries (1997–2024), coded wire tag recoveries (1985–1996), or average fry-to-adult survival estimates multiplied by fry release numbers and estimated exploitation rates (1977–1984).

Table 16.—2025 Prince William Sound wild pink and chum salmon forecast model performance summary. Models selected as the run forecast (lowest MAPE) are bold.

Run	Prediction	MAPE
PWS wild pink		
Exponential	15,993,989	76%
2-year	18,626,351	73%
3-year	18,544,346	80%
5-year	21,881,402	102%
PWS wild chum		
Exponential	367,802	46%
2-year	685,865	40%
3-year	612,700	38%
5-year	488,482	39%

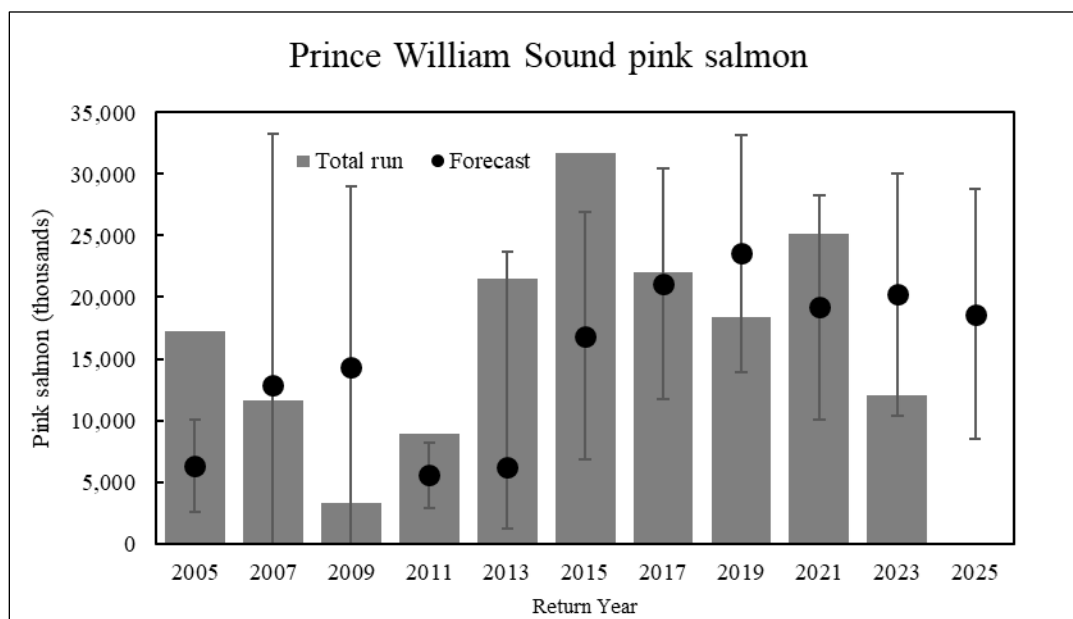


Figure 13.—Total run of Prince William Sound wild pink salmon compared to preseason total run forecasts, 2005–2023, and the 2025 forecast. Error bars represent 80% prediction intervals of forecasts.

PWS Wild Chum Salmon

The 2025 PWS wild chum salmon total run forecast point estimate is *strong* at 613,000 fish (80% prediction interval: 285,000–941,000 fish). This is 18% above the recent 10-year average (2015–2024) PWS wild chum salmon total run of 520,000 fish (Figure 14). Subtracting the 10-year average escapement of 170,000 from the total run forecast results in a harvest point estimate of 443,000 fish (range: 115,000 to 771,000 fish).

Forecast Methods: The 2025 PWS wild chum salmon total run forecast uses the 3-year running average method. Several models were examined for the 2025 PWS wild chum total run forecast, including exponential smoothing and 2-, 3-, and 5-year running averages of past total runs (Table 16). For 2025, the 3-year running average outperformed the other models by having the

lowest MAPE, mean absolute squared error (MASE), and median symmetrical accuracy. The 80% prediction intervals were calculated from the mean squared error of the retrospective forecast predictions.

Total wild run of chum salmon by year was estimated as the total wild (non-hatchery) contribution to commercial harvests combined with the stream escapement index. The stream escapement index is calculated as the area under the curve of weekly aerial escapement surveys adjusted for estimates of stream life. Hatchery and wild stock contributions were estimated using pre-hatchery average natural runs (1998–2003) or thermally marked otolith estimates (2004–2024) for each district in PWS.

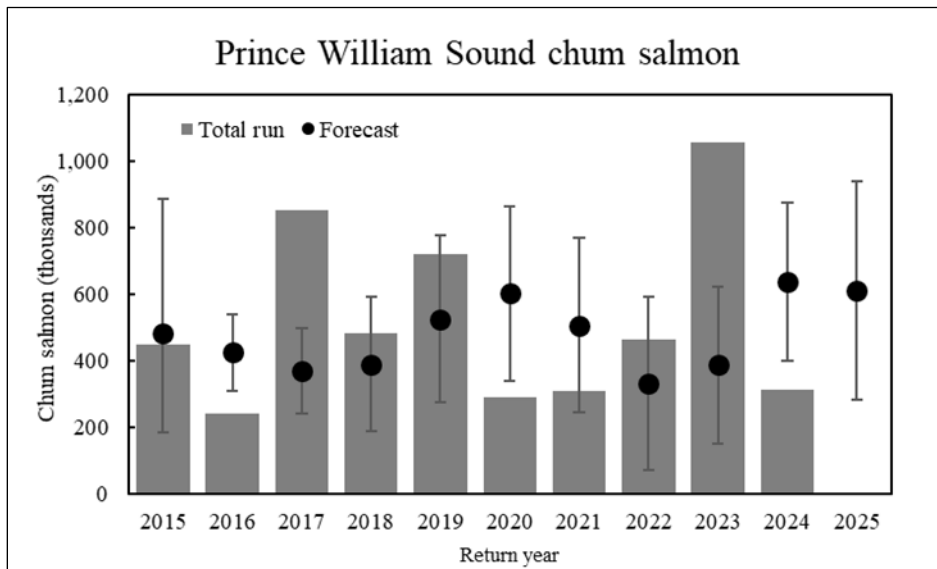


Figure 14.—Total run of Prince William Sound wild chum salmon compared to preseason total run forecasts, 2015–2024, and the 2025 forecast. Error bars represent 80% prediction intervals of forecasts.

Forecast by Jenni Morella, Area Finfish Research Biologist, ADF&G, Cordova.

UPPER COOK INLET

Sockeye Salmon

The Upper Cook Inlet sockeye salmon total run forecast of 6.93 million fish (Table 17, Figure 15) is predicted to be *excellent*. The categorical ranges of sockeye salmon total run strength were developed from the 20th, 40th, 60th, and 80th percentiles of historical runs (Table 18). Forecasts of salmon fisheries are inherently uncertain and are primarily used to gauge the general magnitude of expected runs and guide early-season management strategies.

Table 17.—Forecast of the 2025 Upper Cook Inlet sockeye salmon run, escapement, and harvest in millions of fish. Forecast range is indicated in parentheses.

Production component	Forecast estimate
Total run	6.93 (5.41–8.45)
Escapement	2.00
Available harvest	4.93

Table 18.—Categorical ranges of Upper Cook Inlet sockeye salmon runs 1986 to 2024 and the 2025 forecast in bold.

Category	Range (million)	Percentile
Poor	Less than 4.2	Less than 20th
Weak	4.2 to 5.2	20th to 40th
Average	5.2 to 5.9	40th to 60th
Strong	5.9 to 6.5	60th to 80th
Excellent	Greater than 6.5	Greater than 80th

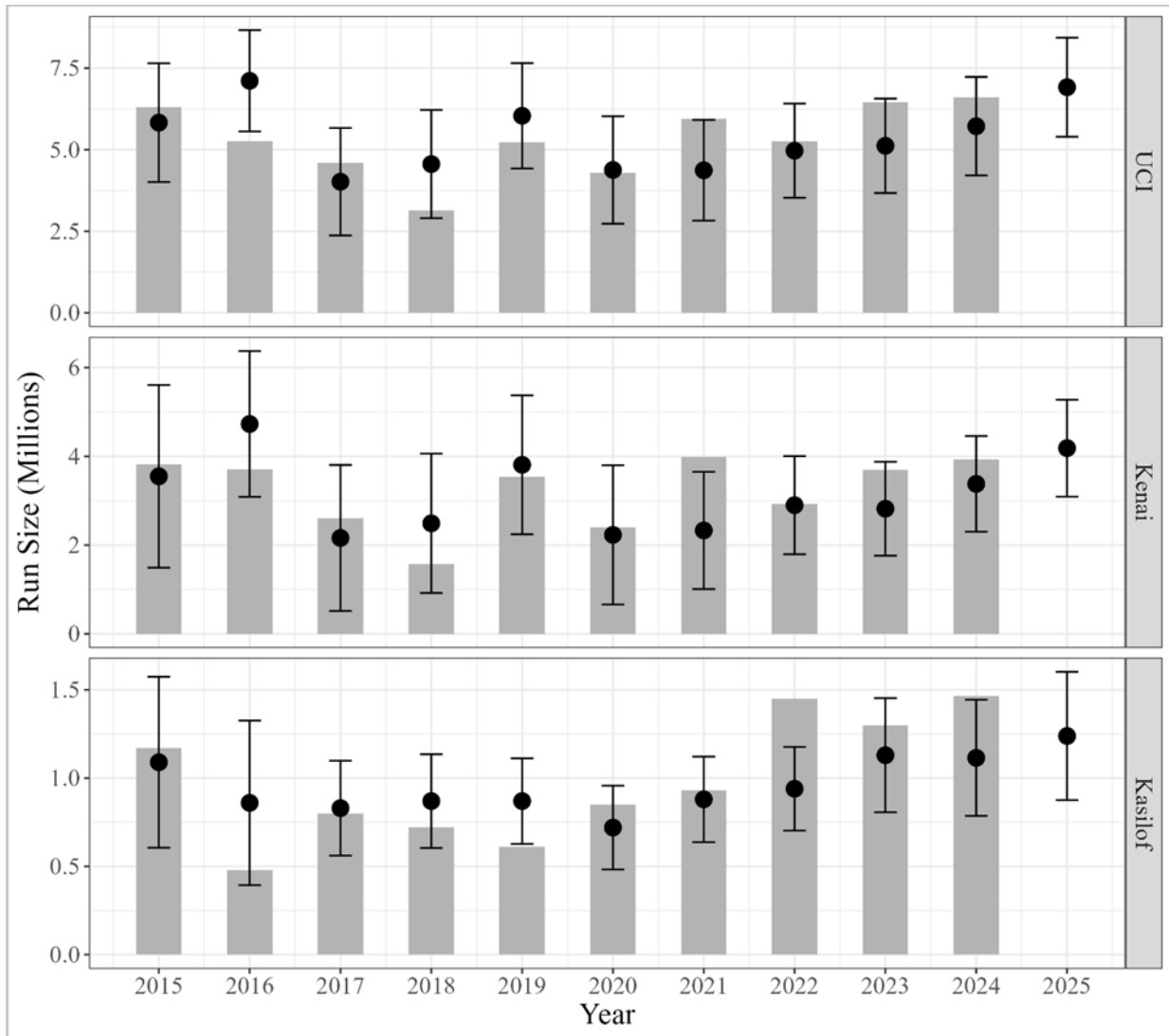


Figure 15.—Estimated total runs (grey bars) of Upper Cook Inlet (UCI; top panel), Kenai River (middle panel), and Kasilof River (bottom panel) sockeye salmon compared to total run forecasts (black points), 2015–2024 and 2025 forecast. Error bars represent 80% confidence intervals of forecasts.

Forecast Methods: The major sockeye salmon systems in Upper Cook Inlet (UCI) are the Kenai, Kasilof, and Susitna Rivers, and Fish Creek. Five model types were evaluated to forecast the total run of sockeye salmon to the Kenai and Kasilof Rivers in 2025: (1) brood-year spawners, (2) emigrating smolt, (3) fall fry, (4) sibling returns, and (5) average returns (Table 19). Forecast model performance was assessed using the mean arctangent absolute percent error (MAAPE) between the forecasts and actual runs over the past 10 years (2015–2024; Table 20). The top 3 models with the lowest MAAPE and statistically significant parameters were selected for each age class, and a weighted hybrid model approach was applied (Table 19). Model weights were assigned based on the MAAPE of each selected model, with a lower MAAPE receiving a greater weight towards the forecast estimate. Weighted forecast estimates were summed across age classes for stock specific run estimates.

For Susitna River sockeye salmon, returns of age-0.3, -1.2, -1.3, -2.2, and -2.3 fish in 2025 were forecast using mean return per spawner by age class spanning brood years 2006–2020. Spawners

for brood years 2019–2021 were estimated by subtracting the projected commercial harvest (average harvest rate of 42% from 2007–2015) from the forecast total run.

The 2025 forecast for Fish Creek sockeye salmon was estimated using the recent 5-year (2020–2024) average of total runs to the system. Total run estimates for Fish Creek sockeye salmon in 2021 and 2022 are assumed to be biased low because weir operations were stopped early, and escapement estimates for these years were not expanded to account for unmonitored passage.

The sockeye salmon forecast for unmonitored systems in UCI was estimated as 17% of the aggregate forecast for the 4 monitored stocks. Unmonitored stocks include Crescent River, Big River, McArthur River, Chilligan River, Coal Creek, Cottonwood Creek, Wasilla Creek, Eagle River, Packers Creek, and many other smaller systems. The fraction of the total run destined for unmonitored systems was calculated using genetic estimates of the stock composition of offshore test fishery harvests.

The estimated available harvestable surplus of sockeye salmon was calculated by subtracting the aggregate escapement from the total run forecast for all stocks. Aggregate escapement was estimated as the sum of the midpoints of the escapement goal ranges for each monitored sockeye salmon-producing system and the escapement into unmonitored systems.

Table 19.—Description of models used to forecast returns of sockeye salmon to the Kasilof and Kenai Rivers, 2025.

Model	Description
5-year moving average	Unweighted average of the previous 5 years' returns for the specified age class.
Exponential smoothing	An exponential function used to assign exponentially decreasing weights over time for the specified age class.
Standard Ricker	Regression between the ratio of recruits of the specified age class and spawners from the same brood year and the number of spawners from the same brood year.
Sibling	Regression between the returns of a specified age class and the most recent returns from the same brood year.
Fry	Regression between the returns of a specified age class and the aggregate total abundance of sockeye salmon fry in Kenai and Skilak Lakes from the same brood year.
Fry CFSWT	Regression between the returns of a specified age class and the aggregate total abundance of age-0 sockeye salmon fry and their average total weight in Kenai and Skilak Lakes from the same brood year.
S AR1	Autoregressive integrated moving average (ARIMA) analysis on the returns of the specified age class with a covariate for the abundance of spawners from the same brood year.
Smolt	Regression between the returns of a specified age class and the abundance of smolt from the same brood year.
Smolt AR1	Autoregressive integrated moving average (ARIMA) analysis on the returns of the specified age class with a covariate for the abundance of smolt from the same brood year.

Note: Log transformation may be needed to meet assumptions of a linear regression.

Table 20.—Kenai and Kasilof River sockeye salmon forecast estimates, 2025.

River	Age Class	Model	MAAPE	Weight	Prediction	Weighted Prediction	Subtotal	Total
Kenai	1.2	5-Year moving average	47.41	0.36	728,136	260,604	608,294	
		Exponential smoothing	51.40	0.33	666,229	219,927		
		Standard Ricker	54.39	0.31	409,512	127,762		
	1.3	Log sibling	18.45	0.44	3,931,242	1,722,860	2,581,558	
		Log R vs log fry CFSWT	27.00	0.30	1,095,289	328,057		
		5-Year moving average	30.83	0.26	2,023,528	530,641		
	2.2	Log R vs log S AR1	22.22	0.36	230,597	82,898	192,430	
		Log sibling	23.46	0.34	161,843	55,100		
		Log sibling AR1	26.62	0.30	181,407	54,432		
	2.3	Fry	41.10	0.38	1,383,688	523,161	804,324	
		Exponential smoothing	44.60	0.35	525,733	183,173		
		5-Year moving average	56.81	0.27	358,288	97,990		
Kasilof	1.2	Exponential smoothing	35.30	0.37	670,532	244,791	554,144	
		5-Year moving average	39.96	0.32	611,557	197,188		
		Log R vs log S AR1	41.23	0.31	358,933	112,165		
	1.3	Log sibling	34.78	0.41	446,377	183,052	423,994	
		Exponential smoothing	46.01	0.31	529,014	163,962		
		Log R vs log smolt	50.94	0.28	274,950	76,980		
	2.2	Log sibling	31.88	0.40	216,259	86,106	216,602	
		Log R vs log S AR1	41.30	0.31	252,996	77,755		
		Log sibling AR1	43.10	0.29	179,085	52,740		
	2.3	Log R vs log smolt AR1	55.41	0.34	53,448	18,088	44,032	
		Sibling	55.67	0.34	58,652	19,757		
		Exponential smoothing	57.75	0.32	19,052	6,187		

Forecast Results and Discussion: In 2025, a run of approximately 6.93 million sockeye salmon is forecast to return to UCI with an estimate of 4.93 million available for harvest (Table 17). Based on the absolute percentage error (APE) for the recent 10-year (2015–2024) forecast UCI runs compared with the estimated runs, there is an 80% probability that the 2025 UCI forecast range falls between 5.41 million and 8.45 million fish (Table 17). This UCI forecast is excellent compared to historical total run estimates from 1986 to present (Table 18). Forecast error for UCI has ranged from 27% below forecast to 45% above forecast over the last 10 years with a MAPE of 18% (Figure 16).

The Kenai River sockeye salmon forecast is approximately 4.19 million fish (Table 20). The 2025 Kenai River forecast is 302,400 (7.5%) fish more than the historical (1986–2024) average run of 3.88 million and 968,000 (26%) fish more than the recent 10-year (2015–2024) average run of 3.22 million (Figure 16). Forecast error for the Kenai River has ranged from 42% below forecast to 59% above forecast over the last 10 years with a MAPE of 21% (Figure 16).

The Kasilof River sockeye salmon run forecast is approximately 1.24 million fish (Table 20). Over the past 10 years, the Kasilof River forecast has experienced a MAPE of 25%, with errors ranging from 35% under forecast to 79% over forecast. The 2025 forecast is 311,000 fish (28%) greater than the historical (1986–2024) average run of 930,000 fish and 261,000 fish (23.5%) greater than the recent 10-year (2015–2024) average run.

Approximately 404,800 and 105,000 sockeye salmon are forecast to return to the Susitna River and Fish Creek, respectively, in 2025. The 2025 Susitna River sockeye salmon forecast is approximately 22,600 fish (6%) below the historical (2002–2024) average run of 430,000 fish but approximately 29,500 fish (7.5%) greater than the recent 10-year (2015–2024) average run of 380,000 fish. The 2025 Fish Creek sockeye salmon forecast is approximately 2,600 fish greater than the recent 10-year average run size (2.5%) of 102,400 fish.

Forecast by Kyle Gatt, ADF&G Area Research Biologist, Soldotna, and Jack Erickson, ADF&G, Regional Research Coordinator, Anchorage.

LOWER COOK INLET

Pink Salmon

The Lower Cook Inlet (LCI) wild pink salmon commercial harvest in 2025 is predicted to be *strong*, with a point estimate of 1,050,000 fish and a range of 305,000–3,620,000 fish (80% confidence interval). The categorical ranges of wild pink salmon harvest were formulated from the 20th, 40th, 60th, and 80th percentiles of historical harvests over the 65-year period 1960–2024 (Table 21). Since adopting new forecast methods in 2018, the point forecasts have, on average, overforecast the harvest by 18%. However, forecasts have ranged from 88% below the observed harvest in 2020 to >6,000% above the observed harvest in 2024 (Figure 16), highlighting the uncertainties that surround all salmon forecasts (Adkison and Peterman 2000) and pink salmon in particular (Adkison 2002; Haeseker et al. 2005).

Table 21.—Categorical ranges of pink salmon commercial harvest (wild run) and this year’s forecast in bold.

Category	Range (thousands)	Percentile
Poor	Less than 219	Less than 20th
Weak	219 to 367	20th to 40th
Average	367 to 964	40th to 60th
Strong	964 to 1,661	60th to 80th
Excellent	Greater than 1,661	Greater than 80th

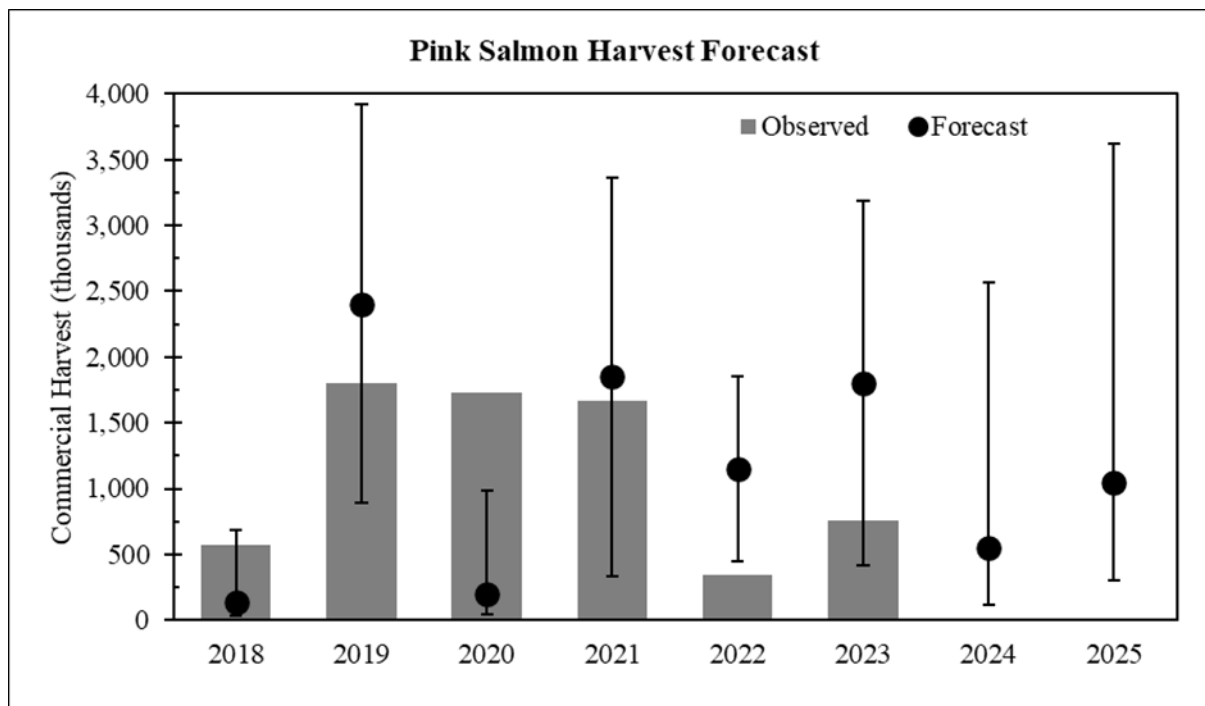


Figure 16.—Annual observed commercial harvest of wild pink salmon in LCI compared to preseason harvest forecasts, 2018–2025. Error bars represent 80% confidence intervals of forecasts. Note: 2024 harvest was 8,525 fish.

Forecast Methods: A naïve forecast based on historical odd- and even-year pink salmon harvests (1960–2024) was used to predict the 2025 wild pink salmon commercial harvest in LCI. Several trend forecast models were evaluated to determine which best fit the data, including exponential smoothing and 2-, 3-, and 5-year running averages of historical harvests. Exponential smoothing and running average models using log-transformed and non-logged data inputs produced point forecasts ranging from 80,000 to 2,086,000. The exponential smoothing model using odd-year log-transformed data inputs outperformed the other models this year based on comparison of several performance metrics (e.g., bias, MPE, mean square error [MSE], and MAPE). The 80% confidence intervals were calculated from the mean square error [MSE] of the retrospective forecast predictions.

Forecast Discussion: The 2025 harvest forecast of 1,050,000 pink salmon has a forecast range of 305,000–3,620,000 fish. If realized, a harvest of 1,050,000 pink salmon would be approximately 50% lower than the recent 5-year average harvest of 2,085,000 fish for odd-year runs between 2015 and 2023.

Prior to 2018, the odd-year LCI pink salmon forecast consisted of a wild stock total run forecast based on a logarithmic regression of total run and escapement from over 50 years of observations on 10 index streams, and a hatchery production forecast provided by Cook Inlet Aquaculture Association (CIAA). However, results from otolith sampling in LCI during 2014–2017 suggested high proportions of hatchery-marked fish in harvest (Otis and Hollowell 2023) and escapement (Otis et al. 2018) samples may confound the viability of spawner-recruit based run forecasts. Consequently, the decision was made to produce a harvest forecast in lieu of the total run forecast.

Because pink salmon exhibit a 2-year life cycle, comparisons of run size are typically stratified by odd and even years to account for dominance of one line over the other. In LCI overall, dominance of one line is typically short lived, lasting 2–7 generations before the opposing line becomes dominant. Odd/even year dominance is not synchronous across individual stocks and line dominance can persist up to 9 generations (18 years) in some stocks. Overall, odd- and even-year runs across all stocks and years are only moderately dissimilar (odd-year runs ~21% higher). However, areawide total runs exhibit 4–14 year periods during which production from one line is 19–58% higher than the other, on average. So, despite the relative parity between odd- and even-year pink salmon runs in LCI over broad time scales, we continue to stratify run size comparisons by odd and even years to account for these short-term dominance cycles. The odd-year brood line has been dominant in LCI since 2011 (7 generations).

In 2023, the parent year for the upcoming 2025 return, LCI experienced strong escapements to most pink salmon index streams. Only 2 of 18 stocks (11%) with SEGs failed to achieve the low end of their respective escapement goal ranges. Twelve stocks (67%) exceeded their SEG range, including several major producers in the Outer and Kamishak districts, some by a substantial amount (e.g., Dogfish, Windy Left, Island, Sunday, and Brown’s Peak Creeks). It is difficult to predict shifts in dominance between odd- and even-year lines of pink salmon before they occur. After a decade of odd-year dominance, an unexpectedly large 2020 run suggested a shift to even-year dominance was forthcoming. However, the 2021 and 2023 total runs were both greater than the preceding even-year run and the 2024 total run was extremely poor, implying odd-year dominance is persisting. If so, the 2025 run of wild pink salmon is likely to produce a harvest close to the point forecast of 1,050,000 fish.

To produce this wild stock commercial harvest forecast, we excluded hatchery cost-recovery harvests (HCR) from our analysis, but it should be noted that hatchery-origin pink and sockeye salmon (discussed in the next section) occur in Southern District harvests (Otis and Hollowell 2023), so a component of this wild stock harvest forecast derives from hatchery production. Additional commercial and HCR harvests of pink and sockeye salmon may be expected from hatchery runs in and around hatchery special harvest areas. LCI hatchery forecasts are available from Cook Inlet Aquaculture Association².

Other Salmon Species

Forecast Methods: Naïve forecasts based on historical commercial harvests (1960–2024) were used to predict the 2025 harvest of other, less abundant salmon species in LCI. Several trend forecast models were evaluated to determine which best fit the data, including exponential smoothing and 2-, 3-, and 5-year running averages of past commercial harvests. Log-transformed and non-logged historical harvest datasets were evaluated with each model and the final forecast for each species was selected based on comparison of several performance metrics (e.g., bias, MSE, root mean square error [RMSE], MPE, MAPE, and mean absolute scaled error [MASE]).

Forecast Discussion: Prior to 2018, the recent 5-year average commercial harvest was used to forecast the LCI sockeye, chum, coho, and Chinook salmon harvest for the coming year. Beginning in 2018, for consistency, we transitioned to the same trend forecast methods we used for LCI pink salmon. Context for the relative strength of the current harvest forecast and how it compares to the recent 10-year average harvest for each species is provided in Table 22. Context for the relative performance of sockeye and chum salmon forecasts since 2018 is provided in Figures 17 and 18, respectively.

Table 22.—Preliminary forecast of the 2025 Lower Cook Inlet (LCI) commercial harvest (wild run) of other, less abundant salmon species.

Salmon Species	Model	Forecast (point)	Forecast (range)	% above/below 10-yr average	Forecast category
Sockeye	5-year running average	156,800	59,700–253,800	8% higher	Strong
Chum	Exponential Smoothing	15,800	3,000–84,400	75% lower	Poor
Coho	2-year running average	250	40–1,640	95% lower	Poor
Chinook	Exponential smoothing	160	60–410	69% lower	Weak

² Cook Inlet Aquaculture Association: <http://www.ciaa.net.org>

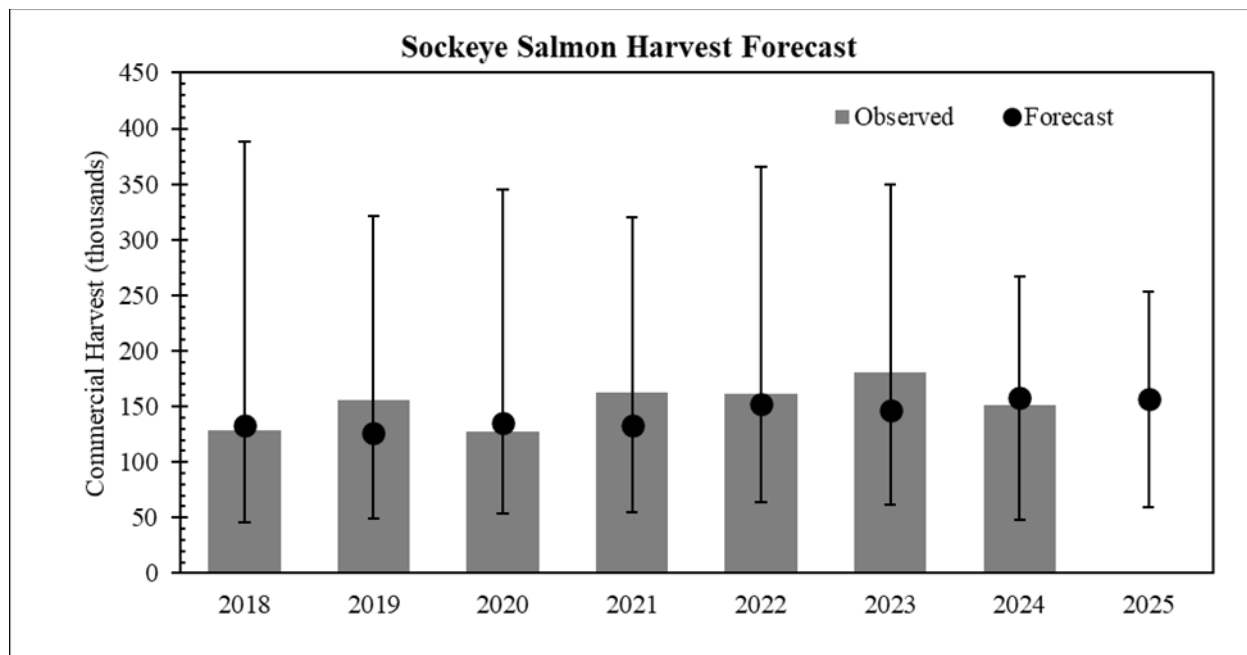


Figure 17.—Annual observed commercial harvest of sockeye salmon in LCI compared to preseason harvest forecasts, 2018–2025. Error bars represent 80% confidence intervals of forecasts.

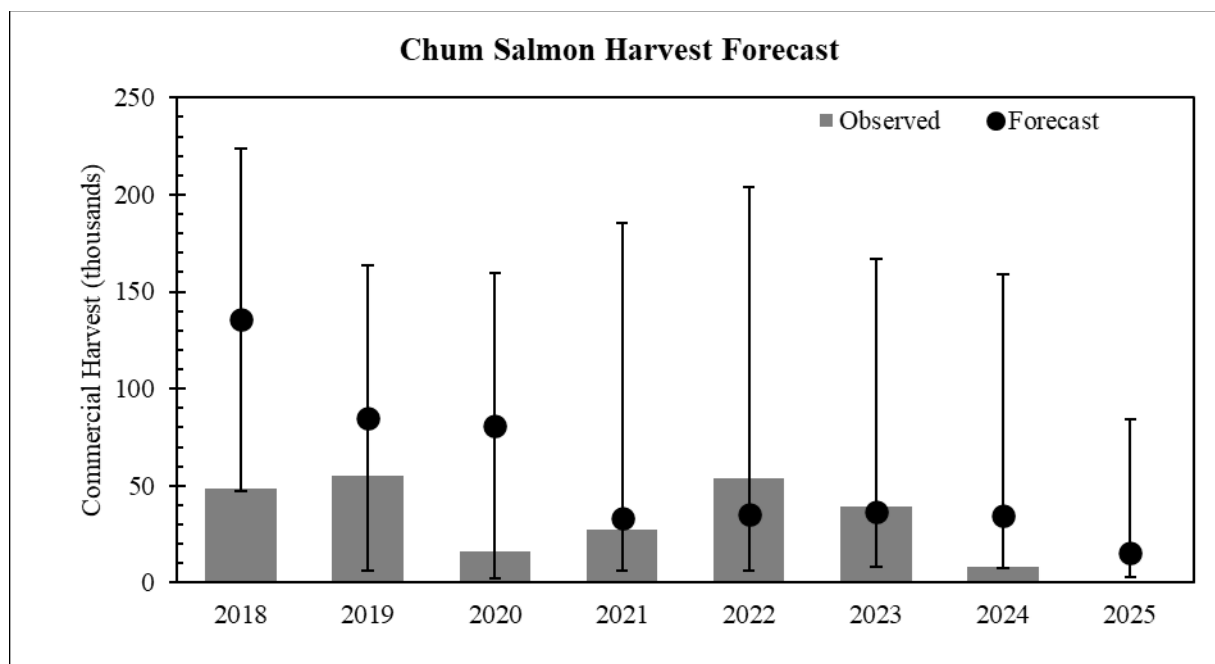


Figure 18.—Annual observed commercial harvest of chum salmon in LCI compared to preseason harvest forecasts, 2018–2025. Error bars represent 80% confidence intervals of forecasts.

Forecast prepared by Edward O. Otis, Area Finfish Research Biologist, Homer; and Glenn J. Hollowell, Area Finfish Management Biologist, Homer.

BRISTOL BAY AREA

Sockeye Salmon

The sockeye salmon total run forecast for Bristol Bay in 2025 is predicted to be *strong* with a point estimate of 51.38 million fish and a range of 37.08 to 65.67 million fish (80% confidence interval; Table 23). Categorical ranges of sockeye salmon total run strength were formulated from percentiles of total runs from 1961 to 2023 (Table 24). Since 2005, our preseason forecasts have underforecast the actual run by 15% on average, ranging from 36% below in 2014 to 21% above in 2011 (Figure 19).

Table 23.—Preliminary forecast of the 2025 Bristol Bay area sockeye salmon run.

Total production:	Forecast (millions)	Forecast range (millions)
Total run	51.38	37.08–65.67
Escapement	14.98	
Total harvestable surplus	36.40	
Bristol Bay harvestable surplus	34.84	
South Peninsula	1.55	
Inshore run	49.83	

Table 24.—Categorical ranges of sockeye salmon total run and this year’s forecast in bold.

Category	Range (millions)	Percentile
Poor	Less than 20	Less than 20th
Weak	20 to 29	20th to 40th
Average	29 to 42	40th to 60th
Strong	42 to 53	60th to 80th
Excellent	Greater than 53	Greater than 80th

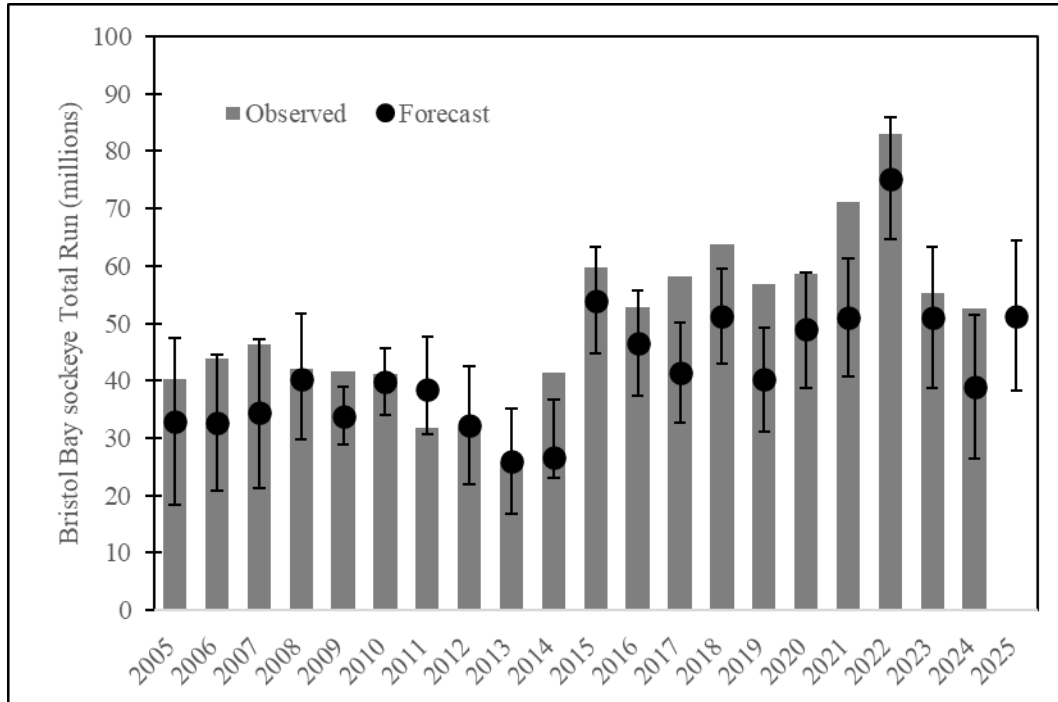


Figure 19.—Annual observed total run of sockeye salmon in Bristol Bay compared to preseason total run forecasts, 2005–2025. Error bars represent 80% confidence intervals of forecasts.

Forecast Methods: The 2025 Bristol Bay sockeye salmon forecast is the sum of individual predictions from 9 river systems (Kvichak, Alagnak, Naknek, Egegik, Ugashik, Wood, Igushik, Nushagak, and Togiak Rivers) and 4 age classes (ages 1.2, 1.3, 2.2, and 2.3). Adult escapement and return data from brood years 1972–2020 were used in the analyses for most rivers.

Forecasts for each age class returning to a river system were derived from models based on the relationship between adult returns of that age class and either total returns or sibling returns from the same brood years. The average return over the last 5 years was also considered as a forecast model, and in certain cases, competing models were averaged in a weighted hybrid model approach. In general, models with statistically significant parameters and/or the best past performance metrics were chosen. Performance was evaluated using mean absolute deviation, MAPE, MAAPE, and MPE between forecast and observed returns measured across the most recent 3- and 5-year time frames.

Where practical, the department will manage escapements proportional to the run size and relative to the historical record (5 AAC 06.355(d)(1)). In the Nushagak District, escapement goals for the

Wood and Nushagak Rivers will be modified by the *Nushagak District King Salmon Stock of Concern Management Plan* (5 AAC 06.391). Based on this regulation, the upper ends of the optimal escapement goal ranges (OEGs) in 2025 will be 2.95 and 2.49 million sockeye salmon for the Wood and Nushagak Rivers, respectively, and escapements are projected as the 75th percentile of the OEGs. For the other systems in Bristol Bay, escapements are projected as the 75th percentile of the escapement goal range if the forecast is in the upper range of historical run sizes (Ugashik, Igushik, and Togiak Rivers in 2025; Table 24) and as the 25th percentile of the escapement goal range if the forecast is in the lower range of historical run sizes (Kvichak, Naknek, Egegik Rivers in 2025; Table 24). Because it is passively managed, the Alagnak River exploitation rate is assumed to be the same as that of the Kvichak River and therefore, the escapement is projected to be the total run forecast minus expected harvestable surplus.

Over the past 5 years, an average of 3% of the Bristol Bay return is thought to have been harvested in the South Peninsula fisheries in June. Preseason harvestable surplus projections are provided to aid the industry in planning for the upcoming season.

Forecast Results: A total of 51.38 million sockeye salmon (with a range of 37.08 to 65.67 million) are expected to return to Bristol Bay in 2025 (Table 24). This is 16% smaller than the most recent 10-year average of 61.23 million fish and 38% greater than the long-term average of 37.07 million fish (1963–2024). All systems are expected to meet their spawning escapement goals in 2025. The forecast range is the upper and lower values of the 80% confidence interval for the total run forecast. The confidence bounds were calculated from the difference between actual runs and run forecasts from 2005 through 2024.

A run of 51.38 million sockeye salmon would allow for a potential harvestable surplus of 36.40 million fish: 34.84 million fish in Bristol Bay and 1.55 million fish in the South Peninsula June fishery. A Bristol Bay inshore harvest of this size is 15% less than the most recent 10-year average harvest of 40.91 million (ranging from 31.61 to 60.23 million), and 50% greater than the long-term average harvest of 23.27 million fish (1963–2024).

Age-specific forecasts for the 2025 run consist of 14.43 million age-1.2 fish (28% of the total run), 4.69 million age-2.2 fish (9% of the total run), 30.06 million age-1.3 fish (59% of the total run), and 2.18 million age-2.3 fish (4% of the total run; Table 24).

Table 25.—Forecast of total run, escapement, and harvest of major age classes of sockeye salmon returning to Bristol Bay (BB) river systems in 2025.

		Millions of sockeye salmon								
		Forecasted production by age class				Forecasted		South		
DISTRICT / River		1.2	2.2	1.3	2.3	Total	Escapement	Surplus	Peninsula ^a	BB Inshore
NAKNEK-KVICHAK										
	Kvichak	2.12	0.78	4.14	0.34	7.38	4.00	3.16	0.22	7.16
	Alagnak	1.68	0.23	2.88	0.18	4.97	2.70	2.13	0.15	4.82
	Naknek	0.78	0.31	2.19	0.37	3.66	1.10	2.45	0.11	3.55
	Total	4.58	1.32	9.22	0.89	16.01	7.80	7.73	0.48	15.53
EGEGIK										
		1.00	2.42	2.20	0.83	6.46	1.10	5.16	0.20	6.26
UGASHIK										
		1.62	0.52	4.54	0.23	6.90	1.18	5.52	0.21	6.70
NUSHAGAK										
	Wood	3.95	0.28	3.24	0.16	7.63	2.38	5.01	0.23	7.40
	Igushik	0.73	0.02	2.02	0.01	2.79	0.34	2.37	0.08	2.70
	Nushagak	2.23	0.13	8.15	0.05	10.58 ^b	1.96	8.31	0.32	10.26
	Total	6.91	0.42	13.42	0.22	21.00	4.68	15.68	0.63	20.36
TOGIAC										
		0.31	0.01	0.68	0.01	1.01 ^c	0.23	0.75	0.03	0.98
BRISTOL BAY										
		14.43	4.69	30.06	2.18	51.38	14.98	34.84	1.55	49.83
		28%	9%	59%	4%	100%				

Note: This table is a summary. Slight differences may appear due to rounding.

^a Projected harvest is based on the current 5-year running average exploitation rate of 3.0%.

^b Nushagak River forecast total includes approximately 25,000 age-1.4 fish.

^c Forecasts for Kulukak, Kanik, Osviak, and Matogak River systems are not included. These systems contribute approximately 50,000 sockeye salmon to Togiak District harvest each year.

Forecast Discussion: Forecasting future salmon returns is inherently difficult and uncertain. The department has used similar methods since 2001 to produce Bristol Bay sockeye salmon forecasts, which have performed well when applied to Bristol Bay as a whole. In the last 20 years, our forecasts have on average underforecast the run by 15%, ranging from 36% below the actual run in 2014 to 21% above the actual run in 2011. Forecasted harvestable surplus has had a mean absolute percentage error of 17% since 2005.

Individual river forecasts have greater uncertainty compared to the baywide forecast. Since 2005, we have on average underforecast returns to the Alagnak (-15%), Togiak (-6%), Kvichak (-37%), Wood (-22%), Nushagak (-22%), Igushik (-7%), Ugashik (-11%), and Naknek (-4%) Rivers, and overforecast returns to Egegik River (7%). Overforecasting returns to some rivers while underforecasting returns to other rivers means that the overall Bristol Bay forecast is often more accurate than the forecast for any individual river.

Forecast by Stacy Vega, Bristol Bay Research Biologist ADF&G, Anchorage.

ARCTIC-YUKON-KUSKOKWIM REGION

ALL SALMON

The department prepares formal run forecasts annually for the following stocks in the AYK Region: Kuskokwim River Chinook salmon, Yukon River Chinook salmon, Canadian-origin Yukon River Chinook salmon, Yukon River summer chum salmon, and Yukon River fall chum salmon. A summary of forecast methods and 2025 run size projections are provided.

The Kuskokwim River Chinook salmon run forecast is based on the estimated run size from the prior year (midpoint), and uncertainty is represented as the recent 7-year average percent error between forecasted and actual run estimates. The preliminary 2025 Kuskokwim River Chinook salmon forecast is for a range of 136,000–217,000 fish. The drainagewide Chinook salmon escapement goal is 65,000–120,000 fish. If the run returns as projected, the drainagewide escapement goal is expected to be achieved and subsistence harvest opportunity provided. There is currently no commercial buyer in the Kuskokwim Area, and commercial harvest of Chinook salmon is not expected during the 2025 season.

Separate forecasts are prepared for Canadian-origin Yukon River Chinook salmon and total run. The Canadian-origin run forecast is produced by a U.S./Canada Joint Technical Committee and presented to the Yukon River Panel. The 2025 forecast is based on projections of brood year returns, sibling relationships, and juvenile abundance estimates from marine trawl surveys conducted in the northeastern Bering Sea. A Bayesian integrated forecast was produced, such that the model components that best fit the observed historical run sizes are given more weight. The final 2025 Canadian-origin Chinook salmon forecast is for a run size of 24,000–37,000, which is less than the minimum border passage objective of 71,000 as agreed to by ADF&G and Canada Department of Fisheries and Oceans. The Canadian-origin forecast is used to develop the drainagewide Chinook salmon outlook (all stocks), based on the assumption that the Canadian-origin run represents approximately 42% of the total run. The preliminary drainagewide forecast for Yukon River Chinook salmon is 58,000–88,000. Yukon River Chinook salmon is classified as a stock of yield concern under the *Policy for the Management of Sustainable Salmon Fisheries*. Harvest of Yukon River Chinook salmon is not expected during the 2025 season in any fishery.

The 2025 Yukon River summer chum salmon run forecast is developed by projecting the age-5 return based on sibling relationships and expanding the age-5 return estimate to total run based on the recent 10-year average age at maturity. The drainagewide summer chum salmon forecast is for a below-average run size of 550,000–1,800,000 fish (80% confidence interval). The drainagewide BEG for this stock is 500,000–1,200,000 fish. If the 2025 summer chum salmon run is near the lower end of the forecast range, there would be no surplus available above escapement and subsistence harvest needs for a directed summer chum salmon commercial fishery. However, if the run size exceeds 750,000 fish, commercial fishery opportunity would be warranted as per the *Yukon River Summer Chum Salmon Management Plan*. Regardless of run size, fishing opportunities for summer chum salmon in 2025 will probably be limited by the low predicted Chinook salmon run size and be restricted to selective gear types (dip nets, beach seines, human-operated fish wheels) with release of Chinook salmon required.

The 2025 Yukon River fall chum salmon run forecast is based on drainagewide projections of brood year returns and age class proportions (age-3 through age-6) using a spawner-recruit analysis. The drainagewide fall chum salmon forecast point estimate is 217,000 fish, which is

below the average run size of 950,000 fish. The forecast range is 114,000–322,000 fall chum salmon, based on the upper and lower values of the 80% confidence bounds of the total run forecast. Confidence bounds are calculated using deviation of the run projection point estimates and the observed returns from 1987–2024. The drainagewide SEG is 300,000–600,000 fall chum salmon. The upper end of the forecast is slightly above the lower end of the minimum drainagewide escapement goal. However, the forecast run based on spanner-recruit has been overestimated for the last 8 years, with the last 5 observed runs all ranking the lowest on record. Surplus to support directed fishing opportunity in 2025 is not anticipated. However, if the run is larger than forecast, thus exceeding the lower end of the escapement goal, there is potential for some limited subsistence fisheries. Given the forecast run size and associated uncertainties, commercial fisheries would likely remain closed.

The department does not produce formal run forecasts for other salmon stocks returning to the Kuskokwim Area or Yukon Area, or any salmon stocks returning to the Norton Sound Area or Kotzebue Management Area. Consequently, commercial harvest outlooks for these stocks are typically based upon available parent-year spawning escapement indicators, age composition information, recent year trends, and the level of commercial harvest that can be expected given the fishery management plans in place and commercial processing capacity. Currently, southeastern Norton Sound Chinook salmon stocks are classified as *stocks of yield concern*. In general, low Chinook and chum salmon abundance is anticipated throughout the AYK Region in 2025, and Yukon River coho salmon are also expected to be poor. Low run sizes for multiple species throughout the AYK Region combined with (currently) no commercial processor registered in the Kuskokwim Management Area is expected to result in well below average commercial harvest (Table 26).

Forecasts by Sean Larson, Kuskokwim Area Research Biologist, ADF&G, Anchorage/Bethel; Bonnie Borba, Yukon Area Research Biologist, ADF&G, Fairbanks; and Fred West, Yukon Area Research Biologist, ADF&G, Anchorage.

Table 26.—The 2025 Arctic-Yukon-Kuskokwim area all-salmon commercial harvest outlook by management area, in thousands of fish.

Management area	Salmon species					
	Chinook	Sockeye	Coho	Pink	Summer chum	Fall chum
Kuskokwim River	0	0	0	0	0	0
Kuskokwim Bay	0	0	0	0	0	0
Kuskokwim Area total ^a	0	0	0	0	0	0
Yukon	0	0	0	0	0–1,000	0
Norton Sound	0	0	10–25	5–25	5–15	0
Kotzebue Sound	0	0	0	0	0	50–150

^a There is not currently a confirmed commercial buyer in the Kuskokwim area for 2025.

WESTWARD REGION

KODIAK MANAGEMENT AREA

Pink Salmon

The 2025 Kodiak Management Area (KMA) predicted pink salmon harvest is expected to be in the *excellent* category with a point estimate of 31.8 million fish, combining the wild stock and Kitoi Bay Hatchery harvest estimates (Table 27 and 28). Harvest categories were delimited from the 20th, 40th, 60th, and 80th percentiles of historical commercial harvest in the KMA from 1988 to 2024.

Forecast Methods: The KMA wild stock pink salmon harvest forecast is derived from a total run forecast minus the estimated KMA escapement (5.0 million). The total run estimates were derived from a combination of weir counts, aerial survey index, and harvest estimates.

For the 2025 KMA wild stock pink salmon forecast, a generalized Ricker model (Quinn and Deriso 1999) was fit to the odd-year KMA returns from 1987 to 2023 utilizing aerial survey indexed escapement and Karluk and Ayakulik Rivers and Dog Salmon Creek pink salmon escapement counts for the spawner index. Three additional terms were included in this generalized Ricker model: 1) Western Gulf of Alaska May SST, 2) previous year (lag-1) Westward Region pink salmon return, 3) and an environmental composite created from a series of forecast indices affecting pink salmon returns. The environmental variables used to create the composite included monthly mean air temperature, total precipitation, and peak precipitation total run correlation anomalies from August to June of the predominantly freshwater life history.

In constructing and evaluating the regression model standard regression diagnostic procedures were used. Forecast range was estimated using the 80% confidence intervals of the absolute percent error of the composite model hindcast estimates.

Table 27.—The 2025 Kodiak Management Area (KMA) predicted pink salmon harvest categories.

KMA harvest category	Range (millions)	Percentile
Poor	Less than 8.2	Less than 20th
Weak	8.2 to 14.1	21st to 40th
Average	14.1 to 20.7	41st to 60th
Strong	20.7 to 27.6	61st to 80th
Excellent	Greater than 27.6	81st to 100th

Table 28.—Preliminary forecast of the 2025 Kodiak Management Area (KMA) pink salmon run.

Total production	Forecast estimate (millions)	Forecast range (millions)
KMA Wild stock total run	26.0	15.9–36.1
KMA Escapement goal ^a	5.0	—
KMA Wild stock harvest	21.0	10.9–31.1
Kitoi Bay Hatchery harvest ^b	10.8	8.0–13.6
Total KMA Pink salmon harvest	31.8	18.9–44.7

Note: Column numbers may not total or correspond exactly with numbers in text due to rounding.

^a The 2025 estimated escapement is within the range of the odd-year aggregate escapement goals for the Kodiak Archipelago (3.0–5.0 million) and the Mainland District (0.25–1.0 million). Escapement ranges are not figured in the harvest range calculations for Kodiak pink salmon.

^b This figure is the total expected return (11.2 million) minus the broodstock collection goal of 0.43 million fish; the Kitoi Bay Hatchery cost recovery goal for 2025 has been set to 1.45 million fish.

The 2025 Kitoi Bay Hatchery pink salmon forecast was prepared by evaluating pink salmon survivals from even brood years 1995 through 2021 when releases from the facility were in excess of 100 million fry (excludes 2016). These brood years are particularly important to the forecasting model because all pink fry were released on the same day in order to saturate the release area with fry (predator satiation). This release strategy has proven to significantly improve fry to adult survival.

The pink return to Kitoi Bay Hatchery is characterized by a dominant odd-year return, with exceptional marine survival observed every fourth year since the first releases in 1977 (excluding 1997). The primary forecasting consideration for 2025 is the 4-year cyclical return, which represents the stronger of the 2 odd-year cycles. The midpoint estimate of 11.2 million reflects a marine survival of 5.6% for the 199 million fry released (.65 g). The average survival was calculated using the last 4-year cyclical returns (parent class 2007, 2011, 2015, and 2019). The range was calculated as (+/-) 25% of the predicted marine survival.

Forecast Discussion: The 2025 KMA wild stock pink salmon total run (26.0 million) is predicted to be an excellent return (Figure 20). With strong escapement indicators, above-average environmental conditions, average SST, and a very weak previous year return (2024) indicate a strong return in 2025 is likely to occur.

The 2025 Kitoi Bay Hatchery pink salmon production is expected to be 11.2 million fish (Figure 21). The brood stock collection goal is 425,000 fish, resulting in a total hatchery harvest projection of about 10.8 million fish. The Kitoi Bay Hatchery cost recovery goal for 2025 has been set to 1.45 million fish.

Confidence in the 2025 forecast estimate is only fair considering the unpredictable nature of pink salmon returns. Despite the strength of the forecast model, the authors recognize that return corollaries are often fleeting due to the dynamic nature of the Gulf of Alaska. This forecast level will allow for an initial weekly fishing period length of 105 hours (4.5 days) for most of the KMA during the initial general pink salmon fisheries (beginning July 6, 2025). Harvests during the initial period provide important data to assess run strength, and subsequent fishing periods will be adjusted by section and district to ensure escapement goals will be met.

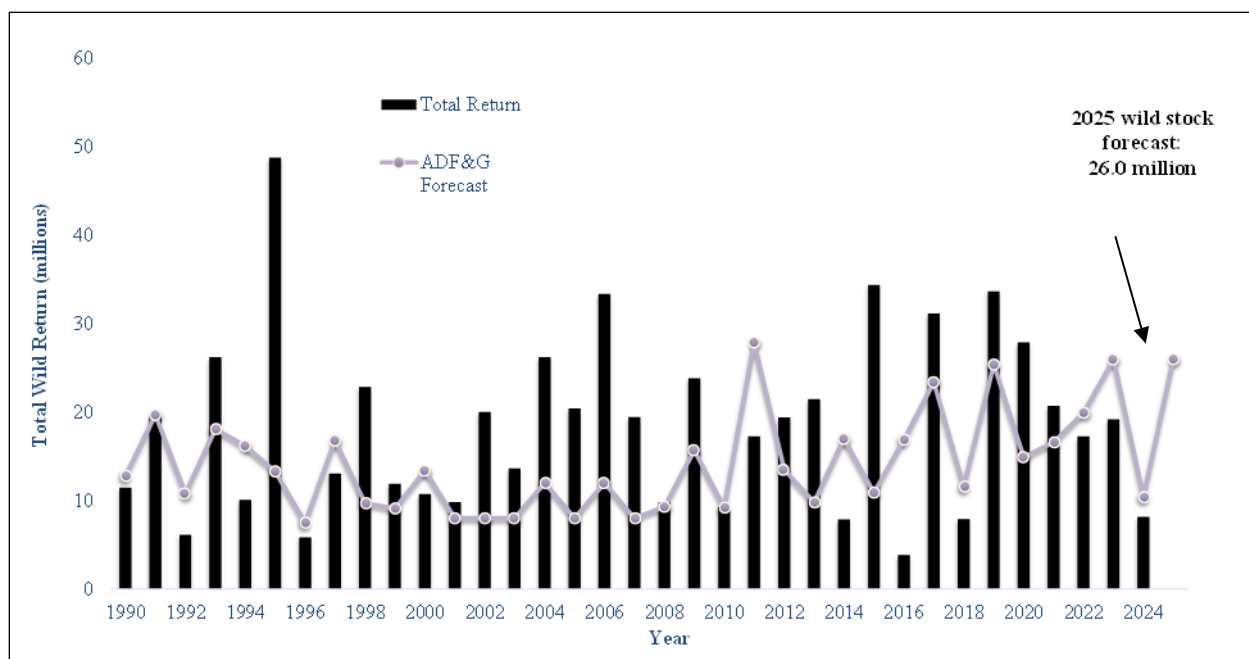


Figure 20.—Kodiak pink salmon wild stock total return compared to ADF&G forecasts, 1990 to 2024, and 2025 forecast.

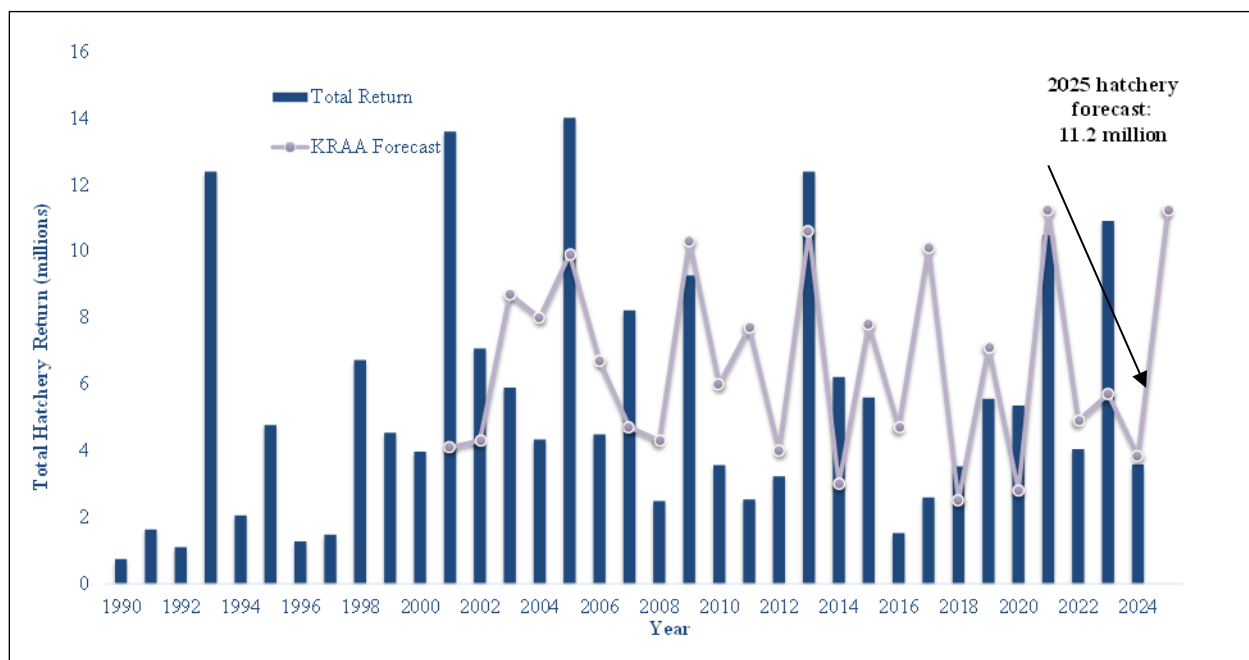


Figure 21.—Kitoi Bay Hatchery pink salmon total return compared to Kodiak Regional Aquaculture Association (KRAA) forecasts, 1990 to 2024, and 2025 forecast.

Forecast by M. Birch Foster, Finfish Research Biologist, Westward Region; and Lauren Deal, Kitoi Bay Hatchery Manager, Kodiak Regional Aquaculture Association.

Ayakulik River

Sockeye Salmon

Forecast Methods: The 2025 Ayakulik River sockeye salmon forecast (Table 29) was prepared primarily by investigating simple linear regression models utilizing recent outmigration year age-class relationships. In constructing and evaluating each of the regression models, standard regression diagnostic procedures were used. Prediction intervals (80%) for the regression estimates were calculated using the variances of the regression models. The age classes that could not be estimated with one of these models were estimated using pooled medians, and the 10th and 90th percentiles of the data were used to calculate the prediction interval of the median estimates.

Table 29.—Preliminary forecast of the 2025 Ayakulik River sockeye salmon run.

Total production	Forecast estimate (thousands)	Forecast range (thousands)
Total run estimate	711	353–1,109
Escapement goal ^a	300	200–400
Harvest estimate	411	—

^a The escapement estimate is the sum of the upper end of escapement goals for the early (140,000–280,000) and late run (60,000–120,000).

The ocean-age-2 sockeye salmon were predicted from prior year ocean-age-1 returns (1992–present). Ocean-age-3 sockeye salmon were predicted from prior year ocean-age-2 returns (1992–present). Ocean-age-1 and all other age classes were predicted by the 20-year median return. Regression and median estimates were summed to estimate the total Ayakulik sockeye salmon run for 2025. The range was estimated as the sum of the individual 80% prediction intervals.

Forecast Discussion: The 2025 Ayakulik forecast of 711,000 sockeye salmon is about 199,000 more fish than the actual 2024 run estimate of approximately 512,000, and about 68,000 fish more than the most recent 10-year average of approximately 643,000 fish. The 2025 run is estimated to be composed of approximately 63% ocean-age-2 fish. The Ayakulik sockeye runs in 2021 and 2022 were the largest since 1998 and we are anticipating another strong run in 2025, but confidence is only fair. The projected harvest of 411,000 fish is based on the achievement of the mid-point of the combined escapement goal ranges (300,000 fish). Ayakulik is managed based on both early- and late-run (post July 15) components. Based on brood-year escapement proportions from what will be the major contributing brood years (2019–2022), approximately 68% of the total run will occur in the early portion of the run.

Forecast by M. Birch Foster, Finfish Research Biologist, Westward Region.

Karluk River

Sockeye Salmon

Forecast Methods: The 2025 Karluk River sockeye salmon forecast (Table 30) was prepared primarily by investigating escapement, sibling age class relationships, moving average, and exponential smoothing methods.

Table 30.—Preliminary forecast of the 2025 Karluk River sockeye salmon run.

Total production		Forecast estimate (thousands)	Forecast range (thousands)
Early Run	Total run estimate	95	0–311
	Escapement goal ^a	95	150–250
	Harvest estimate	0	–
Late Run	Total run estimate	674	195–1,189
	Escapement goal ^a	325	200–450
	Harvest estimate	349	–
Total Karluk River System	Total run estimate	769	195–1,500
	Escapement goal ^a	420	350–700
	Harvest estimate	349	–

Note: Column numbers may not total or correspond exactly with numbers in text due to rounding.

^a The escapement estimates are based on the midpoints of the early-run and late-run escapement goals and summed for the total run, less if the total run is estimated to be below the lower end.

Karluk River sockeye salmon production demonstrates strong positive autocorrelation (i.e., good years followed by good years and bad years followed by bad years). As a result, recent year trends have been a good prognosticator of the near future at Karluk River. Utilizing sibling age-class relationships and regression models has been the preferred method since 1994. An exponential smoothing model was also fit to recent year early- and late-run total returns. Parameter estimates (early run $\alpha = 0.82$; and late run $\alpha = 0.46$) placed heavy weight on recent returns and outperformed 2-, 3-, and 5-year moving averages. However, the sibling age class forecasts for the early, late, and total runs outperformed the exponential smoothing model hindcast estimates (1994 to present) in terms of MSE. Therefore, the sibling age class methods were used for the 2025 forecast.

For the early run, ocean-age-2 sockeye salmon returns were predicted based on the abundance of the prior-year return of ocean-age-1 and ocean-age-3 sockeye salmon returns were predicted based on the abundance of the prior-year return of ocean-age-2 sockeye salmon (2005–present). The ocean-age-1 and -4 return predictions were calculated using their pooled median contribution since 2005.

For the late run, ocean-age-2 sockeye salmon returns were predicted based on the abundance of the prior-year return of ocean-age-1 sockeye salmon (2005–present). The ocean-age-1, -3, and -4 return predictions were calculated using their pooled median contribution since 2005.

Regression and median estimates were summed to estimate the total Karluk sockeye salmon run for 2025. The range was estimated as the overall 80% prediction intervals. The combined early- and late-run 80% prediction interval was calculated by summing the lower prediction bounds and upper prediction bounds of the two runs.

Forecast Discussion: The total 2025 sockeye salmon run to the Karluk River using sibling relationships is predicted to be approximately 769,000 fish. The early run is expected to be approximately 95,000 fish, which is about 143,000 fish below the recent 10-year average (239,000 fish) and 23,000 fish above the 2024 run (72,000 fish). The late run is expected to be approximately 674,000 fish which is 220,000 fish below the recent 10-year average (893,000 fish) and 245,000 fish less than the 2024 run (919,000 fish). The combined exponential smoothing estimates for the early and late runs is 1,102,000 fish for comparison.

The projected harvest estimate for the early run (0 fish) is based on the run projection being below the lower end of the early-run escapement goal range. The projected harvest estimate for the late run (349,000 fish) is based on achievement of the mid-point of the late-run escapement goal. The Karluk sockeye salmon run is expected to be predominated by ocean-age-2 fish (70%). The overall confidence in the Karluk sockeye salmon forecast is fair.

Forecast by M. Birch Foster, Finfish Research Biologist, Westward Region.

Alitak District (Frazer Lake and Upper Station)

Sockeye Salmon

Forecast Methods: The 2025 sockeye salmon run to the Alitak District (Table 31) was forecast with linear regression models using ocean-age-class relationships by system from recent outmigration years and environmental indices. In constructing and evaluating each of the regression models, standard regression diagnostic procedures were used. Prediction intervals (80%) for the regression estimates were calculated using the variances of the regression models. The age classes that could not be estimated with one of these models were estimated using pooled medians and the 10th and 90th percentiles of the data were used to calculate the prediction interval of the median estimate.

Table 31.—Preliminary forecast of the 2025 Alitak District sockeye salmon run.

Total production		Forecast estimate (thousands)	Forecast range (thousands)
Early Upper Station	Total run estimate	55	14–117
	Escapement goal ^a	55	43–93
	Harvest estimate ^b	0	—
Late Upper Station	Total run estimate	326	173–626
	Escapement goal	186	120–265
	Harvest estimate ^b	140	—
Frazer Lake	Total run estimate	203	25–404
	Escapement goal ^c	143	95–190
	Harvest estimate ^b	60	—
Total Alitak District	Total run estimate	584	212–1,147
	Escapement goal	384	258–548
	Harvest estimate ^b	200	—

Note: Column numbers may not total or correspond exactly with numbers in text due to rounding.

^a The Upper Station early run is managed for a biological escapement goal (BEG) of 43,000–93,000 fish. The Upper Station early-run forecast of 55, 000 fish, although within the BEG range, falls below the targeted midpoint of 65,000 fish precluding harvest.

^b The harvest of Upper Station-bound sockeye salmon is concurrent with the harvest of Frazer Lake-bound sockeye salmon and predominantly occurs within the Alitak District.

^c The Frazer Lake escapement goal (75,000–170,000) is increased here by an additional 20,000 fish, which is the 20-year median of the number of fish that escape the Dog Salmon weir but do not ascend the fish pass to reach Frazer Lake. The value of 143,000 fish is S_{MSY} plus 20,000 fish.

Upper Station early-run ocean-age-2 returns were forecasted using prior year ocean-age-1 returns (outmigration years: 2005–2023). Upper Station early-run ocean-age-2 salmon were used to predict log-transformed ocean-age-3 fish data from outmigration years 1992 to 2022. Upper

Station late-run ocean-age-2 and log-transformed ocean-age-3 salmon returns were each predicted using their respective prior year ocean-age-1 and -2 returns (late-run outmigration years: 1998–2023 and 1998–2022 respectively). Upper Station early- and late-run ocean-age-1 and -4 returns were calculated using the pooled median contributions by stock and ocean age from the last 10 years. Frazer ocean-age-2 salmon were predicted using prior year log-transformed ocean-age-1 returns (outmigration years: 2003–2023). Frazer ocean-age-3 salmon were predicted using prior year ocean-age-2 returns (outmigration years: 2006–2022). Frazer ocean-age-1 and -4 returns were calculated using the pooled median contributions from the last 15 years.

Regression and median estimates were summed to estimate the total Alitak District sockeye salmon run for 2025. The combined Alitak District prediction interval was calculated by summing the lower and upper prediction bounds of the three runs.

Forecast Discussion: The 2025 sockeye salmon run to the Alitak District is expected to be approximately 584,000 fish, approximately 117,000 less fish than the recent 10-year average run (701,000 fish) and 30,000 fish more than the 2024 run. The Upper Station early run is expected to be approximately 55,000 fish, which is less than the recent 10-year average run (90,000 fish). The Upper Station late run is expected to be approximately 326,000 fish, which is more than the recent 10-year average run (321,000 fish). The Frazer Lake run is expected to be approximately 203,000 fish, which is less than the recent 10-year average (291,000 fish). The 2025 Alitak District sockeye salmon run should be composed of approximately 69% ocean-age-2 fish, 22% ocean-age-3 fish, and 9% ocean-age-1 fish. Overall, our confidence in the forecast is fair based on the strength of the regression models and the large prediction intervals.

The projected harvest estimate of 200,000 fish is based on achieving the escapement that on average is expected to produce maximum sustained yield (i.e., S_{MSY}) estimates for Upper Station early and late runs and S_{MSY} plus an additional 20,000 fish (20-year median of the number of fish that pass through the Dog Salmon weir but do not ascend the Frazer Lake fish pass) for the Frazer run. S_{MSY} is an estimate of the escapement that has the largest expectation of subsequent surplus production.

Forecast by Heather Finkle, Finfish Research Biologist, Westward Region.

Spiridon Lake

Sockeye Salmon

Forecast Methods: The 2025 Spiridon Lake sockeye salmon forecast (Table 32) is estimated by multiplying the estimated number of smolt released for both net pen and lake combined, by the mean smolt to adult survival proportions by ocean age and mean return by age proportion of each age class for completed BY 2013–2017. The Spiridon Lake and Telrod Cove range estimates the 95% confidence intervals created using Goodman’s variance equation (Goodman 1960).

Table 32.—Preliminary forecast of the 2025 Spiridon Lake sockeye salmon run.

Total production	Forecast estimate (thousands)	Forecast range (thousands)
Total run estimate	169	0–353
Spiridon	109	4–213
Telrod Cove net pen	60	0–140
Escapement goal	n/a	—

Note: Column numbers may not total or correspond exactly with numbers in text due to rounding. “n/a” = not applicable.

Forecast Discussion: Sockeye salmon are prevented from returning to Spiridon Lake due to barrier falls blocking upstream migrations in the outlet creek (Telrod Creek). Therefore, all returning adult sockeye salmon are available for harvest, primarily in the Central Section of the Northwest Kodiak District and in the Spiridon Bay Special Harvest Area in Telrod Cove. The point estimate forecast of 169,000 fish in 2025 is less than the 2024 forecast (184,000). A significant increase in the harvest of sockeye salmon bound for Telrod Cove during common property fishing within the Spiridon Bay Section (254-41) was recently discovered. Production estimates now incorporate the combined harvests of Spiridon Bay and Telrod Cove sections to better approximate the run strength and total enhancement production.

The majority of the 2025 run will probably be composed of 2 major age classes: 52.6% age-1.2 fish and 32.5% age-1.3 fish. All other age classes will make up 14.9% of the remaining run estimate. If realized, this run will be below the recent 10-year average (2015 to 2024) run of 227,000 fish. The peak of the Spiridon Lake sockeye salmon run timing through the westside fishery is typically early to mid-July.

The 2025 season will mark the 13th year of adult returns to Telrod Cove originating from net-pen releases. Of the returning fish to Telrod Cove in 2024, age-1.1, -1.2, -1.3, and -1.4 sockeye salmon will be apportioned between lake and net pen production through evaluation of differential otolith marks created during hatchery incubation.

Forecast by Westley Landry-Murphy, Fisheries Biologist, Kodiak Regional Aquaculture Association.

CHIGNIK AREA

Sockeye Salmon

Forecast Methods: Simple linear regressions models using age class relationships were used to forecast the 2025 Chignik River sockeye salmon total run (Table 33).

Each regression model was assessed with standard regression diagnostic procedures. Data were log transformed to address non-normality or unequal variance. Prediction intervals (80%) for the regression estimates were calculated using the variances of the regression models. Age class returns were not estimated with statistical models; instead, they were estimated using pooled medians with data from the last 15 years. Median prediction intervals were calculated from the 10th and 90th percentiles of the data.

The 2025 total Chignik River sockeye salmon run was forecasted by modeling returns of stock components and summing those results by age. Early-run prior year log transformed ocean-age-2 returns predicted early-run log transformed ocean-age-3 returns using data from the 2000

outmigration year to the present. Late-run age-2.3 returns were predicted from age-2.2 returns using data from brood year 2000 to the present.

The regression and median estimates were summed to estimate the total Chignik River sockeye salmon run for 2025. The total run 80% prediction interval was calculated by summing the lower and upper prediction bounds of the 2 runs.

Table 33.—Preliminary forecast of the 2025 Chignik area sockeye salmon run.

Total production		Forecast estimate (thousands)	Forecast range (thousands)
Early run	Total run estimate	593	384–1,984
	Optimal escapement goal ^a	350	300–400
	Harvest estimate ^b	243	—
Late run	Total run estimate	741	173–1,426
	Optimal escapement goal ^a	300	240–360
	Harvest estimate ^b	441	—
Total Chignik System	Total run estimate	1,334	557–3,410
	Biological escapement goal ^c	625	450–800
	Combined OEG	650	540–760
	Chignik Area harvest estimate ^b	684	—
	SEDM ^d	0	—
	Cape Igvak ^d	0	—

Note: Column numbers may not total or correspond exactly with numbers in text due to rounding.

^a During the February 2023 Board of Fisheries meeting, the board instituted optimal escapement goals (OEGs) for the early run (300,000 to 400,000 sockeye salmon) and late run (240,000 to 360,000 sockeye salmon).

^b Harvest represents the surplus achieved beyond the midpoint of the escapement goal. An additional 73,000 fish transiting through the Chignik Management Area (CMA) are forecasted to be harvested beyond the 684,000 fish that are considered Chignik-bound fish in the CMA.

^c The biological escapement goal is for the single, total run goal. Targeting the midpoints of the early- and late-run OEGs is estimated to reduce the harvest by 25,000 fish.

^d Based on historical run size and timing, a harvestable surplus of Chignik River system sockeye salmon is not forecast to exceed 600,000 fish by July 5 in the Chignik Management Area; therefore, as outlined in regulation, no commercial fisheries were forecasted in Cape Igvak during the regulatory timeframe through July 25, and thus the harvest of Chignik-bound sockeye salmon in the Cape Igvak and Southeastern District Mainland (SEDM) sections is expected to be zero.

Forecast Discussion: The 2025 Chignik River sockeye salmon total run is forecasted to be 1.33 million fish, which is 19,000 fish more than the 10-year average run of 1.32 million fish and almost 354,000 fish more than the 2024 total run of 980,000 million fish. The Chignik River sockeye salmon total run is predicted to be composed of approximately 72% ocean-age-3 and 27% ocean-age-2 fish with the early and late runs accounting for 44% and 56% of the total run, respectively.

The projected 2025 harvest estimate of almost 684,000 Chignik-bound fish is based on achievement of the midpoints of the OEG ranges. The harvest estimate does not include other stocks caught while transiting through the CMA, which is estimated to be an additional 73,000 fish. Harvestable surplus of Chignik-bound sockeye salmon in the CMA was forecast to not exceed 600,000 fish by July 25, which would preclude commercial fisheries in the Cape Igvak Section of the Kodiak Management Area and the Southeastern District Mainland (SEDM) of the Alaska Peninsula Management Area.

The wide confidence intervals around the point estimate of the 2025 forecasts reflect the uncertainty inherent in the forecast models. Given the unreliability of sibling relationships used for forecasting the run and the variability in returns, the 2025 forecast may over-, or underestimate returns if environmental variables, which are unknown at this time, remain spurious. Due to the range of variation in the relationships used in these forecasts and their historical accuracy, our confidence in the forecast is fair.

Forecast by Heather Finkle, Finfish Research Biologist, Westward Region.

ALASKA PENINSULA/ALEUTIAN ISLANDS AREA

Bear Lake (Late Run)

Sockeye Salmon

Forecast Methods: The 2025 forecast of the Bear Lake sockeye salmon late run (Table 34) was prepared using simple linear regressions of sibling age classes. Models were evaluated with standard regression diagnostics. Prediction intervals (80%) for the regression estimates were calculated using variances of the regression models. Age classes that could not be estimated with models were estimated using pooled medians and the 10th and 90th percentiles of the data to calculate the 80% prediction interval of the medians.

Table 34.—Preliminary forecast of the 2025 Bear Lake (late run) sockeye salmon run.

Total production	Forecast estimate (thousands)	Forecast range (thousands)
Total run estimate	389	143-558
Escapement goal ^a	156	117-195
Harvest estimate	233	—

^a The escapement estimate is the midpoint of the escapement goal range (117,000–195,000) in 2025.

Ocean-age-3 sockeye salmon were predicted from prior year ocean-age-2 returns from outmigration years 2002 to present. The remaining age-class components of the run were predicted by calculating median returns from the most recent 10 years.

Regression and median estimates were summed to estimate the total Bear Lake late-run sockeye salmon run for 2025. The forecast range was estimated using the sum of the 80% prediction intervals and 10th and 90th percentiles intervals for each age class forecasted.

Forecast Discussion: The 2025 Bear Lake late-run forecast of 389,000 sockeye salmon is roughly 8,000 fish more than the 10-year average of approximately 381,000 fish and 49,000 fish more than the 2024 run of 339,000 sockeye salmon. The greater returns of ocean-age-2 fish in 2024 compared to 2023 is the primary reason for this anticipated larger run. The 2025 late run is expected to be composed of 4% ocean-age-1, 61% ocean-age-2, and 36% ocean-age-3 fish. The projected harvest of 195,000 fish is based on achieving the midpoint (156,000 fish) of the late-run escapement goal range and adequate run strength. Bear River late-run sockeye salmon returns have shown a general decline in the total run since 1990. The wide range around the point forecast is a result of large fluctuations in age-2.2 and age-2.3 sockeye salmon returns. Based on uncertainty associated with the variable predictive capabilities of sibling age class, our confidence in this forecast is fair.

Forecast by Mary Beth Loewen, Finfish Research Biologist, Westward Region and Heather Finkle, Finfish Research Biologist, Westward Region.

Nelson River

Sockeye Salmon

Forecast Methods: The 2025 Nelson River sockeye salmon run (Table 35) was forecasted using simple linear regression of ocean-age-class and parental escapement data from the past 20 years. Standard regression diagnostics were used to evaluate each model. Prediction intervals (80%) for regression estimates were calculated using the variances of the regression models. Age classes that could not be estimated with modeling were estimated using pooled medians, and the 10th and 90th percentiles of the data were used to calculate the prediction interval of the median estimates.

Table 35.—Preliminary forecast of the 2025 Nelson River sockeye salmon run.

Total production	Forecast estimate (thousands)	Forecast range (thousands)
Total run estimate	430	190–659
Escapement goal ^a	158	97–219
Harvest estimate	272	—

Note: Column numbers may not total or correspond exactly with numbers in text due to rounding.

^a The escapement estimate is the midpoint of the escapement goal range (97,000–219,000) in 2025.

The returns of ocean-age-2 fish were predicted using the number of ocean-age-1 fish returning in 2024. The ocean-age-1, -3, and -4 returns were calculated from median estimates for each ocean age class using run data from the previous 15 years.

Regression and median estimates were summed to estimate the total Nelson River sockeye salmon run for 2025. The range was estimated as an overall 80% prediction interval and calculated from the sum of the 80% prediction intervals for each age class forecasted.

Forecast Discussion: The 2025 Nelson River forecast of 430,000 sockeye salmon is about 11,000 fish less than the most recent 10-year average (2015–2024) of 441,000 fish and is approximately 529,000 fish less than the 2024 run of about 959,000 fish. The 2025 run should be composed mainly of ocean-age-2 (77%) and ocean-age-3 (17%) fish. The projected harvest of 272,000 fish is based on achieving the midpoint (158,000 fish) of the escapement goal range.

The Nelson River sockeye salmon run has been notoriously unpredictable. Therefore, confidence in this forecast is fair.

Forecast by Mary Beth Loewen, Finfish Research Biologist, Westward Region

South Alaska Peninsula Aggregate

Pink Salmon

The 2025 South Alaska Peninsula predicted pink salmon harvest is expected to be in the *strong/excellent* category with a point estimate of 10.6 (8.9–14.6) million fish (Tables 36 and 37). Harvest categories were calculated from the 20th, 40th, 60th, and 80th percentiles of historical post-June commercial harvest on the South Alaska Peninsula from 1984 to 2024.

Table 36.—Preliminary forecast of the 2025 South Alaska Peninsula aggregate pink salmon run.

Total production	Forecast estimate (millions)	Forecast range (millions)
Total run estimate ^a	14.6	10.6–18.6
Escapement goal ^b	4.0	–
Post-June harvest estimate	10.6	6.6–14.6

^a Post-June harvest and escapement.

^b The escapement estimate is the upper end of the aggregate goal range (1.75–4.0 million) in 2025.

Table 37.—The 2025 South Alaska Peninsula pink salmon harvest categories, calculated from the 20th, 40th, 60th, and 80th percentiles of historical post-June commercial harvest on the South Alaska Peninsula from 1984 to 2024.

South Peninsula harvest category	Range (millions)	Percentile
Poor	Less than 1.8	Less than 20th
Weak	1.8 to 4.0	20th to 40th
Average	4.0 to 7.0	40th to 60th
Strong	7.0 to 10.0	60th to 80th
Excellent	Greater than 10.0	80th to 100th

Forecast Methods: The South Alaska Peninsula pink salmon harvest forecast is derived from a total run forecast minus the estimated escapement (4.0 million). The total run estimates were derived from a combination of aerial survey index, and harvest estimates.

For the 2025 pink salmon forecast, a generalized Ricker model (Quinn and Deriso 1999) was fit to the odd-year returns from 1987 to 2023 utilizing aerial survey indexed escapement for the spawner index. Three additional terms were included in this generalized Ricker model: Western Gulf of Alaska May-June SST, and two environmental composites created from a series of forecast indices affecting pink salmon returns using climate data from (1) Kodiak and (2) Cold Bay. The environmental variables used to create the composite included monthly mean air temperature, total precipitation and peak precipitation total run correlation anomalies from August to June of predominantly freshwater life history.

In constructing and evaluating the regression model standard regression diagnostic procedures were used. Forecast range was estimated using the 80% confidence intervals of the absolute percent error of the composite model hindcast estimates.

Forecast Discussion: June harvest of pink salmon has been omitted from the South Alaska Peninsula aggregate pink salmon forecast due to the variability of pink salmon harvest that occurs during the June fishery and the origin of these fish are unknown. The 5-year odd-year average harvest of pink salmon in June is approximately 3.1 million fish, with a range of 0.2–9.0 million fish.

The estimated 2025 South Alaska Peninsula pink salmon total harvest (10.6 million fish) is predicted to be *strong/excellent*. Spawning escapement in 2023 was excellent. There were favorable fresh water spawning environmental conditions for adults and winter and spring temperature rearing conditions for juvenile pinks also appear favorable. Since South Peninsula pink salmon forecasting began in 2011, odd-year forecasts tend to be more accurate than those of

the even-year cycle. Accounting for uncertainty surrounding environmental conditions, confidence in the forecast is fair.

Forecast by M. Birch Foster, Finfish Research Biologist, Westward Region.

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